

Electrical Machines PYQ Answer Bank

All 203 in-syllabus questions answered in exam-writing format

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1. Basic rotating-machine principles and energy conversion

Syllabus area: Weeks 2-3

Question 1

Original: 2A | EM-I ELE 205 Makeup, 05 January 2015

From fundamentals of force acting on conductors, derive an expression for the torque developed in a DC motor. 3

Answer 1

The force on a conductor of active length l carrying current I in a magnetic field B is $F = BIl$. In a DC motor the armature conductors lie under alternate north and south poles; commutation reverses the conductor current at the proper instant, so the developed force continues to produce torque in the same mechanical direction.

Let Z be the total armature conductors, A the number of parallel paths, P the number of poles, ϕ the useful flux per pole, and I_a the armature current. Current in each conductor is I_a/A . The average flux density under one pole is $B = \phi/(\text{pole area})$. Instead of carrying the pole area explicitly, the standard DC-machine result is obtained from power conversion.

Back emf of a DC armature is $E = (P \phi Z N)/(60 A)$, where N is speed in rpm. The converted mechanical power is $P_m = E I_a$. Also $P_m = T \omega = T (2 \pi N/60)$. Equating both expressions:

$$T (2 \pi N/60) = [(P \phi Z N)/(60 A)] I_a$$

$$\text{Therefore } T = (P Z / 2 \pi A) \phi I_a = k \phi I_a.$$

Final: The torque developed in a DC motor is directly proportional to flux per pole and armature current, $T = k \phi I_a$.

2. AC windings, winding factor, harmonics and generated EMF

Syllabus area: Weeks 3-4

Question 2

Original: 5A | EM-I ELE 2103 End Sem, 27 November 2018

Develop a single layer wave winding table for the stator winding of a three phase AC machine with 36 slots and 4 poles. Assume RBY sequence. (04)

Answer 2

For 36 slots and 4 poles, slots per pole = $36/4 = 9$ and slot angle $\beta = 180/9 = 20$ electrical degrees. For a three-phase winding, $q = S/(3P) = 36/(3 \times 4) = 3$ slots per pole per phase.

For a full-pitch single-layer winding, coil pitch = slots per pole = 9 slots. Therefore the coil pairs are 1-10, 2-11, 3-12, ..., 27-36, 28-1, 29-2, ..., 36-9. For a single-layer winding the number of coils is $S/2 = 18$.

Using RBY phase sequence and 60-degree phase belts, one valid slot allocation is:

Slots 1-3: R+, 4-6: B-, 7-9: Y+, 10-12: R-, 13-15: B+, 16-18: Y-, 19-21: R+, 22-24: B-, 25-27: Y+, 28-30: R-, 31-33: B+, 34-36: Y-.

The corresponding full-pitch coil groups are:

R phase: 1-10, 2-11, 3-12, 19-28, 20-29, 21-30.

B phase: 4-13, 5-14, 6-15, 22-31, 23-32, 24-33.

Y phase: 7-16, 8-17, 9-18, 25-34, 26-35, 27-36.

Connect each phase group in series with coil polarities chosen so that emfs in that phase add. The three phase terminals are then taken in R-B-Y order, giving balanced phase emfs displaced by 120 electrical degrees.

Final: A valid single-layer full-pitch wave winding has 18 coils, coil pitch 9 slots, $q = 3$, $\beta = 20$ electrical degrees, and the R/B/Y coil groups listed above.

Question 3

Original: 1A | EM-II ELE 204 End Sem, 13 May 2014

Define the following: (i) Integral and Fractional slot winding (ii) Full pitch and Fractional pitch winding (02)

Answer 3

Use the winding geometry first, then apply the pitch, distribution, and emf relations with all angles in electrical degrees.

Step 1: Integral-slot winding: slots per pole per phase $m = S / (P \times \text{phases})$ is an integer. Example: $m = 3$.

Step 2: Fractional-slot winding: m is not an integer. It is used to reduce cogging, harmonics and sometimes to suit machine design constraints.

Step 3: Full-pitch winding: coil span is one pole pitch, i.e. 180 electrical degrees. It gives maximum fundamental coil emf.

Step 4: Fractional-pitch or short-pitch winding: coil span is less than 180 electrical degrees by chording angle α . It slightly reduces fundamental emf but can suppress selected harmonics. Distributed coil emfs are phasor-added

Final: Full-pitch winding: coil span is one pole pitch, i.e. 180 electrical degrees. It gives maximum fundamental coil emf.; Fractional-pitch or short-pitch winding: coil span is less than 180 electrical degrees by chording angle α . It slightly reduces fundamental emf but can suppress selected harmonics. Distributed coil emfs are phasor-added

Question 4

Original: 1B | EM-II ELE 204 End Sem. 13 May 2014

A 3 phase alternator has 2 slots per pole per phase and coil span of 5 slot pitches. The flux density wave of alternator consists of a fundamental and a 25% third harmonic. Calculate the percentage increase in the phase voltage due to harmonic. (04)

Answer 4

Use the winding geometry first, then apply the pitch, distribution, and emf relations with all angles in electrical degrees.

Step 1: compute slots per pole/phase $m = 2$, hence slots/pole = 6 and $\beta = 180/6 = 30^\circ$.

Step 2: Coil span = 5 slot pitches, so $\alpha = 1$ slot pitch = 30° .

Step 3: $K_{w1} = K_{d1} K_{p1} = 0.93301$; $K_{w3} = K_{d3} K_{p3} = 0.50000$.

Step 4: for a flux-density harmonic problem, $E_3/E_1 = (B_{\max 3} K_{w3})/(B_{\max 1} K_{w1})$.

Step 5: $E_3/E_1 = 0.25 \times 0.50000/0.93301 = 0.13397$.

Step 6: Resultant phase voltage = $\sqrt{(E_1^2 + E_3^2)} = 1.0089 E_1$.

Step 7: Percentage increase = 0.89347%.

Final: For a flux-density harmonic problem, $E_3/E_1 = (B_{\max 3} K_{w3})/(B_{\max 1} K_{w1})$.; $E_3/E_1 = 0.25 \times 0.50000/0.93301 = 0.13397$.; Resultant phase voltage = $\sqrt{(E_1^2 + E_3^2)} = 1.0089 E_1$.; Percentage increase = 0.89347%.

Question 5

Original: 1A | EM-II ELE 204 Makeup. 08 July 2014

A 3 phase, 50 Hz, 1000 rpm, star connected alternator has 72 armature slots with 6 conductors per slot and the coil span is 10 slots. The average air-gap flux per pole is 0.26 Wb. Calculate: (i) Distribution and Pitch factors of the winding (ii) Number of turns per phase and (iii) phase and line value of emf induced. (04)

Answer 5

Use the winding geometry first, then apply the pitch, distribution, and emf relations with all angles in electrical degrees.

Step 1: compute slots per pole = $72/6 = 12.000$, $\beta = 180/(\text{slots/pole}) = 15.000^\circ$, $m = S/(3P) = 4.0000$.

Step 2: the chording angle $\alpha = 30.000^\circ$. $K_d = \sin(m \beta/2)/(m \sin(\beta/2)) = 0.95766$; $K_p = \cos(\alpha/2) = 0.96593$; $K_w = 0.92503$.

Step 3: the series turns per phase $T_{ph} = S \times \text{conductors/slot} / (2 \times 3) = 72.000$ turns.

Step 4: $E_{ph} = 4.44 f \Phi T_{ph} K_w = 3,844.3$ V; for star connection the line voltage $E_L = \sqrt{3} E_{ph} = 6,658.5$ V.

Final: The series turns per phase $T_{ph} = S \times \text{conductors/slot} / (2 \times 3) = 72.000$ turns.; $E_{ph} = 4.44 f \Phi T_{ph} K_w = 3,844.3$ V; for star connection the line voltage $E_L = \sqrt{3} E_{ph} = 6,658.5$ V.

Question 6

Original: 2A | EM-II ELE 204 Makeup. 08 July 2014

A 3 phase, 50 Hz, 750 rpm alternator has its armature winding short pitched by two slots resulting in a coil span of 144° electrical Determine (i) total number of armature slots (ii) minimum order of harmonic emf that can be suppressed. (03)

Answer 6

First find the number of poles from synchronous speed. $N_s = 120 f/P$, so $P = 120 \times 50/750 = 8$ poles.

The coil span is 144 electrical degrees. Therefore the short-pitch angle is $\alpha = 180 - 144 = 36$ electrical degrees. The winding is short pitched by two slots, so one slot angle $\beta = 36/2 = 18$ electrical degrees.

Slots per pole = $180/\beta = 180/18 = 10$ slots/pole. Hence total armature slots = $P \times \text{slots per pole} = 8 \times 10 = 80$.

For harmonic elimination by short pitching, $K_{pn} = \cos(n \alpha/2)$. The harmonic is suppressed when $K_{pn} = 0$, i.e. $n \alpha/2 = 90^\circ$. Thus $n = 180/\alpha = 180/36 = 5$.

Final: Total armature slots = 80; the lowest suppressed harmonic is the 5th harmonic.

Question 7

Original: 1A | EM-II ELE 204 Makeup, 09 July 2015

Discuss the advantages of adopting short pitched windings for the armature of a synchronous machine. (02)

Answer 7

Use the winding geometry first, then apply the pitch, distribution, and emf relations with all angles in electrical degrees.

Short-pitched windings are used when the coil span is less than a full pole pitch.

Step 1: Reduces selected harmonic emfs. If $K_{pn} = \cos(n \alpha / 2)$ is zero, the n th harmonic is eliminated.

Step 2: Improves waveform of generated voltage by reducing harmonic content.

Step 3: Reduces length of end connections, saving copper and reducing copper loss.

Step 4: Reduces leakage reactance and improves cooling around end windings.

Step 5: Makes the machine quieter and reduces pulsating/stray losses due to harmonics. The tradeoff is that the fundamental emf is reduced by $K_p = \cos(\alpha / 2)$, so winding factor must be considered in the emf equation.

Final: Reduces selected harmonic emfs. If $K_{pn} = \cos(n \alpha / 2)$ is zero, the n th harmonic is eliminated.; Makes the machine quieter and reduces pulsating/stray losses due to harmonics. The tradeoff is that the fundamental emf is reduced by $K_p = \cos(\alpha / 2)$, so winding factor must be considered in the emf equation.

Question 8

Original: 1B | EM-II ELE 204 Makeup, 09 July 2015

For a 3-phase, 50 Hz, 10 pole alternator with 90 slots with a 60° phase spread with a coil span of 140° , obtain the pitch and distribution factors for fundamental, 3rd and 5th harmonic emfs. (03)

Answer 8

Use the winding geometry first, then apply the pitch, distribution, and emf relations with all angles in electrical degrees.

Step 1: compute slots per pole = $90/10 = 9$, $\beta = 180/9 = 20^\circ$, $m = 90/(10 \times 3) = 3$.

Step 2: Coil span = 140° , hence $\alpha = 40^\circ$.

Step 3: Fundamental: $K_{p1} = 0.93969$, $K_{d1} = 0.95980$, $K_{w1} = 0.90191$.

Step 4: 3rd harmonic: $K_{p3} = 0.50000$, $K_{d3} = 0.66667$, $K_{w3} = 0.33333$.

Step 5: 5th harmonic: $K_{p5} = -0.17365$, $K_{d5} = 0.21757$; magnitude of $K_{w5} = 0.03778$.

Final: Coil span = 140° , hence $\alpha = 40^\circ$.; Fundamental: $K_{p1} = 0.93969$, $K_{d1} = 0.95980$, $K_{w1} = 0.90191$.; 3rd harmonic: $K_{p3} = 0.50000$, $K_{d3} = 0.66667$, $K_{w3} = 0.33333$.; 5th harmonic: $K_{p5} = -0.17365$, $K_{d5} = 0.21757$; magnitude of $K_{w5} = 0.03778$.

Question 9

Original: 1C | EM-II ELE 204 Makeup, 09 July 2015

A 3 phase, 6 pole, 1000 rpm star connected alternator has an air-gap diameter of 40 cm and a core length of 90 cm. The armature consists of 54 slots with 4 conductors per slot. The flux density in the air-gap is given by $B(\theta) = 0.25 \times \sin \theta + 0.16 \times \sin 3\theta + 0.07 \times \sin 5\theta$. The winding factors for fundamental, 3rd and 5th harmonic are 0.95, 0.58 and 0.14 respectively. Determine resultant phase and line emfs. (05)

Answer 9

Use the winding geometry first, then apply the pitch, distribution, and emf relations with all angles in electrical degrees.

Step 1: $f = PN/120 = 6 \times 1000/120 = 50$ Hz. Total turns/phase = $54 \times 4/(2 \times 3) = 36$.

Step 2: for $B_1 = 0.25$ T, average flux/pole = $2 B_{\max} D L/P = 0.03$ Wb.

Step 3: $E_1 = 4.44 f \Phi_1 T_{ph} K_{w1} = 227.77$ V.

Step 4: using $E_n/E_1 = (B_{\max,n} K_{wn})/(B_{\max,1} K_{w1})$: $E_3 = 89.00$ V and $E_5 = 9.40$ V.

Step 5: Resultant phase emf = $\sqrt{(E_1^2 + E_3^2 + E_5^2)} = 244.72$ V.

Step 6: In a star connection, triplen harmonics do not appear in line voltage. Therefore $E_L = \sqrt{3} \sqrt{(E_1^2 + E_5^2)} = 394.85$ V.

Final: For $B_1 = 0.25$ T, average flux/pole = $2 B_{\max} D L/P = 0.03$ Wb.; $E_1 = 4.44 f \Phi_1 T_{ph} K_{w1} = 227.77$ V.; Resultant phase emf = $\sqrt{(E_1^2 + E_3^2 + E_5^2)} = 244.72$ V.; In a star connection, triplen harmonics do not appear in line voltage. Therefore $E_L = \sqrt{3} \sqrt{(E_1^2 + E_5^2)} = 394.85$ V.

Question 10

Original: 3A | EM-II ELE 2202 End Sem, 23 April 2018

A 3 phase, 10 pole, star connected alternator runs at 600 rpm. It has 120 stator slots with 8 conductors per slot. The conductors of each phase are connected in series. If the winding is short chorded by two slots, determine the rms value of phase and line electromotive forces if the flux per pole is 56 mWb. (04)

Answer 10

Use the winding geometry first, then apply the pitch, distribution, and emf relations with all angles in electrical degrees.

Step 1: compute slots per pole = $120/10 = 12.000$, $\beta = 180/(\text{slots/pole}) = 15.000^\circ$, $m = S/(3P) = 4.0000$.

Step 2: the chording angle $\alpha = 30.000^\circ$. $K_d = \sin(m\beta/2)/(m \sin(\beta/2)) = 0.95766$; $K_p = \cos(\alpha/2) = 0.96593$; $K_w = 0.92503$.

Step 3: the series turns per phase $T_{ph} = S \times \text{conductors/slot} / (2 \times 3) = 160.00$ turns.

Step 4: $E_{ph} = 4.44 f \Phi T_{ph} K_w = 1,840.0$ V; for star connection the line voltage $E_L = \sqrt{3} E_{ph} = 3,187.0$ V.

Final: The series turns per phase $T_{ph} = S \times \text{conductors/slot} / (2 \times 3) = 160.00$ turns.; $E_{ph} = 4.44 f \Phi T_{ph} K_w = 1,840.0$ V; for star connection the line voltage $E_L = \sqrt{3} E_{ph} = 3,187.0$ V.

Question 11

Original: 4B | EM-II ELE 2202 End Sem, 24 April 2017

A three-phase, 50 Hz, 1,000 rpm alternator has 108 slots. The armature is lap connected with a coil span of 160° . Calculate pitch & distribution factors for fundamental, 5th harmonic & 7th harmonic. (03)

Answer 11

Use the winding geometry first, then apply the pitch, distribution, and emf relations with all angles in electrical degrees.

Step 1: $N_s = 1000 \text{ rpm}$, $f = 50 \text{ Hz} \rightarrow P = 120f/N_s = 6 \text{ poles}$.

Step 2: compute slots per pole $= 108/6 = 18$, $\beta = 10^\circ$, $m = 108/(6 \times 3) = 6$, coil span $= 160^\circ$ so $\alpha = 20^\circ$.

Step 3: Fundamental: $K_{p1} = 0.98481$, $K_{d1} = 0.95614$, $K_{w1} = 0.94162$.

Step 4: 5th: $K_{p5} = 0.64279$, $K_{d5} = 0.19718$, $K_{w5} = 0.12675$.

Step 5: 7th: $K_{p7} = 0.34202$, $K_{d7} = -0.14529$; magnitude of $K_{w7} = 0.04969$.

Final: Compute slots per pole $= 108/6 = 18$, $\beta = 10^\circ$, $m = 108/(6 \times 3) = 6$, coil span $= 160^\circ$ so $\alpha = 20^\circ$.; Fundamental: $K_{p1} = 0.98481$, $K_{d1} = 0.95614$, $K_{w1} = 0.94162$.; 5th: $K_{p5} = 0.64279$, $K_{d5} = 0.19718$, $K_{w5} = 0.12675$.; 7th: $K_{p7} = 0.34202$, $K_{d7} = -0.14529$; magnitude of $K_{w7} = 0.04969$.

Question 12

Original: 3C | EM-II ELE 2202 End Sem. 29 April 2019

In a star-connected alternator calculate the percentage of 5th harmonics induced in the line voltage if the average flux per pole of fifth harmonic components is 1% of that of the fundamental. Neglect the pitch and distribution factors. (02)

Answer 12

Use the winding geometry first, then apply the pitch, distribution, and emf relations with all angles in electrical degrees.

Step 1: for harmonic emf, $E_n/E_1 = n \Phi_n/\Phi_1$ when pitch and distribution factors are neglected.

Step 2: Given $\Phi_5 = 0.01 \Phi_1$, $E_{5\text{phase}}/E_{1\text{phase}} = 5 \times 0.01 = 0.05$.

Step 3: In a star connection, the 5th harmonic appears in line voltage also; the $\sqrt{3}$ factor cancels in the percentage ratio.

Final: fifth-harmonic line-voltage component = 5% of the fundamental line voltage.

Question 13

Original: 2B | EM-II ELE 2202 Makeup. 02 July 2016

A three phase, 50 Hz, 1000 rpm, star connected alternator has an air-gap area of 314 cm² per pole. The armature winding has a phase spread of 60° and is accommodated in 4 slots per pole per phase with 6 conductors per slot. The coils are short pitched with a span of 165° . The flux density distribution is given by $B_{\max} = 0.9 \sin + 0.35 \sin 3$. Determine, (a) Winding factors for fundamental and third harmonic components. (b) Induced emf per phase. 6M

Answer 13

Use the winding geometry first, then apply the pitch, distribution, and emf relations with all angles in electrical degrees.

Step 1: $P = 120f/N_s = 6 \text{ poles}$. Given 4 slots/pole/phase $\rightarrow m = 4$, slots/pole $= 12$ and $\beta = 15^\circ$.

Step 2: Coil span $= 165^\circ$, hence $\alpha = 15^\circ$.

Step 3: $K_{w1} = K_{d1} K_{p1} = 0.94947$; $K_{w3} = K_{d3} K_{p3} = 0.60355$.

Final: $P = 120f/N_s = 6 \text{ poles}$. Given 4 slots/pole/phase $\rightarrow m = 4$, slots/pole $= 12$ and $\beta = 15^\circ$.; Coil span $= 165^\circ$, hence $\alpha = 15^\circ$.; $K_{w1} = K_{d1} K_{p1} = 0.94947$; $K_{w3} = K_{d3} K_{p3} = 0.60355$.

Question 14

Original: 3A | EM-II ELE 2202 Makeup, 13 June 2019

A 3-phase, 50 Hz, 1000 rpm, star connected alternator has 72 armature slots with 6 conductors per slot and the coil span is 10 slots. The average air-gap flux per pole is 0.26Wb. Calculate the distribution and pitch factors of the winding, number of turns per phase and line value of emf induced. (05)

Answer 14

Use the winding geometry first, then apply the pitch, distribution, and emf relations with all angles in electrical degrees.

Step 1: compute slots per pole = $72/6 = 12.000$, $\beta = 180/(\text{slots/pole}) = 15.000^\circ$, $m = S/(3P) = 4.0000$.

Step 2: the chording angle $\alpha = 30.000^\circ$. $K_d = \sin(m\beta/2)/(m\sin(\beta/2)) = 0.95766$; $K_p = \cos(\alpha/2) = 0.96593$; $K_w = 0.92503$.

Step 3: the series turns per phase $T_{ph} = S \times \text{conductors/slot} / (2 \times 3) = 72.000$ turns.

Step 4: $E_{ph} = 4.44 f \Phi T_{ph} K_w = 3,844.3$ V; for star connection the line voltage $E_L = \sqrt{3} E_{ph} = 6,658.5$ V.

Final: The series turns per phase $T_{ph} = S \times \text{conductors/slot} / (2 \times 3) = 72.000$ turns.; $E_{ph} = 4.44 f \Phi T_{ph} K_w = 3,844.3$ V; for star connection the line voltage $E_L = \sqrt{3} E_{ph} = 6,658.5$ V.

Question 15

Original: 3B | EM-II ELE 2202 Makeup, 16 June 2017

A 3-phase, 50 Hz, 1000 rpm, star connected alternator has 72 armature slots with 6 conductors per slot and the coil span is 10 slots. The average air-gap flux per pole is 0.26Wb. Calculate the distribution and pitch factors of the winding, number of turns per phase and line value of emf induced. (04)

Answer 15

Use the winding geometry first, then apply the pitch, distribution, and emf relations with all angles in electrical degrees.

Step 1: compute slots per pole = $72/6 = 12.000$, $\beta = 180/(\text{slots/pole}) = 15.000^\circ$, $m = S/(3P) = 4.0000$.

Step 2: the chording angle $\alpha = 30.000^\circ$. $K_d = \sin(m\beta/2)/(m\sin(\beta/2)) = 0.95766$; $K_p = \cos(\alpha/2) = 0.96593$; $K_w = 0.92503$.

Step 3: the series turns per phase $T_{ph} = S \times \text{conductors/slot} / (2 \times 3) = 72.000$ turns.

Step 4: $E_{ph} = 4.44 f \Phi T_{ph} K_w = 3,844.3$ V; for star connection the line voltage $E_L = \sqrt{3} E_{ph} = 6,658.5$ V.

Final: The series turns per phase $T_{ph} = S \times \text{conductors/slot} / (2 \times 3) = 72.000$ turns.; $E_{ph} = 4.44 f \Phi T_{ph} K_w = 3,844.3$ V; for star connection the line voltage $E_L = \sqrt{3} E_{ph} = 6,658.5$ V.

Question 16

Original: 3A | EM-II ELE 2202 Makeup, 19 June 2018

For a synchronous generator, derive suitable expressions for (i) Pitch factor (ii) Distribution factor (04)

Answer 16

Use the winding geometry first, then apply the pitch, distribution, and emf relations with all angles in electrical degrees.

Pitch factor and distribution factor reduce the ideal emf because the coil sides may not be exactly 180 electrical degrees apart and the phase belt is distributed over several slots. Pitch factor, K_p : In a full-pitch coil, the emfs in the two coil sides are directly opposite in space and add arithmetically. If the coil is short-pitched by chording angle α , the resultant coil emf is the phasor sum of two equal emfs separated by α .

Final: Pitch factor, K_p : In a full-pitch coil, the emfs in the two coil sides are directly opposite in space and add arithmetically. If the coil is short-pitched by chording angle α , the resultant coil emf is the phasor sum of two equal emfs separated by α .

Question 17

Original: 1B | EM-II ELE 2225 End Sem, 09 May 2024

A 3-phase, 32-pole, 50Hz alternator used in a hydro-generating station has a star-connected winding with 288 slots and 8 conductors per slot. The coil is short pitched by 40° . The flux per pole is 0.08 Wb. Estimate the phase and line values of induced emfs. Assume the total turns per phase are series connected. 4

Answer 17

Use the winding geometry first, then apply the pitch, distribution, and emf relations with all angles in electrical degrees.

Step 1: compute slots per pole $= 288/32 = 9.0000$, $\beta = 180/(\text{slots/pole}) = 20.000^\circ$, $m = S/(3P) = 3.0000$.

Step 2: the chording angle $\alpha = 40.000^\circ$. $K_d = \sin(m\beta/2)/(\sin(\beta/2)) = 0.95980$; $K_p = \cos(\alpha/2) = 0.93969$; $K_w = 0.90191$.

Step 3: the series turns per phase $T_{ph} = S \times \text{conductors/slot} / (2 \times 3) = 384.00$ turns.

Step 4: $E_{ph} = 4.44 f \Phi T_{ph} K_w = 6,150.9$ V; for star connection the line voltage $E_L = \sqrt{3} E_{ph} = 10,654$ V.

Final: The series turns per phase $T_{ph} = S \times \text{conductors/slot} / (2 \times 3) = 384.00$ turns.; $E_{ph} = 4.44 f \Phi T_{ph} K_w = 6,150.9$ V; for star connection the line voltage $E_L = \sqrt{3} E_{ph} = 10,654$ V.

Question 18

Original: 1B | EM-II ELE 2225 Makeup, 26 June 2024

A 3-phase alternator has 2 slots per pole per phase and a coil span of 5 slot pitch. The flux density wave of alternator consists of a fundamental and a 25% third harmonic. Calculate the percentage increase in the phase voltage due to harmonic. 3

Answer 18

Use the winding geometry first, then apply the pitch, distribution, and emf relations with all angles in electrical degrees.

Step 1: compute slots per pole/phase $m = 2$, hence slots/pole $= 6$ and $\beta = 180/6 = 30^\circ$.

Step 2: Coil span $= 5$ slot pitches, so $\alpha = 1$ slot pitch $= 30^\circ$.

Step 3: $K_{w1} = K_{d1} K_{p1} = 0.93301$; $K_{w3} = K_{d3} K_{p3} = 0.50000$.

Step 4: for a flux-density harmonic problem, $E_3/E_1 = (B_{\max 3} K_{w3})/(B_{\max 1} K_{w1})$.

Step 5: $E_3/E_1 = 0.25 \times 0.50000/0.93301 = 0.13397$.

Step 6: Resultant phase voltage $= \sqrt{(E_1^2 + E_3^2)} = 1.0089 E_1$.

Step 7: Percentage increase $= 0.89347 \%$.

Final: For a flux-density harmonic problem, $E_3/E_1 = (B_{\max 3} K_{w3})/(B_{\max 1} K_{w1})$.; $E_3/E_1 = 0.25 \times 0.50000/0.93301 = 0.13397$.; Resultant phase voltage $= \sqrt{(E_1^2 + E_3^2)} = 1.0089 E_1$.; Percentage increase $= 0.89347 \%$.

Question 19

Original: 1C | EM-II ELE 2251 End Sem, 31 May 2023

Determine the distribution factor corresponding to the fifth harmonic component of generated voltage in a three-phase, 50 Hz, AC generator with 54 slots & 6 poles. Also, comment on the effects of the fifth harmonic component in the generated voltage. (03)

Answer 19

Use the winding geometry first, then apply the pitch, distribution, and emf relations with all angles in electrical degrees.

Step 1: compute slots per pole = $54/6 = 9$, $\beta = 180/9 = 20^\circ$.

Step 2: compute slots per pole/phase $m = 54/(6 \times 3) = 3$.

Step 3: for the 5th harmonic: $K_{d5} = \sin(3 \times 5 \times 20/2)/(3 \sin(5 \times 20/2)) = 0.21757$.

Step 4: the 5th harmonic is a negative-sequence harmonic in a 3-phase alternator; it produces waveform distortion and can cause additional losses/heating. It does not cancel from line voltage like triplen harmonics.

Final: For the 5th harmonic: $K_{d5} = \sin(3 \times 5 \times 20/2)/(3 \sin(5 \times 20/2)) = 0.21757$; the 5th harmonic is a negative-sequence harmonic in a 3-phase alternator; it produces waveform distortion and can cause additional losses/heating. It does not cancel from line voltage like triplen harmonics.

Question 20

Original: 2B | EM-II ELE 2251 End Sem, 31 May 2023

Consider a 3-phase, 8 pole, 50 Hz, star-connected alternator. The average flux in the machine is 40 mWb /pole. The armature has 72 slots with 10 conductors in each slot. Calculate the induced voltage per phase. The coils are short-pitched by 2 slots. (04)

Answer 20

Use the winding geometry first, then apply the pitch, distribution, and emf relations with all angles in electrical degrees.

Step 1: compute slots per pole = $72/8 = 9.0000$, $\beta = 180/(\text{slots/pole}) = 20.000^\circ$, $m = S/(3P) = 3.0000$.

Step 2: the chording angle $\alpha = 40.000^\circ$. $K_d = \sin(m \beta/2)/(m \sin(\beta/2)) = 0.95980$; $K_p = \cos(\alpha/2) = 0.93969$; $K_w = 0.90191$.

Step 3: the series turns per phase $T_{ph} = S \times \text{conductors/slot} / (2 \times 3) = 120.00$ turns.

Step 4: $E_{ph} = 4.44 f \Phi T_{ph} K_w = 961.08$ V; for star connection the line voltage $E_L = \sqrt{3} E_{ph} = 1,664.6$ V.

Final: The series turns per phase $T_{ph} = S \times \text{conductors/slot} / (2 \times 3) = 120.00$ turns.; $E_{ph} = 4.44 f \Phi T_{ph} K_w = 961.08$ V; for star connection the line voltage $E_L = \sqrt{3} E_{ph} = 1,664.6$ V.

Question 21

Original: 3A | EM-II ELE 2251 Grade Improvement, 11 August 2021

A 3-phase, 50 Hz, 1000 rpm, star connected alternator has 72 armature slots with 6 conductors per slot and the coil span is 10 slots. The average air-gap flux per pole is 0.26Wb. Calculate the distribution and pitch factors of the winding, number of turns per phase and line value of emf induced. (06)

Answer 21

Use the winding geometry first, then apply the pitch, distribution, and emf relations with all angles in electrical degrees.

Step 1: compute slots per pole = $72/6 = 12.000$, $\beta = 180/(\text{slots/pole}) = 15.000^\circ$, $m = S/(3P) = 4.0000$.

Step 2: the chording angle $\alpha = 30.000^\circ$. $K_d = \sin(m \beta/2)/(m \sin(\beta/2)) = 0.95766$; $K_p = \cos(\alpha/2) = 0.96593$; $K_w = 0.92503$.

Step 3: the series turns per phase $T_{ph} = S \times \text{conductors/slot} / (2 \times 3) = 72.000$ turns.

Step 4: $E_{ph} = 4.44 f \Phi T_{ph} K_w = 3,844.3$ V; for star connection the line voltage $E_L = \sqrt{3} E_{ph} = 6,658.5$ V.

Final: The series turns per phase $T_{ph} = S \times \text{conductors/slot} / (2 \times 3) = 72.000$ turns.; $E_{ph} = 4.44 f \Phi T_{ph} K_w = 3,844.3$ V; for star connection the line voltage $E_L = \sqrt{3} E_{ph} = 6,658.5$ V.

Question 22

Original: 4B-ii | EM-II ELE 2251 Grade Improvement. 11 August 2021

The stator of a 3-phase alternator has nine slots per pole and carries a balanced 3-phase, double-layer winding. The coils are short pitched and the coil pitch is seven slots. Find the distribution factor and pitch factor. (part ii)

Answer 22

There are 9 slots per pole in a balanced three-phase winding, so slots per pole per phase $q = 9/3 = 3$. The slot angle is $\beta = 180/9 = 20$ electrical degrees.

For a distributed phase belt, the distribution factor is $K_d = \sin(q \beta/2)/(q \sin(\beta/2))$.

Substituting $q = 3$ and $\beta = 20^\circ$:

$$K_d = \sin(30^\circ)/(3 \sin(10^\circ)) = 0.9598.$$

The full-pitch coil span is 9 slots. The actual coil pitch is 7 slots, so the coil is short by 2 slots. The chording angle is $\alpha = 2 \times 20 = 40$ electrical degrees.

$$\text{Pitch factor } K_p = \cos(\alpha/2) = \cos(20^\circ) = 0.9397.$$

Final: Distribution factor $K_d = 0.960$ and pitch factor $K_p = 0.940$.

3. Three-phase induction motor fundamentals, equivalent circuit and power flow

Syllabus area: Weeks 6-8

Question 23

Original: 5B | EM-I ELE 205 End Sem, 04 December 2006

A 115 V, 60 Hz, 3 phase star connected, 6 pole induction motor has stator impedance of $(0.07+j0.3) \Omega$ and equivalent rotor impedance at standstill of $(0.08+j0.3) \Omega$. Magnetising branch has $G_0 = 0.022$ and $B_0 = 0.158$. Using the approximate equivalent circuit, at a slip of 2% determine: (i) Rotor current (ii) Stator current (iii) Power input and input power factor (iv) Power output (v) Torque developed (vi) Efficiency of the motor.

Answer 23

Use the induction-motor slip, rotor-frequency, and torque relations, keeping clear which quantities are standstill and which are at running slip.

Step 1: $V_{ph} = 115/\sqrt{3} = 66.40$ V, $s = 0.02$.

Step 2: Rotor branch = $0.08/0.02 + j0.3 = 4+j0.3 \Omega$; stator impedance = $0.07+j0.3 \Omega$.

Step 3: Rotor current = $V_{ph}/(Z_1+Z_2/s) = 16.14$ A.

Step 4: Exciting current = $V_{ph}(G_0 - jB_0) = 66.4(0.022 - j0.158)$ A. Total stator current = 21.65 A.

Step 5: Input pf = 0.805 lagging; input power = 3.471 kW.

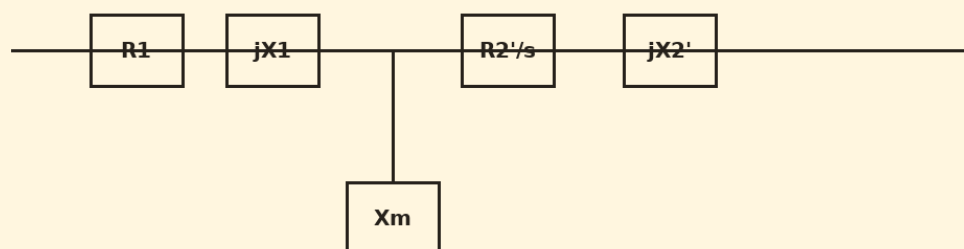
Step 6: Rotor copper loss = 62.5 W; air-gap power = 3.126 kW; mechanical output approx = 3.063 kW.

Step 7: Developed torque is 24.87 N-m; efficiency = 88.2%.

Final: Exciting current = $V_{ph}(G_0 - jB_0) = 66.4(0.022 - j0.158)$ A. Total stator current = 21.65 A.; Input pf = 0.805 lagging; input power = 3.471 kW.; Rotor copper loss = 62.5 W; air-gap power = 3.126 kW; mechanical output approx = 3.063 kW.; Developed torque is 24.87 N-m; efficiency = 88.2%.

Equivalent Circuit

Approximate per-phase induction motor circuit



Question 24

Original: 7A | EM-I ELE 205 End Sem, 04 December 2006

The rotor of a 6-pole, 50 Hz, slip ring induction motor has a resistance of 0.2Ω /phase and runs at 960 rpm on full load. Calculate the approximate resistance/phase to be included in the rotor circuit such that the speed is reduced to 800 rpm for full load torque.

Answer 24

Use the induction-motor slip, rotor-frequency, and torque relations, keeping clear which quantities are standstill and which are at running slip.

Step 1: $N_s = 120f/P = 1000$ rpm.

Step 2: Initial full-load slip = $(1000-960)/1000 = 0.04$. Required slip = $(1000-800)/1000 = 0.20$.

Step 3: for the same torque, R_{total}/s is approximately constant.

Step 4: $R_{total,new} = 0.2 \times 0.20/0.04 = 1.0 \Omega/\text{phase}$.

Step 5: Additional rotor resistance = $1.0 - 0.2 = 0.8 \Omega/\text{phase}$.

Final: $N_s = 120f/P = 1000$ rpm.; Initial full-load slip = $(1000-960)/1000 = 0.04$. Required slip = $(1000-800)/1000 = 0.20$.; $R_{total,new} = 0.2 \times 0.20/0.04 = 1.0 \Omega/\text{phase}$.; Additional rotor resistance = $1.0 - 0.2 = 0.8 \Omega/\text{phase}$.

Question 25

Original: 4B | EM-I ELE 205 End Sem. 08 December 2007

A 10 kW, 50 Hz, 4 pole, 3 phase induction motor has a rotor leakage impedance of $(0.2 + j1.5) \Omega$ per phase at standstill. When delivering full load torque the motor runs at 1440 rpm. Standstill rotor voltage = 60 V per phase. Determine the magnitude of emf injected at the rotor terminals for a speed of a) 1000 rpm b) 1800 rpm. Assume load torque remains constant. (05)

Answer 25

Use the induction-motor slip, rotor-frequency, and torque relations, keeping clear which quantities are standstill and which are at running slip.

Step 1: $N_s = 1500$ rpm, full-load slip = $(1500-1440)/1500 = 0.04$.

Step 2: Full-load rotor current phasor is $I_2 = sE_2/(R + j sX) = 11.009 - j3.303$ A.

Step 3: for a new slip s , injected emf per phase is $E_{inj} = I_2(R + j sX) - sE_2$.

Step 4: at 1000 rpm, $s = 0.3333$: $|E_{inj}| = 16.86$ V per phase, injected mainly in opposition to rotor emf.

Step 5: at 1800 rpm, $s = -0.20$: $|E_{inj}| = 13.79$ V per phase, with reversed slip/generating tendency.

Final: Full-load rotor current phasor is $I_2 = sE_2/(R + j sX) = 11.009 - j3.303$ A.; for a new slip s , injected emf per phase is $E_{inj} = I_2(R + j sX) - sE_2$.; at 1000 rpm, $s = 0.3333$: $|E_{inj}| = 16.86$ V per phase, injected mainly in opposition to rotor emf.; at 1800 rpm, $s = -0.20$: $|E_{inj}| = 13.79$ V per phase, with reversed slip/generating tendency.

Question 26

Original: 3A | EM-I ELE 205 End Sem. 26 November 2012

Explain the principle of rotor emf injection method of controlling speed of slip ring induction motor (03)

Answer 26

Use the induction-motor slip, rotor-frequency, and torque relations, keeping clear which quantities are standstill and which are at running slip.

Rotor emf injection is a speed-control method for slip-ring induction motors. An external voltage of slip frequency is injected into the rotor circuit through slip rings.

Step 1: If the injected voltage opposes the rotor induced emf, rotor current reduces and the motor slows down for the same torque.

Final: Rotor emf injection is a speed-control method for slip-ring induction motors. An external voltage of slip frequency is injected into the rotor circuit through slip rings.

Question 27

Original: 5A | EM-I ELE 205 Makeup, 05 January 2015

A 3 phase, 440 V, 50 Hz, 6 pole star connected induction motor has the following parameters: Stator impedance = $(0.3 + j 0.433) \Omega$ per phase. Rotor impedance = $(0.08 + j 0.16) \Omega$ per phase at stand still condition. Stator to rotor turns ratio = 1.75. Shunt resistance representing rotational loss = 54Ω per phase. Magnetising reactance = $j 8.3 \Omega$ per phase. Use approximate equivalent circuit to determine the following when it draws a current of 65 A, 0.8 pf lagging at rated voltage: (i) Exciting branch current (ii) Equivalent rotor current (iii) Rotational loss (sum of core and mechanical loss) (iv) stator and rotor copper loss (v) shaft output (vi) rotor speed

Answer 27

Use the induction-motor slip, rotor-frequency, and torque relations, keeping clear which quantities are standstill and which are at running slip.

Step 1: Rated phase voltage = $440/\sqrt{3} = 254.0$ V. Input current = 65 at 36.87° lagging A.

Step 2: Air-gap voltage after stator drop: $E_g = V - I(0.3 + j0.433) = 221.8$ at -2.80° V.

Step 3: Exciting branch current = $E_g/54 + E_g/(j8.3) = 27.04$ A approximately.

Step 4: Equivalent rotor current = $I - I_0 = 50.67$ A approximately.

Step 5: using stator/rotor turns ratio 1.75, $R_2' = 0.08(1.75)^2 = 0.245 \Omega$ and $X_2' = 0.49 \Omega$.

Step 6: Rotor branch gives slip $s = 0.0570$. Therefore rotor speed = 943 rpm.

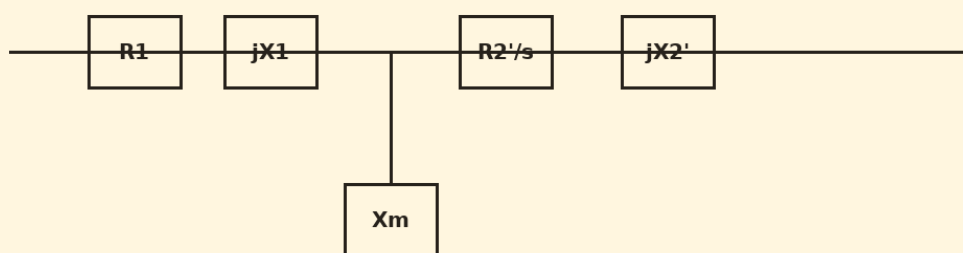
Step 7: Rotational loss = 2.733 kW; stator copper loss = 3.803 kW; rotor copper loss = 1.887 kW.

Step 8: Shaft output = 28.47 kW.

Final: Equivalent rotor current = $I - I_0 = 50.67$ A approximately.; Rotor branch gives slip $s = 0.0570$. Therefore rotor speed = 943 rpm.; Rotational loss = 2.733 kW; stator copper loss = 3.803 kW; rotor copper loss = 1.887 kW.; Shaft output = 28.47 kW.

Equivalent Circuit

Approximate per-phase induction motor circuit



Question 28

Original: 4A | EM-I ELE 205 Makeup, 05 January 2016

A 3 phase, 50 Hz, 12 pole, 420 V Δ connected induction motor has the following equivalent circuit parameters: Stator impedance = $(2.95 + j 6.82) \Omega$ per phase; Stand still rotor impedance referred to stator = $(2.08 + j 4.11) \Omega$ per phase. When running at 4 % slip, determine: a) Electrical input to the motor b) Stator and Rotor copper losses c) Useful torque if rotational losses are 750 W. Neglect exciting branch admittance.

Answer 28

Use the induction-motor slip, rotor-frequency, and torque relations, keeping clear which quantities are standstill and which are at running slip.

Step 1: Δ motor: $V_{ph} = 420$ V. At $s = 0.04$, rotor branch = $2.08/0.04 + j4.11 = 52 + j4.11 \Omega$.

Step 2: Total per-phase impedance = $(2.95 + j6.82) + (52 + j4.11) = 54.95 + j10.93 \Omega$.

Step 3: Phase current = 7.496 A; line current = 12.98 A; pf = 0.981 lagging.

Step 4: Electrical input = 9.264 kW.

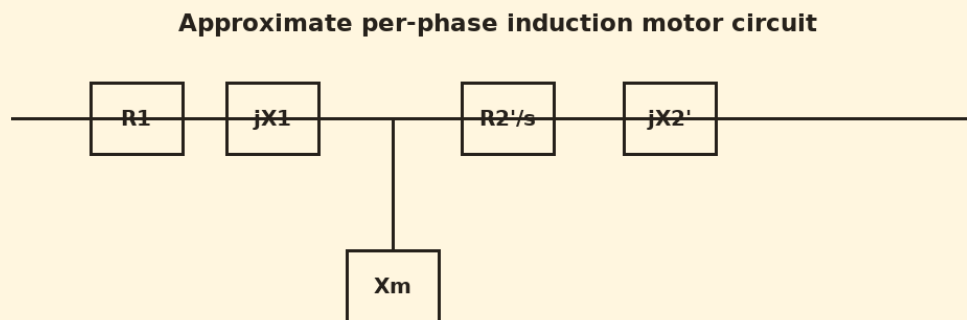
Step 5: Stator copper loss = 497 W; rotor copper loss = 351 W.

Step 6: Mechanical developed power = rotor input - rotor Cu loss = 8.416 kW; shaft output after 750 W rotational loss = 7.666 kW.

Step 7: Useful torque at 480 rpm = 152.5 N-m.

Final: Electrical input = 9.264 kW.; Stator copper loss = 497 W; rotor copper loss = 351 W.; Mechanical developed power = rotor input - rotor Cu loss = 8.416 kW; shaft output after 750 W rotational loss = 7.666 kW.; Useful torque at 480 rpm = 152.5 N-m.

Equivalent Circuit



Question 29

Original: 5B | EM-I ELE 205 Makeup, 08 January 2008

For a 6 pole 3 phase Induction motor has mechanical losses total 2 HP at a speed of 950 RPM on 550 V, 50 Hz mains. The power factor is 0.88. Calculate for this load (i) The rotor copper loss (ii) The total input if the stator losses are 2000 W (iii) The efficiency (iv) The line current (04)

Answer 29

Use the induction-motor slip, rotor-frequency, and torque relations, keeping clear which quantities are standstill and which are at running slip.

Step 1: This problem, exactly as printed in the uploaded paper, does not specify the shaft output or load power. Therefore rotor copper loss, total input, efficiency and line current cannot be uniquely determined.

Step 2: the missing quantity is normally shaft output P_{out} or mechanical power developed.

Step 3: If P_{out} is known, use: $N_s = 1000$ rpm, $s = (1000-950)/1000 = 0.05$.

Step 4: P_{mech} developed = $P_{out} + 2$ HP. Rotor Cu loss = $s/(1-s) \times P_{mech}$ developed.

Step 5: Input = $P_{out} + \text{mechanical losses} + \text{rotor Cu loss} + \text{stator losses}$.

Step 6: the efficiency is P_{out}/Input and line current = $\text{Input}/(\sqrt{3} \times 550 \times 0.88)$.

Final: If P_{out} is known, use: $N_s = 1000$ rpm, $s = (1000-950)/1000 = 0.05$.; P_{mech} developed = $P_{out} + 2$ HP. Rotor Cu loss = $s/(1-s) \times P_{mech}$ developed.; Input = $P_{out} + \text{mechanical losses} + \text{rotor Cu loss} + \text{stator losses}$.; the efficiency is P_{out}/Input and line current = $\text{Input}/(\sqrt{3} \times 550 \times$

0.88).

Question 30

Original: 3B | EM-I ELE 2103 End Sem. 27 November 2018

A 3 phase, 400 V, 6-pole, 50 Hz induction motor develops mechanical power of 20 kW at 985 rpm. The stator losses are equal to 1800 W. Neglect the mechanical losses. Calculate: i) The rotor copper loss & rotor frequency ii) The total input power. (03)

Answer 30

Use the induction-motor slip, rotor-frequency, and torque relations, keeping clear which quantities are standstill and which are at running slip.

Step 1: $N_s = 120f / P = 1000 \text{ rpm}$. Slip = $(1000 - 985) / 1000 = 0.015$.

Step 2: Mechanical power developed = 20kW, mechanical losses neglected.

Step 3: Rotor copper loss = $(s)/(1 - s) \times P_{\text{mech}} = 0.015 / 0.985 \times 20 = 0.305 \text{ kW}$.

Step 4: Rotor frequency = $s f = 0.015 \times 50 = 0.75 \text{ Hz}$.

Step 5: Total input = mechanical power + rotor Cu loss + stator losses = $20 + 0.305 + 1.8 = 22.105 \text{ kW}$.

Final: Mechanical power developed = 20kW, mechanical losses neglected.; Rotor copper loss = $(s)/(1 - s) \times P_{\text{mech}} = 0.015 / 0.985 \times 20 = 0.305 \text{ kW}$.; Rotor frequency = $s f = 0.015 \times 50 = 0.75 \text{ Hz}$.; Total input = mechanical power + rotor Cu loss + stator losses = $20 + 0.305 + 1.8 = 22.105 \text{ kW}$.

Question 31

Original: 2B | EM-I ELE 2123 End Sem. 05 December 2023

A 3-phase, 400 V, 50 Hz, 4 pole induction motor is rated for an output of 15 4 HP at 1,440 rpm. When supplying rated load, its stator losses are 800 W and 5 N-m of torque meets friction and windage losses. Determine the following for this loading: a) Slip b) Rotor copper loss c) Input to the motor d) Efficiency

Answer 31

Use the induction-motor slip, rotor-frequency, and torque relations, keeping clear which quantities are standstill and which are at running slip.

Step 1: $N_s = 1500 \text{ rpm}$, slip = $(1500 - 1440) / 1500 = 0.04$.

Step 2: Shaft output = 15HP = 11.19kW. Friction and windage loss torque = 5Ncdot m, so $P_{\text{fw}} = 2\pi n T/60 = 0.754 \text{ kW}$.

Step 3: Mechanical power developed = $11.19 + 0.754 = 11.944 \text{ kW}$.

Step 4: Rotor copper loss = $(s)/(1 - s) \times P_{\text{mech developed}} = 0.04 / 0.96 \times 11.944 = 0.498 \text{ kW}$.

Step 5: Input power = $P_{\text{mech developed}} + \text{rotor Cu loss} + \text{stator losses} = 11.944 + 0.498 + 0.800 = 13.242 \text{ kW}$.

Step 6: the efficiency is $11.19 / 13.242 = 84.5\%$.

Final: Mechanical power developed = $11.19 + 0.754 = 11.944 \text{ kW}$.; Rotor copper loss = $(s)/(1 - s) \times P_{\text{mech developed}} = 0.04 / 0.96 \times 11.944 = 0.498 \text{ kW}$.; Input power = $P_{\text{mech developed}} + \text{rotor Cu loss} + \text{stator losses} = 11.944 + 0.498 + 0.800 = 13.242 \text{ kW}$.; the efficiency is $11.19 / 13.242 = 84.5\%$.

Question 32

Original: 2C | EM-I ELE 2123 End Sem. 25 November 2024

A 3-phase, 400 V, 50 Hz, 4-pole, star connected induction motor has per-phase 3 stator impedance, $Z_1 = (0.07 + j0.3)\Omega$ and per-phase rotor impedance referred to stator side, $Z_2' = (0.08 + j0.3)\Omega$. The per phase magnetizing reactance is 10Ω and the resistance representing core loss is 50Ω . The slip is 4%. Using approximate equivalent circuit approach, solve for: a) Stator current and power factor b) Torque developed and c) Gross efficiency

Answer 32

Use the induction-motor slip, rotor-frequency, and torque relations, keeping clear which quantities are standstill and which are at running slip.

Step 1: Approximate circuit at $s = 0.04$: rotor branch $Z_2'/s = 0.08/0.04 + j0.3 = 2 + j0.3\Omega$.

Step 2: Exciting branch: $R_c = 50\Omega$, $X_m = 10\Omega$ at $V_{ph} = 400/\sqrt{3} = 230.94\text{ V}$.

Step 3: $I_0 = V/R_c + V/(jX_m)$, and load branch current $I_2 = V/(Z_1 + Z_2'/s)$.

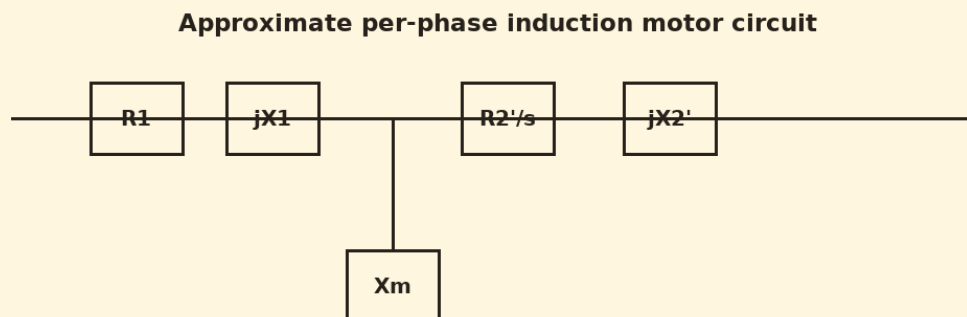
Step 4: Calculated stator current = 119.86 A at $\text{pf} = 0.897$ lagging.

Step 5: Rotor copper loss = 2.756 kW; air-gap power = 68.89 kW.

Step 6: Torque developed = $P_{ag}/\omega_s = 438.6\text{ N}\cdot\text{m}$. Gross efficiency = mechanical developed/input = 88.8%.

Final: $I_0 = V/R_c + V/(jX_m)$, and load branch current $I_2 = V/(Z_1 + Z_2'/s)$.; Calculated stator current = 119.86 A at $\text{pf} = 0.897$ lagging.; Rotor copper loss = 2.756 kW; air-gap power = 68.89 kW.; Torque developed = $P_{ag}/\omega_s = 438.6\text{ N}\cdot\text{m}$. Gross efficiency = mechanical developed/input = 88.8%.

Equivalent Circuit



Question 33

Original: 4B | EM-I ELE 2154 Makeup/GI, 28 July 2021

Consider a 415V, 4pole, 50Hz induction motor operating at 4% slip. The shaft power output is 1.2 kW. The machine has stator losses of 50W and rotational losses of 70W. Draw the power flow diagram with power stages. (05)

Answer 33

The power flow is most conveniently worked backward from shaft output because slip and mechanical losses are known.

Synchronous speed: $N_s = 120 f/P = 120 \times 50/4 = 1500\text{ rpm}$. At 4 percent slip, rotor speed is $N_r = 1500(1 - 0.04) = 1440\text{ rpm}$.

Shaft output power = 1200 W. Rotational losses = 70 W. Therefore mechanical power developed is $P_{mech} = 1200 + 70 = 1270\text{ W}$.

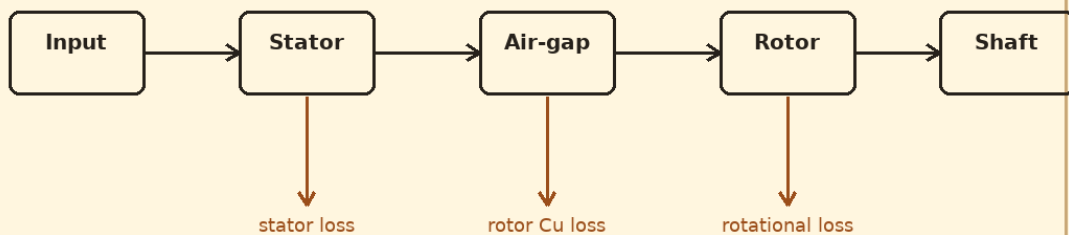
For an induction motor, $P_{\text{mech}} = (1 - s)P_{\text{ag}}$ and rotor copper loss = $s P_{\text{ag}}$. Hence $P_{\text{ag}} = 1270/0.96 = 1322.9 \text{ W}$ and rotor copper loss = $0.04 \times 1322.9 = 52.9 \text{ W}$.

Stator losses are 50 W, so input power is $P_{\text{in}} = P_{\text{ag}} + \text{stator losses} = 1322.9 + 50 = 1372.9 \text{ W}$. Overall efficiency = $1200/1372.9 = 87.4 \text{ percent}$.

The power stages are: $P_{\text{in}} = 1372.9 \text{ W} \rightarrow \text{stator loss} = 50 \text{ W} \rightarrow \text{air-gap power} = 1322.9 \text{ W} \rightarrow \text{rotor copper loss} = 52.9 \text{ W} \rightarrow \text{mechanical power developed} = 1270 \text{ W} \rightarrow \text{rotational loss} = 70 \text{ W} \rightarrow \text{shaft output} = 1200 \text{ W}$.

Final: $P_{\text{ag}} = 1.323 \text{ kW}$, rotor copper loss = 52.9 W, mechanical power developed = 1.270 kW, input = 1.373 kW, efficiency = 87.4 percent.

Power Flow



Question 34

Original: 6A | EM-I ELE 2154 Makeup/GI, 28 July 2021

A 3-phase, 50 Hz, 36 kW, 4 pole induction motor has a full load efficiency of 82 %. The friction & windage losses are one-fourth of no load losses and rotor copper losses equal the iron loss at full load. Determine (a) Total Losses (b) Stator Core Loss (c) Rotor Copper Loss (d) Friction & Windage Loss (05)

Answer 34

Use the induction-motor slip, rotor-frequency, and torque relations, keeping clear which quantities are standstill and which are at running slip.

Step 1: Output = 36 kW, efficiency = 82%, hence input = $36/0.82 = 43.90 \text{ kW}$.

Step 2: Total losses = input - output = 7.902 kW.

Step 3: Let stator core loss = rotor copper loss = X. No-load loss = core + friction, and friction = 1/4 no-load loss, so core = 3 × friction.

Thus total losses = $X + X + X/3 = 7X/3$.

Step 5: $X = 3.386 \text{ kW}$. Therefore stator core loss = 3.386 kW, rotor copper loss = 3.386 kW, and friction-windage loss = 1.129 kW.

Step 6: This allocation assumes stator copper loss is neglected/not separately specified.

Final: Total losses = input - output = 7.902 kW.; Let stator core loss = rotor copper loss = X. No-load loss = core + friction, and friction = 1/4 no-load loss, so core = 3 × friction.; Thus total losses = $X + X + X/3 = 7X/3$.; $X = 3.386 \text{ kW}$. Therefore stator core loss = 3.386 kW, rotor copper loss = 3.386 kW, and friction-windage loss = 1.129 kW.

Question 35

Original: 2B | EM-I ELE 2154 Online End Sem. 27 January 2022

A three-phase, 400V, 4-pole, 50 Hz, three-phase induction motor provides shaft power of 1.2kW. Considering 2.5% of shaft power as friction and windage losses, determine the rotor copper losses, power supplied to the rotor circuit, gross torque developed by the motor, and efficiency of the rotor when the rotor is running at a speed of 1420rpm. (03)

Answer 35

Use the induction-motor slip, rotor-frequency, and torque relations, keeping clear which quantities are standstill and which are at running slip.

Step 1: $N_s = 1500$ rpm; slip = $(1500-1420)/1500 = 0.05333$.

Step 2: Shaft power = 1.2 kW. Friction and windage = 2.5% of shaft power = 30 W.

Step 3: Mechanical power developed = 1230 W.

Step 4: Rotor copper loss = $s/(1-s) \times P_{mech} = 69.3$ W.

Step 5: Rotor input = $P_{ag} = P_{mech} + \text{rotor Cu loss} = 1299.3$ W.

Step 6: Gross torque developed = $P_{ag}/\omega_s = 8.27$ N·m. Rotor efficiency = $1-s = 94.67\%$.

Final: Mechanical power developed = 1230 W.; Rotor copper loss = $s/(1-s) \times P_{mech} = 69.3$ W.; Rotor input = $P_{ag} = P_{mech} + \text{rotor Cu loss} = 1299.3$ W.; Gross torque developed = $P_{ag}/\omega_s = 8.27$ N·m. Rotor efficiency = $1-s = 94.67\%$.

4. Induction motor torque-slip characteristics, starting, speed control and braking

Syllabus area: Weeks 7-9

Question 36

Original: 4A | EM-I ELE 205 End Sem, 03 December 2014

A squirrel cage induction motor, when started by means of a star-delta starter draws 200 % of full load current and develops 44 % of full load torque at starting. If an auto-transformer with 75 % tapping is used, determine: (i) Full load slip (ii) Ratio of starting torque to full load torque (iii) Starting motor current and starting line current as % of full load current. 4

Answer 36

For a star-delta starter, the starting line current and starting torque are one-third of the direct-on-line values for the same motor. Given star-delta starting line current = 2 IfL, the direct-on-line starting current would be 6 IfL. Given star-delta starting torque = 0.44 Tfl, the direct-on-line starting torque is $3 \times 0.44 = 1.32$ Tfl.

Using $T_{st}/T_{fl} = (I_{st}/I_{fl})^2 s_{fl}$ for direct-on-line starting, $1.32 = 6^2 s_{fl}$. Hence $s_{fl} = 1.32/36 = 0.0367 = 3.67$ percent.

For an auto-transformer with tapping $k = 0.75$, motor starting current = $k \times I_{sc} = 0.75 \times 6 I_{fl} = 4.5 I_{fl}$. Supply line current = $k^2 I_{sc} = 0.75^2 \times 6 I_{fl} = 3.375 I_{fl}$.

Starting torque with auto-transformer = $k^2 \times DOL$ starting torque = $0.75^2 \times 1.32 T_{fl} = 0.7425 T_{fl}$.

Final: Full-load slip = 3.67 percent; auto-transformer starting torque = 74.3 percent of full-load torque; motor current = 450 percent and line current = 337.5 percent of full-load current.

Question 37

Original: 4B | EM-I ELE 205 End Sem, 03 December 2014

A 3 phase, 50 Hz, 4 pole, 400 V star connected double cage induction motor has its stator impedance of $(0.5 + j 7)$ per phase and equivalent standstill cage impedances of $(0.2 + j 5)$ and $(5 + j 4)$ per phase. The exciting shunt branch is neglected. At full load speed of 1440 rpm, determine (a) current drawn from the supply (b) copper loss in each cage and (c) total torque developed. 4

Answer 37

At 50 Hz and 4 poles, $N_s = 1500$ rpm. At full-load speed 1440 rpm, slip $s = (1500 - 1440)/1500 = 0.04$. Phase voltage for the star-connected motor is $V_{ph} = 400/\sqrt{3} = 230.94$ V.

At slip s , the two cage branches are represented by $R/s + jX$. Thus cage 1: $Z_{1r} = 0.2/0.04 + j5 = 5 + j5 \Omega$. Cage 2: $Z_{2r} = 5/0.04 + j4 = 125 + j4 \Omega$. Their parallel equivalent is approximately $4.973 + j4.617 \Omega$.

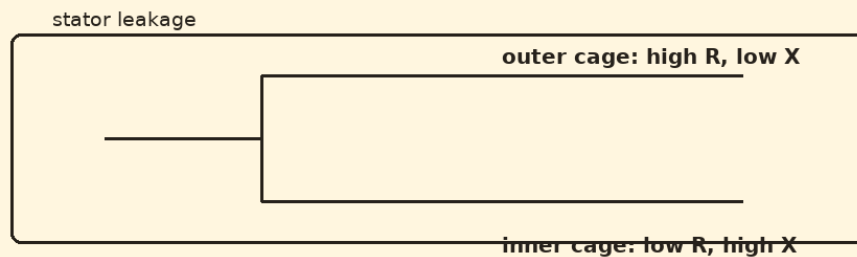
Including stator impedance, total impedance is $(0.5 + j7) + (4.973 + j4.617) = 5.473 + j11.617 \Omega$. Therefore stator current is $I_1 = 230.94/(5.473 + j11.617)$, whose magnitude is 17.98 A.

The cage currents from the rotor terminal voltage are approximately 17.26 A and 0.976 A. Rotor copper losses are $3 I^2 R$: cage 1 gives 178.7 W and cage 2 gives 14.3 W.

Air-gap power = $3[(I_1 c^2 R_1/s) + (I_2 c^2 R_2/s)] = 4.825$ kW. Electromagnetic torque = P_{ag}/ω_s , with $\omega_s = 2\pi \times 1500/60 = 157.1$ rad/s, giving $T = 30.7$ N m.

Final: Supply current = 18.0 A; cage copper losses = 178.7 W and 14.3 W; total developed torque = 30.7 N m.

Double Cage



Outer cage dominates starting; inner cage dominates running

Question 38

Original: 4C | EM-I ELE 205 End Sem, 03 December 2014

Give a technical explanation of crawling in induction motors. 2

Answer 38

Crawling is the tendency of an induction motor to run at a stable speed much lower than its normal full-load speed. It is mainly produced by space harmonics in the air-gap flux wave, especially the seventh harmonic in a three-phase winding.

If the fundamental field has synchronous speed N_s , the seventh harmonic produces a forward rotating field whose synchronous speed is approximately $N_s/7$. The rotor can therefore develop a stable torque near this low speed. The fifth harmonic generally produces a backward torque, while the seventh harmonic can reinforce a forward low-speed operating point.

In the torque-speed curve this appears as a dip near normal acceleration and a secondary torque peak near one-seventh synchronous speed. If the load torque intersects this harmonic torque peak, the motor may crawl instead of accelerating to its normal operating region.

Crawling is reduced by proper choice of stator and rotor slots, short-pitched or distributed windings to suppress harmonics, and skewing of rotor slots. Better winding design reduces the harmonic flux responsible for the low-speed torque.

Final: Crawling is a harmonic-torque phenomenon; the motor may run near $N_s/7$ due to the seventh harmonic, and it is prevented by harmonic-reducing winding and slot design.

Question 39

Original: 5C | EM-I ELE 205 End Sem, 04 December 2006

Explain the working of a star-delta starter for a 3 phase induction motor.

Answer 39

Use the induction-motor slip, rotor-frequency, and torque relations, keeping clear which quantities are standstill and which are at running slip.

Step 1: DOL starting would draw 5 times full-load current, which is high for a squirrel-cage motor and can cause voltage dip and mechanical stress. A star-delta starter is recommended if the motor is delta-rated for normal running and starts under light or moderate load. Star-delta starter principle After acceleration, timer opens star contactor and closes delta contactor for rated running. ``txt DOL starting line current = 5 IFL Star-delta starting line current =  $(1/3) \times \text{DOL current} = 5/3 \text{ IFL} = 1.67 \text{ IFL}$  Star-delta starting torque =  $(1/3) \times \text{DOL starting torque}$  `` Therefore star-delta starting limits supply current effectively. The limitation is reduced starting torque, so it should not be used for heavy-load starts unless the available torque is sufficient.

Final: The limitation is reduced starting torque, so it should not be used for heavy-load starts unless the available torque is sufficient.

Question 40

Original: 6B | EM-I ELE 205 End Sem, 04 December 2006

Explain the phenomena of crawling and cogging in a 3 phase induction motor.

Answer 40

Use the induction-motor slip, rotor-frequency, and torque relations, keeping clear which quantities are standstill and which are at running slip.

Crawling is a low-speed running trouble caused mainly by space harmonics in the air-gap field; cogging is a starting trouble caused by magnetic locking of stator and rotor teeth.

Step 1: Crawling means that the motor runs stably at a speed much lower than normal, commonly near one-seventh of synchronous speed due to the 7th harmonic field producing forward torque.

Step 2: the fifth harmonic tends to produce a backward torque, whereas the seventh harmonic is forward rotating and can create a stable operating point near one-seventh of synchronous speed.

Step 3: Cogging means that the motor refuses to start because stator and rotor teeth magnetically lock, especially when stator and rotor slot numbers have an unsuitable relationship.

Step 4: the usual remedies are skewing rotor slots, proper choice of stator/rotor slot combinations, short-pitched/distributed windings, and using double-cage or deep-bar rotors.

Final: The fifth harmonic tends to produce a backward torque, whereas the seventh harmonic is forward rotating and can create a stable operating point near one-seventh of synchronous speed.

Question 41

Original: 7C | EM-I ELE 205 End Sem, 04 December 2006

At standstill, the equivalent impedance of inner and outer cages of a double cage rotor are $(0.4 + j0.2) \Omega$ and $(2 + j0.4) \Omega$ respectively. Calculate the ratio of the torques produced by the two cages at (i) standstill and (ii) 5% slip.

Answer 41

Use the induction-motor slip, rotor-frequency, and torque relations, keeping clear which quantities are standstill and which are at running slip.

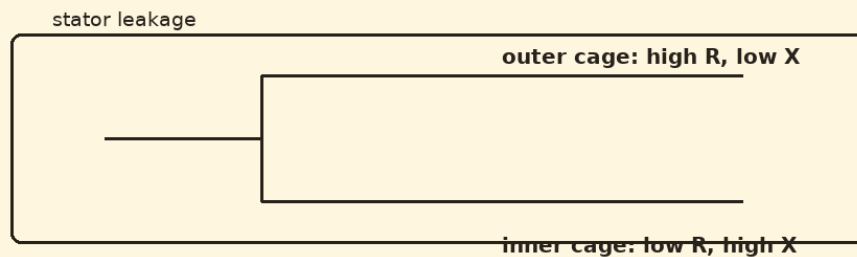
Step 1: Use T proportional to $R / [R^2 + (sX)^2]$. Inner cage: $0.4 + j0.2 \Omega$; outer cage: $2 + j0.4 \Omega$.

Step 2: at standstill: $T_{\text{inner}}/T_{\text{outer}} = 4.16$, i.e., $\text{outer/inner} = 0.240$.

Step 3: at $s = 0.05000$: $T_{\text{inner}}/T_{\text{outer}} = 4.9974$, i.e., $\text{outer/inner} = 0.20010$.

Final: Use T proportional to $R / [R^2 + (sX)^2]$. Inner cage: $0.4 + j0.2 \Omega$; outer cage: $2 + j0.4 \Omega$; at standstill: $T_{\text{inner}}/T_{\text{outer}} = 4.16$, i.e., $\text{outer/inner} = 0.240$; at $s = 0.05000$: $T_{\text{inner}}/T_{\text{outer}} = 4.9974$, i.e., $\text{outer/inner} = 0.20010$.

Double Cage



Outer cage dominates starting; inner cage dominates running

Question 42

Original: 5A | EM-I ELE 205 End Sem, 08 December 2007

A 3 phase induction motor has a starting torque of 150 % & a maximum torque of 250 % of the full load torque. Neglecting stator impedance calculate a) the slip at maximum torque b) full load slip (03)

Answer 42

Use the induction-motor slip, rotor-frequency, and torque relations, keeping clear which quantities are standstill and which are at running slip.

Step 1: $T_{st}/T_{max} = 1.5/2.5 = 0.6$. For induction motor torque, $T/T_{max} = 2/(s/s_m + s_m/s)$.

Step 2: at starting $s = 1$: $0.6 = 2/(1/s_m + s_m)$, giving $s_m = 1/3$ (physical root).

Step 3: $T_{fl}/T_{max} = 1/2.5 = 0.4$. Hence $0.4 = 2/(x + 1/x)$, $x = s_{fl}/s_m$.

Step 4: the low-slip root is $x = 0.2087$. Therefore $s_{fl} = 0.2087 \times 1/3 = 0.0696$.

Final: slip at maximum torque = 0.333; full-load slip = 0.0696 or 6.96%.

Question 43

Original: 6A | EM-I ELE 205 End Sem, 08 December 2007

At standstill, the equivalent impedance of inner and outer cages of a double cage induction motor are $(0.4 + j 0.2) \Omega$ and $(2 + j 0.4) \Omega$ respectively. Calculate the ratio of the torques produced by the two cages at (i) standstill and (ii) 4 % slip (04)

Answer 43

Use the induction-motor slip, rotor-frequency, and torque relations, keeping clear which quantities are standstill and which are at running slip.

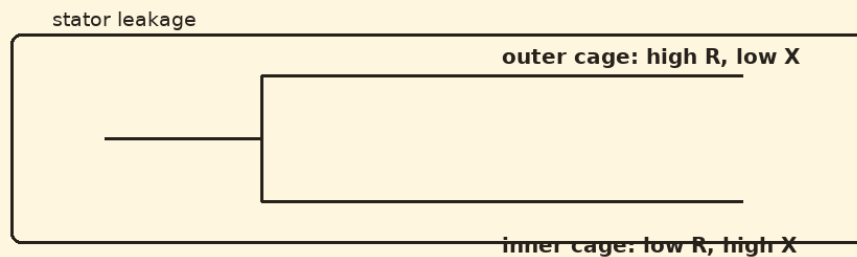
Step 1: Use T proportional to $R/[R^2 + (sX)^2]$. Inner cage: $0.4 + j0.2 \Omega$; outer cage: $2 + j0.4 \Omega$.

Step 2: at standstill: $T_{inner}/T_{outer} = 4.16$, i.e., outer/inner = 0.240.

Step 3: at $s = 0.04000$: $T_{inner}/T_{outer} = 4.9983$, i.e., outer/inner = 0.20007.

Final: Use T proportional to $R/[R^2 + (sX)^2]$. Inner cage: $0.4 + j0.2 \Omega$; outer cage: $2 + j0.4 \Omega$.; at standstill: $T_{inner}/T_{outer} = 4.16$, i.e., outer/inner = 0.240.; at $s = 0.04000$: $T_{inner}/T_{outer} = 4.9983$, i.e., outer/inner = 0.20007.

Double Cage



Outer cage dominates starting; inner cage dominates running

Question 44

Original: 3B | EM-I ELE 205 End Sem, 26 November 2012

A 400 V, 50 Hz, 3 phase, 4 pole star connected slip ring induction motor has its rotor standstill leakage impedance of $(0.4 + j2) \Omega$ per phase. The stator to rotor turns ratio is 1.7. Compute (i) Maximum Torque (ii) Full load torque when the slip is 4%. (iii) The resistance to be included in the rotor circuit to develop 80% of maximum torque at starting. (04)

Answer 44

For the 4-pole, 50 Hz motor, synchronous speed is $N_s = 1500$ rpm and $\omega_s = 157.1$ rad/s. Phase voltage is $400/\sqrt{3} = 230.94$ V. With stator-to-rotor turns ratio 1.7, standstill rotor emf per phase is approximately $E_2 = 230.94/1.7 = 135.85$ V.

For the rotor circuit, $R_2 = 0.4 \Omega$ and $X_2 = 2 \Omega$. The slip at maximum torque is $s_m = R_2/X_2 = 0.4/2 = 0.20$. Maximum torque is $T_{max} = 3 E_2^2 / (2 \omega_s X_2) = 88.1$ N m.

At full-load slip $s = 0.04$, torque is $T = [3 E_2^2 (R_2/s)] / [\omega_s ((R_2/s)^2 + X_2^2)] = 33.9$ N m.

To obtain 80 percent of maximum torque at starting, use $T/T_{max} = 2 R X / (R^2 + X^2) = 0.8$. This gives total rotor resistance $R = 1 \Omega$ or 4Ω . The lower value is normally selected for starting without excessive external resistance.

Final: $T_{max} = 88.1$ N m; full-load torque at 4 percent slip = 33.9 N m; external resistance for 0.8 T_{max} at starting = $1 - 0.4 = 0.6 \Omega$ /phase (alternate high-resistance solution: 3.6Ω /phase).

Question 45

Original: 3C | EM-I ELE 205 End Sem, 26 November 2012

The short circuit line current of a 6 HP Induction Motor is 3.5 times its full load current. An autotransformer starter is used to limit starting line current to twice the full load current. For a full load slip is 2.5%, (i) Percentage tapping of autotransformer. (ii) Estimate the torque at starting in terms of full load torque (iii) Line current drawn from the supply in terms of full load current with autotransformer set to the above tapping. (03)

Answer 45

Let k be the auto-transformer tapping. The direct-on-line short-circuit current is 3.5 times full-load current. Supply current with auto-transformer is k^2 times the DOL short-circuit current.

Given the line current limit is $2 I_{fL}$: $k^2 \times 3.5 = 2$. Hence $k = \sqrt{2/3.5} = 0.756$. Therefore the required tapping is 75.6 percent.

The direct-on-line starting torque relative to full-load torque is $(3.5)^2 s_{fl} = 12.25 \times 0.025 = 0.306 T_{fl}$. With auto-transformer starting, torque is k^2 times this value: $T_{st} = (2/3.5) \times 0.306 T_{fl} = 0.175 T_{fl}$.

The supply line current at this tapping is exactly 2 IfL by design. The current actually applied to the motor winding is $k \times 3.5 I_{fL} = 2.65 I_{fL}$.

Final: Auto-transformer tapping = 75.6 percent; starting torque = 17.5 percent of full-load torque; supply line current = 2 times full-load current.

Question 46

Original: 4B | EM-I ELE 205 End Sem, 26 November 2012

Draw and explain the equivalent circuit of a double cage induction motor and sketch the inner, outer and net torque slip characteristics. (04)

Answer 46

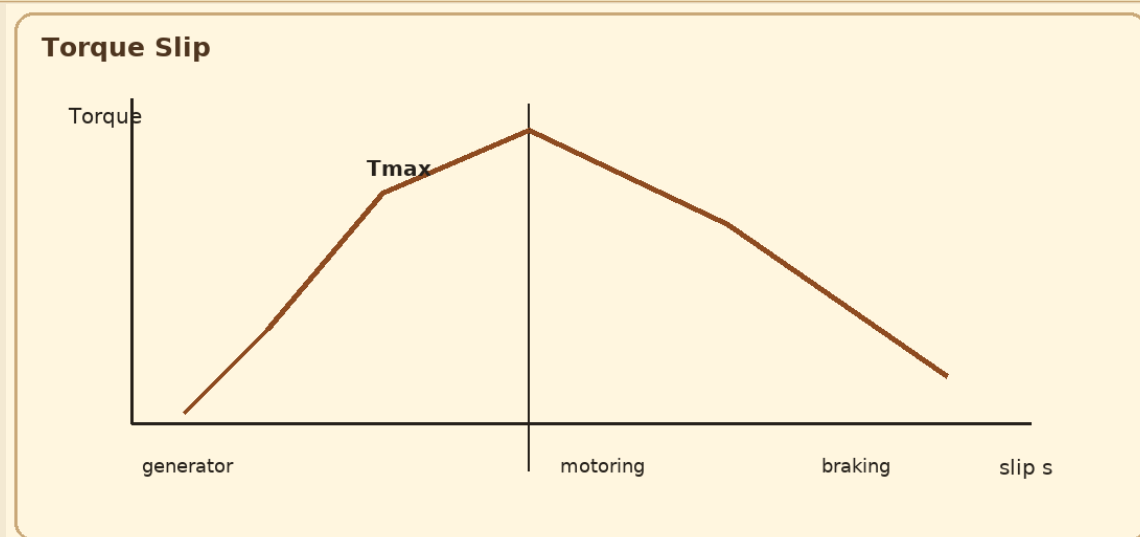
Use the induction-motor slip, rotor-frequency, and torque relations, keeping clear which quantities are standstill and which are at running slip.

Step 1: A double-cage induction motor has two rotor cages. The outer cage has high resistance and low leakage reactance, while the inner cage has low resistance and high leakage reactance. Double-cage rotor equivalent circuit - At starting, rotor frequency is high, so leakage reactance dominates. Current flows mainly in the outer cage because it has lower leakage reactance. Its high resistance gives high starting torque.

Step 2: at normal running, rotor frequency is low, so reactance is small. Current shifts to the low-resistance inner cage, giving good running efficiency.

Thus the double-cage rotor combines high starting torque with good full-load efficiency.

Final: At normal running, rotor frequency is low, so reactance is small. Current shifts to the low-resistance inner cage, giving good running efficiency.; Thus the double-cage rotor combines high starting torque with good full-load efficiency.



Question 47

Original: 3C | EM-I ELE 205 End Sem, 30 November 2010

A 12 pole, 50Hz, 3phase induction motor has the rotor resistance of 0.15Ω per phase and the standstill reactance of 0.25Ω per phase. On full load it is running at a speed of 480rpm. The rotor induced emf per phase at stand-still is observed to be 32V. Calculate (i) Full load torque. (ii) Starting Torque. (iii) Speed at Maximum Torque. (03)

Answer 47

Use the induction-motor slip, rotor-frequency, and torque relations, keeping clear which quantities are standstill and which are at running slip.

Step 1: $N_s = 120f/P = 500$ rpm; full-load slip = $(500-480)/500 = 0.04$.

Step 2: at slip s , $I_2 = sE_2/\sqrt{(R_2^2 + (sX_2)^2)} = 8.514$ A.

Step 3: Rotor copper loss = $3 I_2^2 R_2 = 32.62$ W; air-gap power = rotor Cu loss/ $s = 815.6$ W.

Step 4: Full-load torque = $P_{ag}/\omega_s = 15.58$ N·m.

Step 5: Starting torque = $3 E_2^2 R_2/[\omega_s(R_2^2 + X_2^2)] = 103.54$ N·m.

Step 6: Maximum torque occurs when $s = R_2/X_2 = 0.6$, so speed at maximum torque = $(1-0.6)500 = 200$ rpm.

Final: Rotor copper loss = $3 I_2^2 R_2 = 32.62$ W; air-gap power = rotor Cu loss/ $s = 815.6$ W.; Full-load torque = $P_{ag}/\omega_s = 15.58$ N·m.; Starting torque = $3 E_2^2 R_2/[\omega_s(R_2^2 + X_2^2)] = 103.54$ N·m.; Maximum torque occurs when $s = R_2/X_2 = 0.6$, so speed at maximum torque = $(1-0.6)500 = 200$ rpm.

Question 48

Original: 5A | EM-I ELE 205 End Sem. 30 November 2010

A 3-phase squirrel cage induction motor takes a starting current 6 times the full load current. Estimate the starting torque as a percentage of full load torque if the motor is started (i) direct on line (ii) through a star-delta starter. The full load slip of the motor is 4%. (02)

Answer 48

Use the induction-motor slip, rotor-frequency, and torque relations, keeping clear which quantities are standstill and which are at running slip.

Step 1: DOL starting torque/full-load torque = $(I_{sc}/I_{fl})^2 \times s_{fl} = 6^2 \times 0.04 = 1.44$.

Therefore DOL starting torque = 144% of full-load torque.

Step 3: with a star-delta starter, phase voltage during starting is $1/\sqrt{3}$ of delta running voltage, so starting torque becomes one-third of DOL value.

Step 4: Star-delta starting torque = $144/3 = 48\%$ of full-load torque.

Final: DOL starting torque/full-load torque = $(I_{sc}/I_{fl})^2 \times s_{fl} = 6^2 \times 0.04 = 1.44$.; Therefore DOL starting torque = 144% of full-load torque.; with a star-delta starter, phase voltage during starting is $1/\sqrt{3}$ of delta running voltage, so starting torque becomes one-third of DOL value.; Star-delta starting torque = $144/3 = 48\%$ of full-load torque.

Question 49

Original: 5B | EM-I ELE 205 End Sem. 30 November 2010

The standstill rotor voltage of a 3 phase induction motor is 190V per phase. The motor is running with a slip of 4% and the load torque is proportional to square of the speed. What must be the rotor injected voltage to run the motor with slip of 0.6. The rotor resistance per phase is 0.5 . (03)

Answer 49

Assume rotor leakage reactance is neglected, as is usual for this injected-voltage speed-control calculation. Standstill rotor emf per phase is $E_{20} = 190$ V and rotor resistance is $R_2 = 0.5 \Omega$.

At the initial slip $s_1 = 0.04$, rotor emf per phase is $E_{2s_1} = s_1 E_{20} = 0.04 \times 190 = 7.6$ V. Therefore initial rotor current is $I_{21} = 7.6/0.5 = 15.2$ A.

The load torque is proportional to speed squared. Since rotor speed is proportional to $(1 - s)$,

$$T_2/T_1 = [(1 - s_2)/(1 - s_1)]^2 = (0.4/0.96)^2 = 0.1736.$$

With leakage reactance neglected, induction-motor torque is proportional to I_2^2/s . Therefore

$$I_{22}/I_{21} = \sqrt{((T_2/T_1)(s_2/s_1))} = \sqrt{(0.1736 \times 0.6/0.04)} = 1.614.$$

$$\text{Hence } I_{22} = 1.614 \times 15.2 = 24.5 \text{ A.}$$

At $s_2 = 0.6$, the natural rotor emf is $s_2 E_{20} = 0.6 \times 190 = 114 \text{ V}$. The net voltage needed across rotor resistance is $I_{22} R_2 = 24.5 \times 0.5 = 12.3 \text{ V}$.

Therefore the injected voltage must oppose the rotor emf and have magnitude

$$E_{inj} = 114 - 12.3 = 101.7 \text{ V per phase.}$$

Final: Required injected rotor voltage is approximately 102 V per phase, connected in opposition to the rotor induced emf.

Question 50

Original: 6A | EM-I ELE 205 Makeup. 05 January 2015

Sketch and explain the torque-slip characteristics of a 3 phase slip ring induction motor for different values of rotor resistance. 3

Answer 50

The torque-slip characteristic of a three-phase induction motor shows the variation of electromagnetic torque with slip. At $s = 0$, rotor speed equals synchronous speed, so relative speed is zero, rotor emf is zero, and torque is zero. As slip increases from zero, rotor current increases and torque rises almost linearly in the normal operating region.

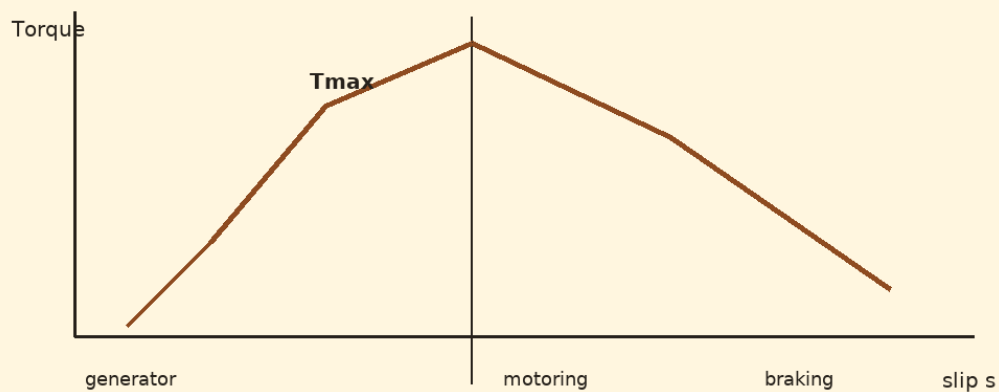
The torque reaches maximum value when the effective rotor resistance equals the standstill leakage reactance term, approximately $R_2/s = X_2$ for the simplified circuit. Hence the slip at maximum torque is proportional to rotor resistance. Beyond this point, reactance dominates and torque falls as slip approaches unity.

At $s = 1$ the motor is at standstill and develops starting torque. For a squirrel-cage motor, rotor resistance is fixed, so the starting torque is limited. In a wound-rotor/slip-ring motor, external rotor resistance can be added, shifting the maximum-torque point toward higher slip and improving starting torque.

The magnitude of maximum torque is approximately independent of rotor resistance if supply voltage and reactance are unchanged, but the slip at which it occurs increases with added rotor resistance. The complete characteristic also includes generating operation for $s < 0$ and plugging/braking operation for $s > 1$.

Final: Rotor resistance mainly shifts the torque-slip curve horizontally; it can raise starting torque and make maximum torque occur at starting, while the maximum torque magnitude remains nearly unchanged.

Torque Slip



Question 51

Original: 6B | EM-I ELE 205 Makeup, 05 January 2015

A 3 phase double cage induction motor has cage impedances of $(2 + j 5) \Omega$ and $(0.6 + j 10) \Omega$ per phase for its outer and inner cage respectively at standstill condition. Determine: (i) the ratio of outer torque to inner torque at starting and when running at 4% slip (ii) slip at which both cages develop equal torque 5

Answer 51

In a double-cage motor, torque contribution of a cage at slip s is proportional to $R/[R^2 + (sX)^2]$, because rotor reactance varies as sX at rotor frequency.

Outer cage: $R = 2 \Omega$, $X = 5 \Omega$. Inner cage: $R = 0.6 \Omega$, $X = 10 \Omega$.

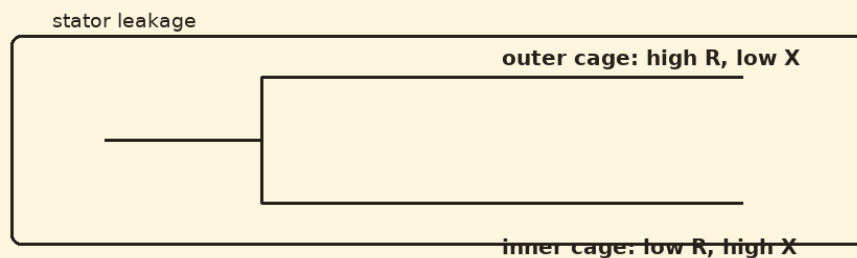
At starting, $s = 1$. Therefore $T_{out}/T_{in} = [2/(2^2 + 5^2)]/[0.6/(0.6^2 + 10^2)] = 11.5$. The high-resistance outer cage therefore gives most of the starting torque.

At running slip $s = 0.04$: $T_{out}/T_{in} = [2/(2^2 + (0.04 \times 5)^2)]/[0.6/(0.6^2 + (0.04 \times 10)^2)] = 0.429$. Thus the low-resistance inner cage carries the main running torque.

For equal torques, solve $2[0.6^2 + (10s)^2] = 0.6[2^2 + (5s)^2]$. This gives $s = 0.0953$.

Final: Outer/inner torque ratio = 11.5 at starting, 0.429 at 4 percent slip; equal torque occurs at about 9.53 percent slip.

Double Cage



Outer cage dominates starting; inner cage dominates running

Question 52

Original: 6C | EM-I ELE 205 Makeup, 05 January 2015

The ratio V/f should be maintained constant during speed control of a 3 phase induction motor. Give reasons. $2 \times \times \times \times \times \times \times \times \times \times \times \times \times \times \times$

Answer 52

Use the induction-motor slip, rotor-frequency, and torque relations, keeping clear which quantities are standstill and which are at running slip.

Step 1: Variable frequency control changes the supply frequency to control synchronous speed $N_s = 120f/P$. To keep air-gap flux nearly constant, voltage is varied in proportion to frequency. $E_1 = 4.44 f N \phi$
 $K_w \Rightarrow \phi \propto V/f$ - If f is reduced but V is kept constant, flux increases and the core may saturate.

Step 2: If f is increased without raising V , flux falls and torque capability reduces.

Step 3: Keeping V/f constant maintains nearly constant flux and approximately constant torque capability up to base speed.

Step 4: Above base speed, voltage reaches its limit; field weakening occurs and constant-power operation is possible. Common strategies are scalar V/f control, closed-loop V/f control, vector control and direct torque control.

Final: Variable frequency control changes the supply frequency to control synchronous speed $N_s = 120f/P$. To keep air-gap flux nearly constant, voltage is varied in proportion to frequency. $E_1 = 4.44 f N \phi K_w \Rightarrow \phi \propto V/f$ - If f is reduced but V is kept constant, flux increases and the core may saturate.

Question 53

Original: 4B | EM-I ELE 205 Makeup, 05 January 2016

Draw the torque slip characteristics of a squirrel cage induction motor and 4M mark the salient points.

Answer 53

The torque-slip characteristic of a three-phase induction motor shows the variation of electromagnetic torque with slip. At $s = 0$, rotor speed equals synchronous speed, so relative speed is zero, rotor emf is zero, and torque is zero. As slip increases from zero, rotor current increases and torque rises almost linearly in the normal operating region.

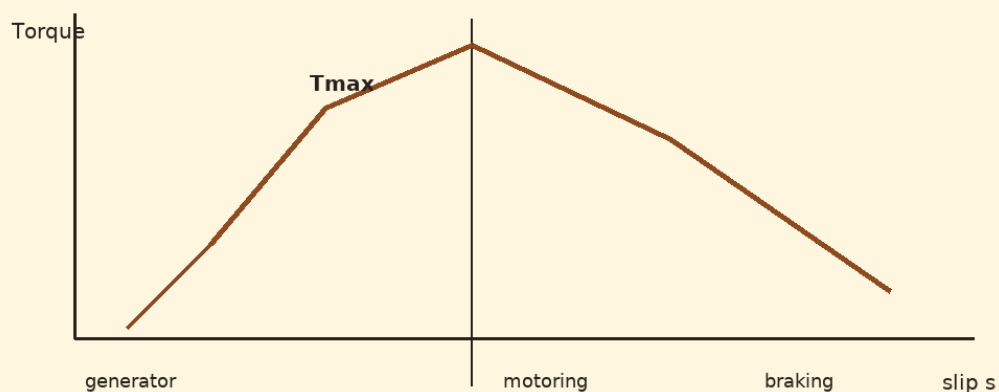
The torque reaches maximum value when the effective rotor resistance equals the standstill leakage reactance term, approximately $R_2/s = X_2$ for the simplified circuit. Hence the slip at maximum torque is proportional to rotor resistance. Beyond this point, reactance dominates and torque falls as slip approaches unity.

At $s = 1$ the motor is at standstill and develops starting torque. For a squirrel-cage motor, rotor resistance is fixed, so the starting torque is limited. In a wound-rotor/slip-ring motor, external rotor resistance can be added, shifting the maximum-torque point toward higher slip and improving starting torque.

The magnitude of maximum torque is approximately independent of rotor resistance if supply voltage and reactance are unchanged, but the slip at which it occurs increases with added rotor resistance. The complete characteristic also includes generating operation for $s < 0$ and plugging/braking operation for $s > 1$.

Final: Rotor resistance mainly shifts the torque-slip curve horizontally; it can raise starting torque and make maximum torque occur at starting, while the maximum torque magnitude remains nearly unchanged.

Torque Slip



Question 54

Original: 3C | EM-I ELE 205 Makeup. 08 January 2008

A 10 pole 50 Hz slip ring induction motor runs at 580 RPM on full load. The rotor resistance per phase is 0.3 Ω . Calculate the additional resistance per phase to be inserted in the rotor circuit if the speed is to be reduced to 500 RPM for full load torque. (03)

Answer 54

Use the induction-motor slip, rotor-frequency, and torque relations, keeping clear which quantities are standstill and which are at running slip.

Step 1: $N_s = 120f/P = 600$ rpm.

Step 2: Initial slip = $(600-580)/600 = 0.03333$. Required slip = $(600-500)/600 = 0.1667$.

Step 3: for constant torque in a slip-ring motor, R_{total}/s is approximately constant.

Step 4: $R_{total,new} = R_2 \times s_2/s_1 = 0.3 \times 0.1667/0.03333 = 1.5 \Omega$.

Step 5: Additional resistance = $1.5 - 0.3 = 1.2 \Omega/\text{phase}$.

Final: Initial slip = $(600-580)/600 = 0.03333$. Required slip = $(600-500)/600 = 0.1667$.; for constant torque in a slip-ring motor, R_{total}/s is approximately constant.; $R_{total,new} = R_2 \times s_2/s_1 = 0.3 \times 0.1667/0.03333 = 1.5 \Omega$.; Additional resistance = $1.5 - 0.3 = 1.2 \Omega/\text{phase}$.

Question 55

Original: 5A | EM-I ELE 205 Makeup. 08 January 2008

Draw and explain the torque-slip characteristic of a 3-phase induction motor. Also explain the effect of rotor resistance on torque slip characteristics. (04)

Answer 55

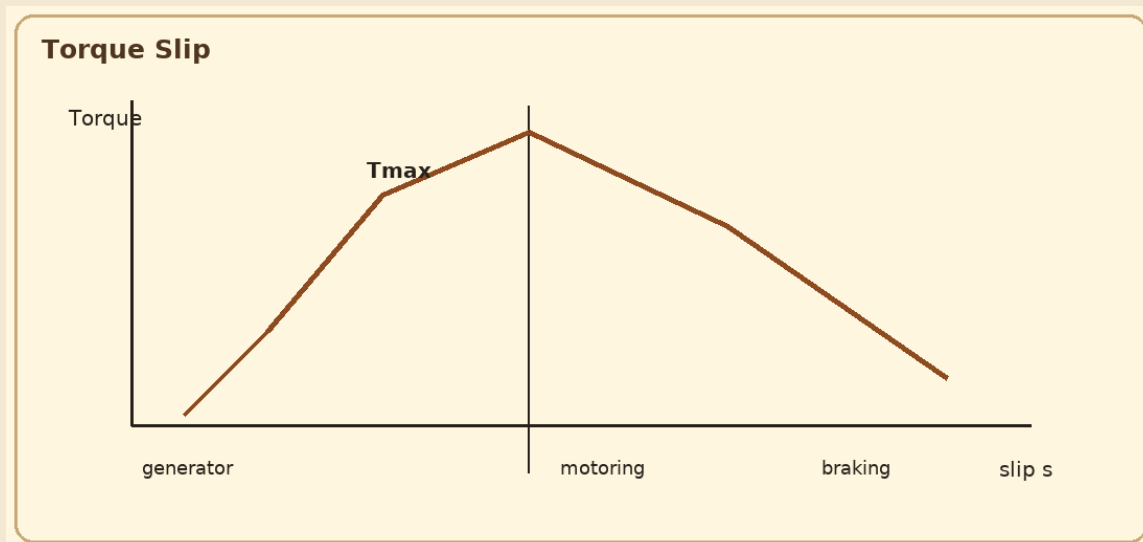
The torque-slip characteristic of a three-phase induction motor shows the variation of electromagnetic torque with slip. At $s = 0$, rotor speed equals synchronous speed, so relative speed is zero, rotor emf is zero, and torque is zero. As slip increases from zero, rotor current increases and torque rises almost linearly in the normal operating region.

The torque reaches maximum value when the effective rotor resistance equals the standstill leakage reactance term, approximately $R_2/s = X_2$ for the simplified circuit. Hence the slip at maximum torque is proportional to rotor resistance. Beyond this point, reactance dominates and torque falls as slip approaches unity.

At $s = 1$ the motor is at standstill and develops starting torque. For a squirrel-cage motor, rotor resistance is fixed, so the starting torque is limited. In a wound-rotor/slip-ring motor, external rotor resistance can be added, shifting the maximum-torque point toward higher slip and improving starting torque.

The magnitude of maximum torque is approximately independent of rotor resistance if supply voltage and reactance are unchanged, but the slip at which it occurs increases with added rotor resistance. The complete characteristic also includes generating operation for $s < 0$ and plugging/braking operation for $s > 1$.

Final: Rotor resistance mainly shifts the torque-slip curve horizontally; it can raise starting torque and make maximum torque occur at starting, while the maximum torque magnitude remains nearly unchanged.



Question 56

Original: 5C | EM-I ELE 205 Makeup, 08 January 2008

What changes can be made on cage rotor construction to improve the starting torque of a three phase induction motor. (02)

Answer 56

Use the induction-motor slip, rotor-frequency, and torque relations, keeping clear which quantities are standstill and which are at running slip.

Starting torque of a cage induction motor can be improved by modifying rotor construction to increase effective rotor resistance at starting while keeping low resistance at running speed.

Step 1: Use deep-bar rotor: skin effect at high rotor frequency forces current near the top of bars, increasing effective resistance at starting.

Step 2: Use double-cage rotor: outer cage provides high starting torque; inner cage provides efficient running.

Final: Starting torque of a cage induction motor can be improved by modifying rotor construction to increase effective rotor resistance at starting while keeping low resistance at running speed.

Question 57

Original: 2C-b | EM-I ELE 2103 End Sem, 27 November 2018

Write a technical note on crawling in three phase induction motors. (part b only, 04)

Answer 57

Crawling is the tendency of an induction motor to run at a stable speed much lower than its normal full-load speed. It is mainly produced by space harmonics in the air-gap flux wave, especially the seventh harmonic in a three-phase winding.

If the fundamental field has synchronous speed N_s , the seventh harmonic produces a forward rotating field whose synchronous speed is approximately $N_s/7$. The rotor can therefore develop a stable torque near this low speed. The fifth harmonic generally produces a backward torque, while the seventh harmonic can reinforce a forward low-speed operating point.

In the torque-speed curve this appears as a dip near normal acceleration and a secondary torque peak near one-seventh synchronous speed. If the load torque intersects this harmonic torque peak, the motor may crawl instead of accelerating to its normal operating region.

Crawling is reduced by proper choice of stator and rotor slots, short-pitched or distributed windings to suppress harmonics, and skewing of rotor slots. Better winding design reduces the harmonic flux

responsible for the low-speed torque.

Final: Crawling is a harmonic-torque phenomenon; the motor may run near $N_s/7$ due to the seventh harmonic, and it is prevented by harmonic-reducing winding and slot design.

Question 58

Original: 3A | EM-I ELE 2103 End Sem. 27 November 2018

Explain how the double cage induction motor is able to develop better starting torque compared with the conventional squirrel-cage induction motor. (03)

Answer 58

A double-cage induction motor improves starting torque by using two rotor cages with different resistance and leakage reactance. The outer cage is near the rotor surface, so it has high resistance and low leakage reactance. The inner cage is deeper, so it has low resistance and high leakage reactance.

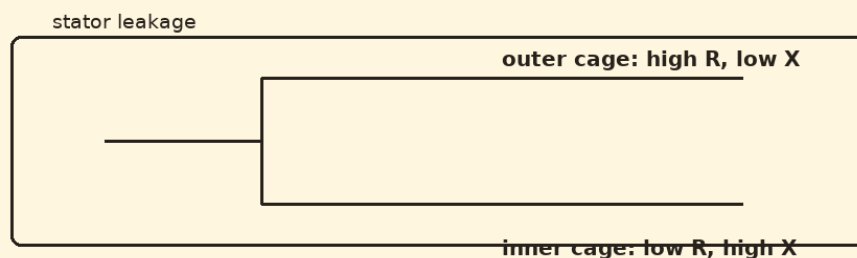
At starting, rotor frequency equals supply frequency. The reactance of the inner cage is then large, so most rotor current flows in the high-resistance outer cage. High rotor resistance gives better starting torque because the rotor current is more nearly in phase with induced emf.

During normal running, slip is small and rotor frequency is low. Since rotor reactance is proportional to slip, the reactance of the inner cage becomes small. Current then shifts mainly to the low-resistance inner cage, giving good efficiency and low running copper loss.

Thus the motor behaves like a high-resistance rotor at starting and a low-resistance rotor at normal speed. This gives better starting torque without sacrificing running efficiency.

Final: Double-cage construction combines high starting torque from the outer cage with efficient running from the inner cage.

Double Cage



Outer cage dominates starting; inner cage dominates running

Question 59

Original: 3C | EM-I ELE 2103 End Sem. 27 November 2018

A 4-pole, 50 Hz, 3-phase induction motor, with star connected rotor, has a rotor resistance of $4.5 \Omega/\text{phase}$ and a standstill leakage reactance of $8.5 \Omega/\text{phase}$. With no external resistance in the rotor circuit, the starting torque of the motor is 85 N-m. If 3Ω resistance were added in each rotor phase, find the following. i) The starting torque. ii) The torque at a slip of 3 %. (04)

Answer 59

Use the induction-motor slip, rotor-frequency, and torque relations, keeping clear which quantities are standstill and which are at running slip.

Step 1: Torque of an induction motor is proportional to R/s divided by $[(R/s)^2 + X^2]$.

Step 2: Initial starting torque is 85 N·m with $R = 4.5 \Omega$, $X = 8.5 \Omega$.

Step 3: After adding 3 Ω /phase, $R_{total} = 7.5 \Omega$.

Step 4: New starting torque = $85 \times [7.5/(7.5^2 + 8.5^2)]/[4.5/(4.5^2 + 8.5^2)] = 101.98 \text{ N}\cdot\text{m}$.

Step 5: at $s = 0.03$, $T = 85 \times [(7.5/0.03)/((7.5/0.03)^2 + 8.5^2)]/[4.5/(4.5^2 + 8.5^2)] = 6.98 \text{ N}\cdot\text{m}$.

Final: Initial starting torque is 85 N·m with $R = 4.5 \Omega$, $X = 8.5 \Omega$.; After adding 3 Ω /phase, $R_{total} = 7.5 \Omega$.; New starting torque = $85 \times [7.5/(7.5^2 + 8.5^2)]/[4.5/(4.5^2 + 8.5^2)] = 101.98 \text{ N}\cdot\text{m}$.; at $s = 0.03$, $T = 85 \times [(7.5/0.03)/((7.5/0.03)^2 + 8.5^2)]/[4.5/(4.5^2 + 8.5^2)] = 6.98 \text{ N}\cdot\text{m}$.

Question 60

Original: 4A | EM-I ELE 2103 End Sem, 27 November 2018

Discuss the variable frequency control strategies for the speed control of 3-phase induction motor. (02)

Answer 60

Use the induction-motor slip, rotor-frequency, and torque relations, keeping clear which quantities are standstill and which are at running slip.

Step 1: Variable frequency control changes the supply frequency to control synchronous speed $N_s = 120f / P$. To keep air-gap flux nearly constant, voltage is varied in proportion to frequency. $E_1 = 4.44 f N \phi K_w \Rightarrow \phi = V/f$ - If f is reduced but V is kept constant, flux increases and the core may saturate.

Step 2: If f is increased without raising V , flux falls and torque capability reduces.

Step 3: Keeping V/f constant maintains nearly constant flux and approximately constant torque capability up to base speed.

Step 4: Above base speed, voltage reaches its limit; field weakening occurs and constant-power operation is possible. Common strategies are scalar V/f control, closed-loop V/f control, vector control and direct torque control.

Final: Variable frequency control changes the supply frequency to control synchronous speed $N_s = 120f / P$. To keep air-gap flux nearly constant, voltage is varied in proportion to frequency. $E_1 = 4.44 f N \phi K_w \Rightarrow \phi = V/f$ - If f is reduced but V is kept constant, flux increases and the core may saturate.

Question 61

Original: 2C | EM-I ELE 2123 End Sem, 05 December 2023

A three-phase squirrel cage induction motor is required to be used as a three- 3 phase generator. Explain how this can be achieved. Draw and explain the torque-slip characteristics for the entire operation indicating salient points.

Answer 61

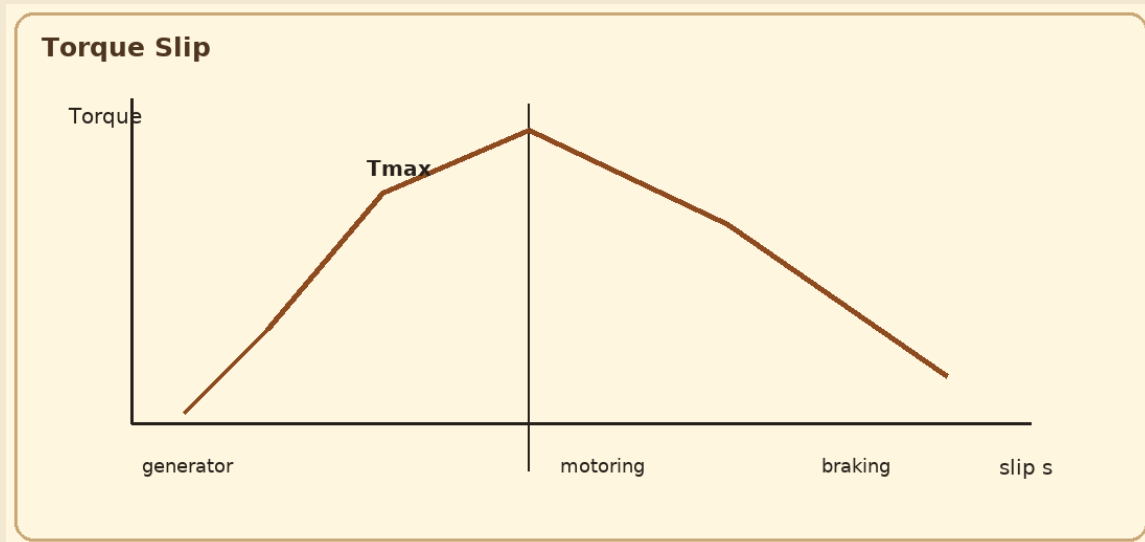
Use the induction-motor slip, rotor-frequency, and torque relations, keeping clear which quantities are standstill and which are at running slip.

An induction generator is an induction machine driven above synchronous speed by a prime mover. In this condition slip becomes negative and power flows from mechanical shaft to the electrical system.

Step 1: for motoring: $N_r < N_s$, slip is positive and the machine absorbs electrical power.

Step 2: for generating: $N_r > N_s$, slip is negative and electromagnetic torque opposes the prime mover while electrical power is delivered.

Final: For motoring: $N_r < N_s$, slip is positive and the machine absorbs electrical power.; for generating: $N_r > N_s$, slip is negative and electromagnetic torque opposes the prime mover while electrical power is delivered.



Question 62

Original: 3B | EM-I ELE 2123 End Sem, 05 December 2023

A 415V, 3 phase, 50 Hz squirrel cage motor draws 5 times its full load current 3 during starting. Which of the following starters you would recommend? a) Direct Online starter b) Star-delta starter Give sufficient reasons. Also, draw the connection diagram of the motor and starter to the supply.

Answer 62

Use the induction-motor slip, rotor-frequency, and torque relations, keeping clear which quantities are standstill and which are at running slip.

Step 1: DOL starting would draw 5 times full-load current, which is high for a squirrel-cage motor and can cause voltage dip and mechanical stress. A star-delta starter is recommended if the motor is delta-rated for normal running and starts under light or moderate load. Star-delta starter principle After acceleration, timer opens star contactor and closes delta contactor for rated running. DOL starting line current = 5 IFL Star-delta starting line current = $(1/3) \times \text{DOL current} = 5/3 \text{ IFL} = 1.67 \text{ IFL}$ Star-delta starting torque = $(1/3) \times \text{DOL starting torque}$ Therefore star-delta starting limits supply current effectively. The limitation is reduced starting torque, so it should not be used for heavy-load starts unless the available torque is sufficient.

Final: The limitation is reduced starting torque, so it should not be used for heavy-load starts unless the available torque is sufficient.

Question 63

Original: 2B | EM-I ELE 2123 End Sem, 25 November 2024

Sketch the torque-slip characteristic of a 3-phase squirrel cage induction motor 4 indicating therein the starting torque, maximum torque and the operating region. Analyze and distinguish how these characteristics can be altered in a wound rotor induction motor with respect to the following aspects: a) Starting torque and its magnitude b) Slip at which the maximum torque occurs c) Magnitude of the maximum torque

Answer 63

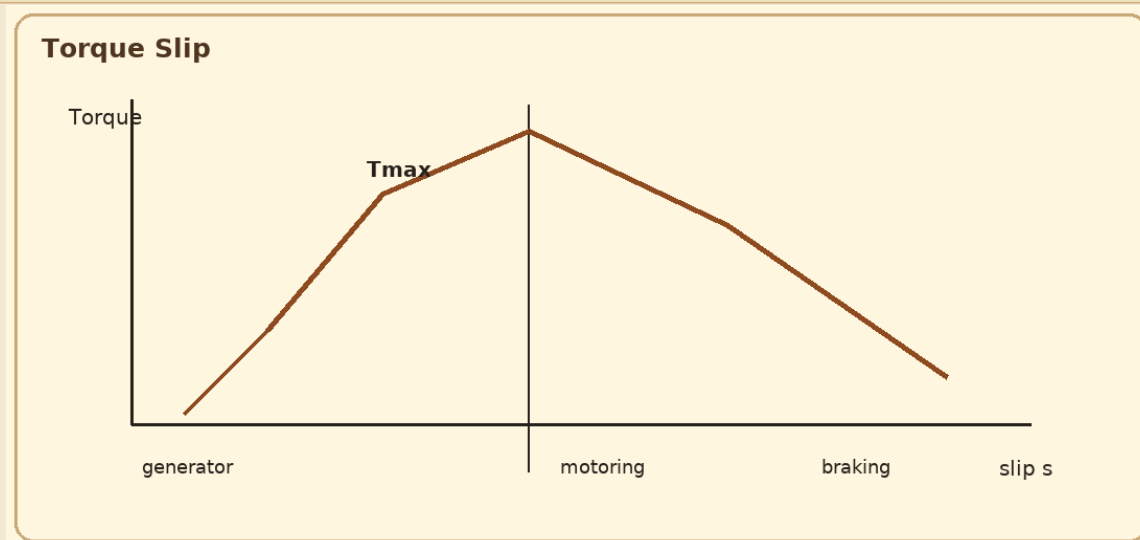
The torque-slip characteristic of a three-phase induction motor shows the variation of electromagnetic torque with slip. At $s = 0$, rotor speed equals synchronous speed, so relative speed is zero, rotor emf is zero, and torque is zero. As slip increases from zero, rotor current increases and torque rises almost linearly in the normal operating region.

The torque reaches maximum value when the effective rotor resistance equals the standstill leakage reactance term, approximately $R_2/s = X_2$ for the simplified circuit. Hence the slip at maximum torque is proportional to rotor resistance. Beyond this point, reactance dominates and torque falls as slip approaches unity.

At $s = 1$ the motor is at standstill and develops starting torque. For a squirrel-cage motor, rotor resistance is fixed, so the starting torque is limited. In a wound-rotor/slip-ring motor, external rotor resistance can be added, shifting the maximum-torque point toward higher slip and improving starting torque.

The magnitude of maximum torque is approximately independent of rotor resistance if supply voltage and reactance are unchanged, but the slip at which it occurs increases with added rotor resistance. The complete characteristic also includes generating operation for $s < 0$ and plugging/braking operation for $s > 1$.

Final: Rotor resistance mainly shifts the torque-slip curve horizontally; it can raise starting torque and make maximum torque occur at starting, while the maximum torque magnitude remains nearly unchanged.



Question 64

Original: 2A | EM-I ELE 2154 Makeup/GI. 28 July 2021

Sketch the complete torque-slip characteristics of induction motor explaining the modes of operation. (04)

Answer 64

Use the induction-motor slip, rotor-frequency, and torque relations, keeping clear which quantities are standstill and which are at running slip.

Step 1: the torque-slip curve shows how electromagnetic torque varies with slip. For an induction motor, useful motoring operation occurs between $s=0$ and $s=1$.

Step 2: at $s=0$, rotor emf and rotor current are zero, so torque is zero.

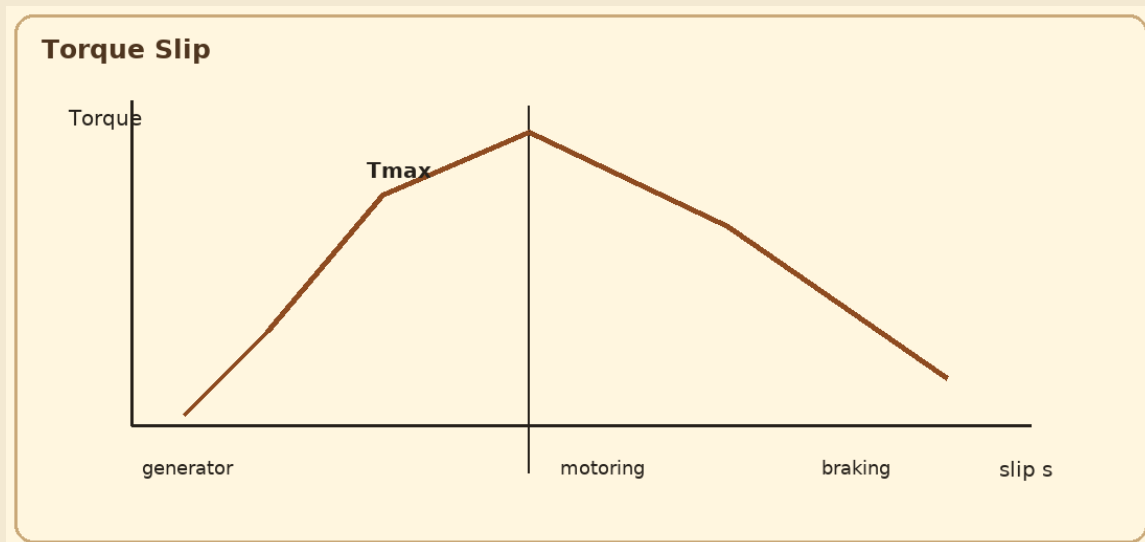
Step 3: As slip increases, torque initially rises almost linearly.

Step 4: Maximum torque occurs when rotor resistance equals effective rotor reactance condition.

Step 5: at $s=1$, the motor is at starting condition and develops starting torque.

Step 6: for $s < 0$, the machine operates as an induction generator; for $s > 1$, it operates in plugging/braking region.

Final: At $s=0$, rotor emf and rotor current are zero, so torque is zero.; As slip increases, torque initially rises almost linearly.; at $s=1$, the motor is at starting condition and develops starting torque.; for $s < 0$, the machine operates as an induction generator; for $s > 1$, it operates in plugging/braking region.



Question 65

Original: 5A | EM-I ELE 2154 Makeup/GI, 28 July 2021

Consider a 415 V, 6-pole, 50 Hz induction motor with 6 connected rotor windings. The voltage measured between the slip rings at standstill is 60 V and when running at full load is 3 V. The rotor resistance and standstill rotor reactance are $0.6 \, \Omega$ and $2.8 \, \Omega$ respectively. Calculate at full load, (a) Speed (b) Rotor current (c) Torque developed & (d) Power developed by the motor (05)

Answer 65

Use the induction-motor slip, rotor-frequency, and torque relations, keeping clear which quantities are standstill and which are at running slip.

Step 1: Rotor slip is obtained directly from slip-ring voltage ratio: $s = 3/60 = 0.05$.

Step 2: for a 6-pole, 50 Hz motor, $N_s = 1000$ rpm, so speed = 950 rpm.

Step 3: at slip 0.05, rotor reactance = $sX_2 = 0.05 \times 2.8 = 0.14 \, \Omega$.

Step 4: Rotor current/phase = $3/\sqrt{(0.6^2 + 0.14^2)} = 4.87$ A.

Step 5: Rotor copper loss = $3I_2^2R_2 = 42.6$ W.

Step 6: Rotor input = rotor Cu loss/ s = 852 W; mechanical power developed = 809 W.

Final: At slip 0.05, rotor reactance = $sX_2 = 0.05 \times 2.8 = 0.14 \, \Omega$; Rotor current/phase = $3/\sqrt{(0.6^2 + 0.14^2)} = 4.87$ A; Rotor copper loss = $3I_2^2R_2 = 42.6$ W; Rotor input = rotor Cu loss/ s = 852 W; mechanical power developed = 809 W.

5. Circle diagram and induction motor tests

Syllabus area: Week 8

Question 66

Original: 3B | EM-I ELE 205 End Sem. 03 December 2014

A 3 phase, 50 Hz, 400 V induction motor has the following test data: No load Test: 400 V, 10 A, 1 kW; Blocked rotor Test: 150 V, 40 A, 4 kW. Equivalent rotor resistance per phase referred to stator is equal to Stator resistance per phase. Draw the circle diagram and determine (a) Line current and operating slip when the shaft power is 40 HP, (b) Maximum power input. 6

Answer 66

First obtain the approximate equivalent-circuit data from the two tests. From the no-load test, $\cos \phi_0 = 1000/(\sqrt{3} \times 400 \times 10) = 0.144$. Hence $I_w = 1.44$ A, $I_m = 9.90$ A, $R_c = 230.9/1.44 = 160 \Omega$, and $X_m = 230.9/9.90 = 23.3 \Omega$.

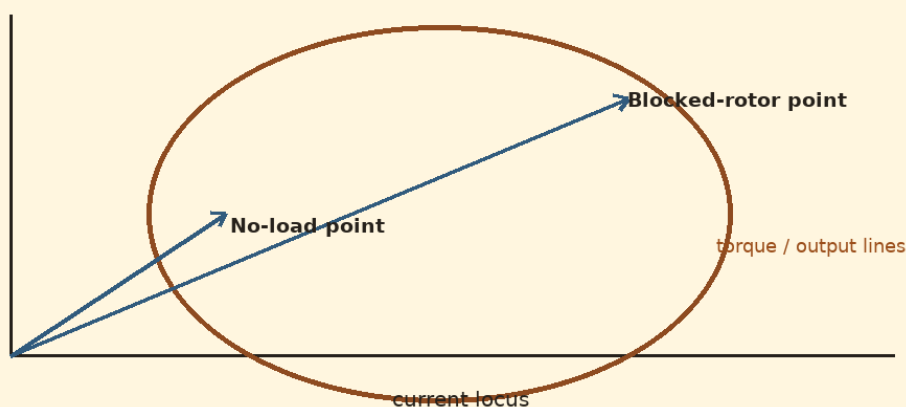
For the blocked-rotor test, $V_{ph} = 150/\sqrt{3} = 86.6$ V and $I_{br} = 40$ A. Therefore $Z_{eq} = 86.6/40 = 2.165 \Omega$ and $R_{eq} = 4000/(3 \times 40^2) = 0.833 \Omega$. Since $R_2' = R_1$, $R_1 = R_2' = 0.4167 \Omega$. The leakage reactance is $X_{eq} = \sqrt{(2.165^2 - 0.833^2)} = 1.998 \Omega$, so $X_1 = X_2' = 0.999 \Omega$ if equally divided.

Using these values for the circle diagram or the corresponding approximate equivalent circuit, the maximum developed shaft-equivalent output from the given data is only about 25.0 kW at slip $s = 0.164$. The specified 40 HP is $40 \times 746 = 29.8$ kW, which is greater than this maximum. Therefore the requested 40 HP operating point does not lie on the physically valid part of the circle.

At the maximum output point, the line current is approximately 66.8 A and the input power is about 36.1 kW. If the question asks for maximum input power from the same circle/equivalent circuit, the peak input is about 39.2 kW at slip $s = 0.273$, with line current about 84.9 A.

Final: No valid line current and slip exist for 40 HP with the stated test data; the data allow only about 25.0 kW maximum output. Maximum input power is approximately 39.2 kW.

Circle Diagram



Question 67

Original: 6A | EM-I ELE 205 End Sem. 04 December 2006

A 415 V, 40 HP, 50 Hz, δ connected induction motor gave the following test data: No-load test: 415 V, 21 A, 1250 W; Blocked rotor test: 100 V, 45 A, 2730 W. Stator copper losses is equal to rotor copper losses. Draw the circle diagram and determine, at rated output: (i) Line current and power factor (ii) Efficiency (iii) Slip (iv) Maximum output and maximum torque.

Answer 67

Start from the induction-motor test or equivalent-circuit data and convert line quantities to per-phase quantities wherever required.

Step 1: Equivalent-circuit form of circle diagram: $R_{01} = 1.348 \, \Omega$, $X_{01} = 3.605 \, \Omega$. With equal standstill stator and rotor Cu losses, $R_1 = R_2' = 0.674 \, \Omega$ and $X_1 = X_2' = 1.803 \, \Omega$.

Step 2: from the no-load test obtain approximate constant loss = 0.953 kW.

Step 3: Maximum shaft-equivalent output = 48.75 kW at $s = 0.149$.

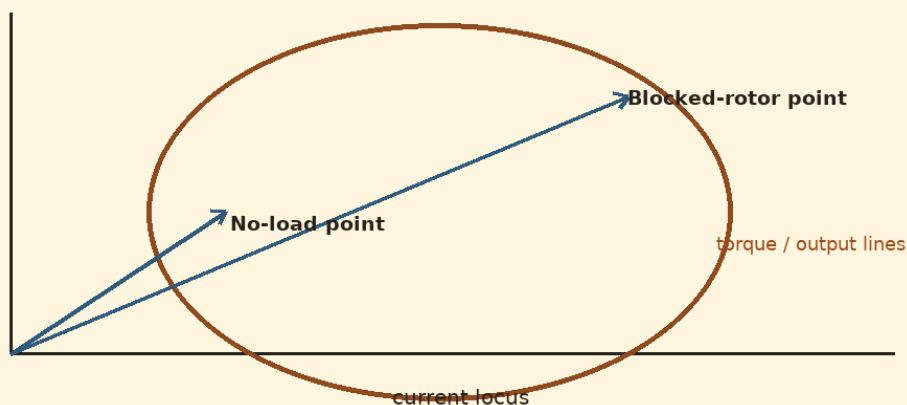
Step 4: Starting developed torque = 149.7 N-m.

Step 5: Maximum developed torque = 378.8 N-m at $s = 0.184$.

Step 6: at rated output 40 HP = 29.84 kW: line current = 182.5 A, $\text{pf} = 0.521$ lagging, slip = 0.371, efficiency = 43.7%. These are analytical equivalents to the circle diagram readings.

Final: Maximum shaft-equivalent output = 48.75 kW at $s = 0.149$.; Starting developed torque = 149.7 N-m.; Maximum developed torque = 378.8 N-m at $s = 0.184$.; at rated output 40 HP = 29.84 kW: line current = 182.5 A, $\text{pf} = 0.521$ lagging, slip = 0.371, efficiency = 43.7%. These are analytical equivalents to the circle diagram readings.

Circle Diagram



Question 68

Original: 3A | EM-I ELE 205 End Sem, 08 December 2007

A 415 V, 40 HP, 50 Hz δ connected induction motor gave the following test data: No-load test : 415 V, 21 A, 1250 W Blocked rotor test : 100 V, 45 A, 2730 W Stator copper losses are equal to rotor copper losses. Draw the circle diagram and determine (i) maximum output (ii) starting torque and (iii) maximum torque (iv) full load efficiency and slip (07)

Answer 68

Start from the induction-motor test or equivalent-circuit data and convert line quantities to per-phase quantities wherever required.

Step 1: Equivalent-circuit form of circle diagram: $R_{01} = 1.348 \, \Omega$, $X_{01} = 3.605 \, \Omega$. With equal standstill stator and rotor Cu losses, $R_1 = R_2' = 0.674 \, \Omega$ and $X_1 = X_2' = 1.803 \, \Omega$.

Step 2: from the no-load test obtain approximate constant loss = 0.953 kW.

Step 3: Maximum shaft-equivalent output = 48.75 kW at $s = 0.149$.

Step 4: Starting developed torque = 149.7 N-m.

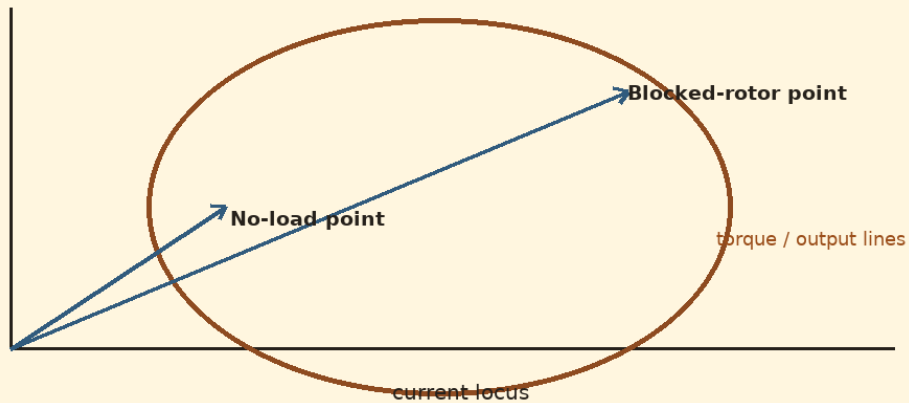
Step 5: Maximum developed torque = 378.8 N-m at $s = 0.184$.

Step 6: at rated output 40 HP = 29.84 kW: line current = 182.5 A, $\text{pf} = 0.521$ lagging, slip = 0.371, efficiency = 43.7%. These are analytical equivalents to the circle diagram readings.

Final: Maximum shaft-equivalent output = 48.75 kW at $s = 0.149$.; Starting developed torque = 149.7 N-m.; Maximum developed torque = 378.8 N-m at $s = 0.184$.; at rated output 40 HP =

29.84 kW: line current = 182.5 A, pf = 0.521 lagging, slip = 0.371, efficiency = 43.7%. These are analytical equivalents to the circle diagram readings.

Circle Diagram



Question 69

Original: 4A | EM-I ELE 205 End Sem, 26 November 2012

A 3 phase, 400V, 50 Hz, 6 pole star connected induction motor has the following test data: No load Test: 400V, 9 A, 1250 W (Line Value) Blocked Rotor test: 200V, 50A, 6930 W (Line Value) Draw the circle diagram and obtain the values of operating power factor, slip and efficiency at rated current of 30 A. Assume stator and rotor copper losses to be equal. (06)

Answer 69

Start from the induction-motor test or equivalent-circuit data and convert line quantities to per-phase quantities wherever required.

Step 1: the blocked-rotor readings give $R_{01} = 0.924 \, \Omega$ and $X_{01} = 2.116 \, \Omega$ per phase.

Step 2: with equal stator and rotor Cu losses: $R_1 = R_2' = 0.462 \, \Omega$ and $X_1 = X_2' = 1.058 \, \Omega$.

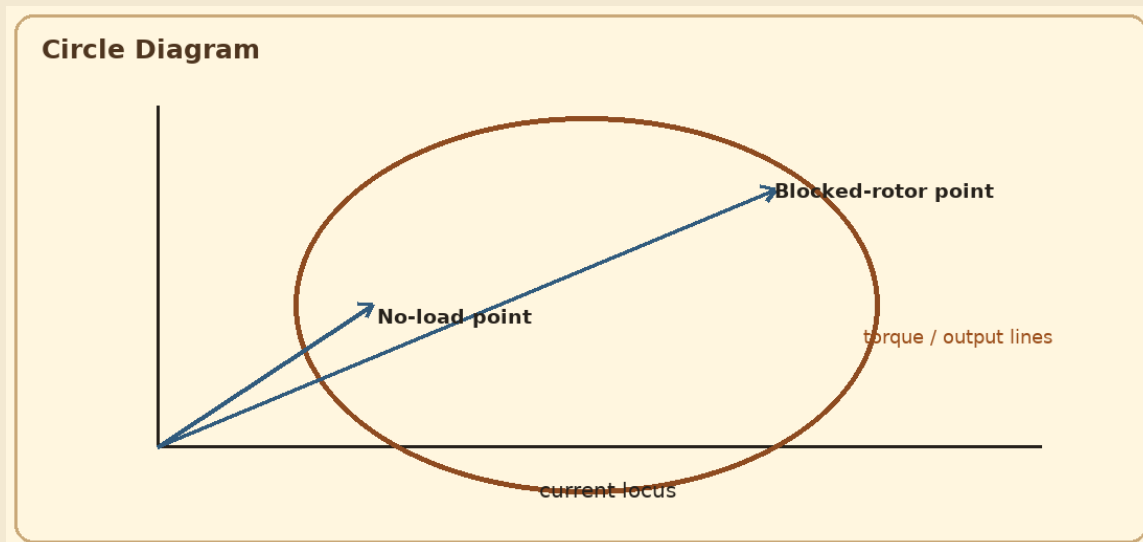
Step 3: the no-load readings give $R_c = 128 \, \Omega$, $X_m = 26.19 \, \Omega$ and constant loss is approximately 1.138 kW.

Step 4: Solving for rated current 30 A gives slip $s = 0.05435$.

Step 5: Operating power factor = 0.874 lagging; speed = 945.6 rpm.

Step 6: Input power = 18.16 kW; shaft-equivalent output = 14.03 kW; efficiency = 77.3%.

Final: The no-load readings give $R_c = 128 \, \Omega$, $X_m = 26.19 \, \Omega$ and constant loss is approximately 1.138 kW.; Solving for rated current 30 A gives slip $s = 0.05435$.; Operating power factor = 0.874 lagging; speed = 945.6 rpm.; Input power = 18.16 kW; shaft-equivalent output = 14.03 kW; efficiency = 77.3%.



Question 70

Original: 4B | EM-I ELE 205 End Sem, 30 November 2010

A 3phase, δ connected, 25kW, 440V, 6 pole 50Hz, induction motor gave the following test data. No load test : 440V, 10A, 1300 W Blocked rotor test : 96V, 36A, 1600 W From the circle diagram for full load condition calculate the (i) line current (ii) power factor (iii) slip (iv) torque and (v) efficiency (vi) maximum output. The rotor copper loss is equal to stator copper loss at standstill. (07)

Answer 70

Start from the induction-motor test or equivalent-circuit data and convert line quantities to per-phase quantities wherever required.

Step 1: using the equivalent-circuit form of the circle diagram: $R_{01} = 1.2346 \, \Omega$ and $X_{01} = 4.4508 \, \Omega$ per phase.

Step 2: with equal standstill stator and rotor Cu losses: $R_1 = R_2' = 0.6173 \, \Omega$; $X_1 = X_2' = 2.225 \, \Omega$.

Step 3: from the no-load test obtain approximate constant loss = 1.238 kW.

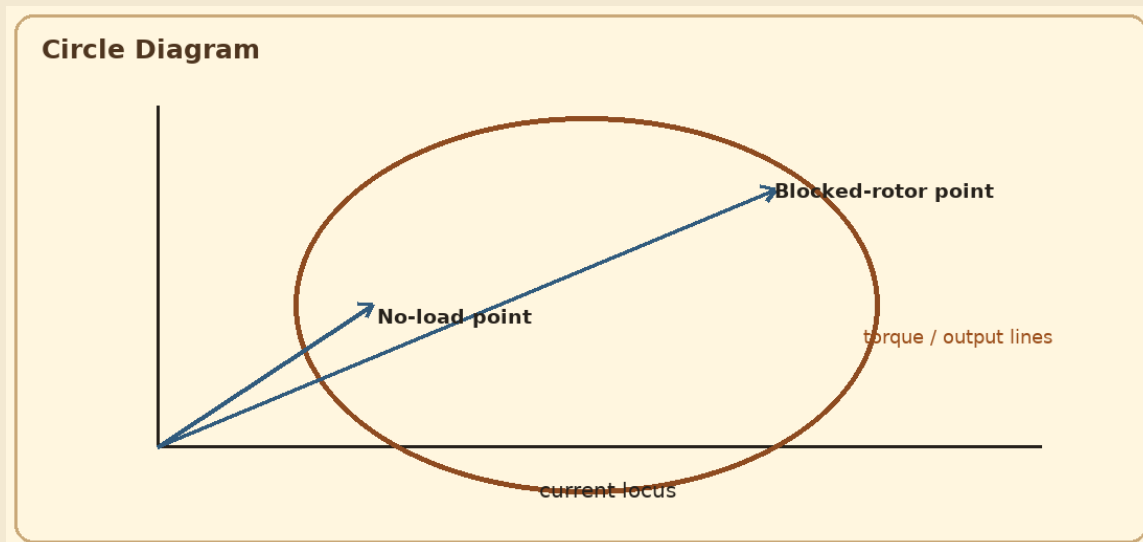
Step 4: at full-load shaft output 25 kW, solved slip is $s = 0.0323$.

Step 5: the line current is 42.5 A; power factor = 0.904 lagging; speed = 967.7 rpm.

Step 6: Developed torque is 258.95 N·m; shaft torque = 246.7 N·m; efficiency = 85.4%.

Step 7: Maximum output from the same approximate circle/equivalent circuit is about 48.37 kW at $s = 0.118$.

Final: At full-load shaft output 25 kW, solved slip is $s = 0.0323$; the line current is 42.5 A; power factor = 0.904 lagging; speed = 967.7 rpm.; Developed torque is 258.95 N·m; shaft torque = 246.7 N·m; efficiency = 85.4%; Maximum output from the same approximate circle/equivalent circuit is about 48.37 kW at $s = 0.118$.



Question 71

Original: 5B | EM-I ELE 205 Makeup, 05 January 2015

Draw the sketch of circle diagram of an Induction motor and define various phasors involved in it. Identify the length representing the starting torque. Justify your statement. 4

Answer 71

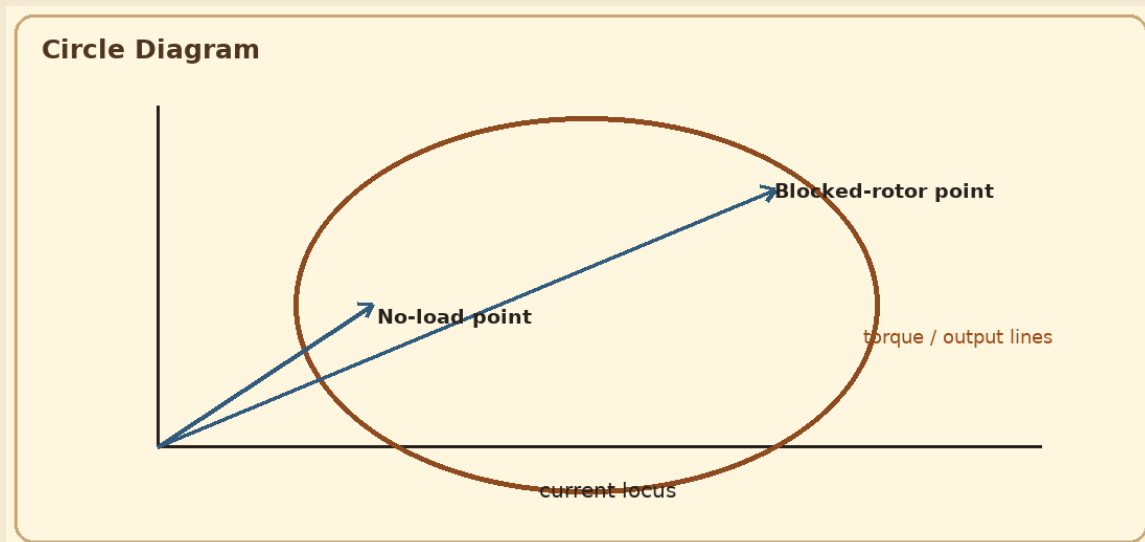
The induction motor circle diagram is drawn from the no-load and blocked-rotor current points. Take voltage as reference. Plot the no-load current vector OA at its no-load power-factor angle and the blocked-rotor current vector OB at its blocked-rotor power-factor angle, after correcting the blocked-rotor current to rated voltage if necessary.

The circle through the no-load point and blocked-rotor point represents the locus of stator current as slip changes. Important phasors are: OA = no-load current, OB = blocked-rotor current, the horizontal projection = active current component, the vertical projection = reactive current component, and the chord from no-load point to an operating point represents the load component of current.

The output line, torque line, and slip line are obtained by separating stator copper loss, rotor copper loss, and mechanical output on the power scale. The air-gap power is proportional to electromagnetic torque because $T = P_{ag}/\omega_s$.

At starting, slip $s = 1$ and rotor copper loss equals air-gap power minus stator copper loss. Hence the vertical intercept corresponding to the blocked-rotor point, measured on the torque scale from the torque line, represents starting torque.

Final: Starting torque is represented by the torque-scale intercept through the blocked-rotor point, because at $s = 1$ the circle diagram ordinate is proportional to starting air-gap power and hence starting torque.



Question 72

Original: 5A | EM-I ELE 205 Makeup, 05 January 2016

A 3 phase, 400V, 50 Hz, 6 pole star connected induction motor has the following test data: No load Test : 400V, 9 A, 1250 W (Line Value) Blocked Rotor test : 200V, 50A, 6930 W (Line Value) Draw the circle diagram and obtain the values of operating power factor, slip and efficiency at rated current of 30 A. Assume stator and rotor copper losses to be equal.

Answer 72

Start from the induction-motor test or equivalent-circuit data and convert line quantities to per-phase quantities wherever required.

Step 1: the blocked-rotor readings give $R_{01} = 0.924 \, \Omega$ and $X_{01} = 2.116 \, \Omega$ per phase.

Step 2: with equal stator and rotor Cu losses: $R_1 = R_2' = 0.462 \, \Omega$ and $X_1 = X_2' = 1.058 \, \Omega$.

Step 3: the no-load readings give $R_c = 128 \, \Omega$, $X_m = 26.19 \, \Omega$ and constant loss is approximately 1.138 kW.

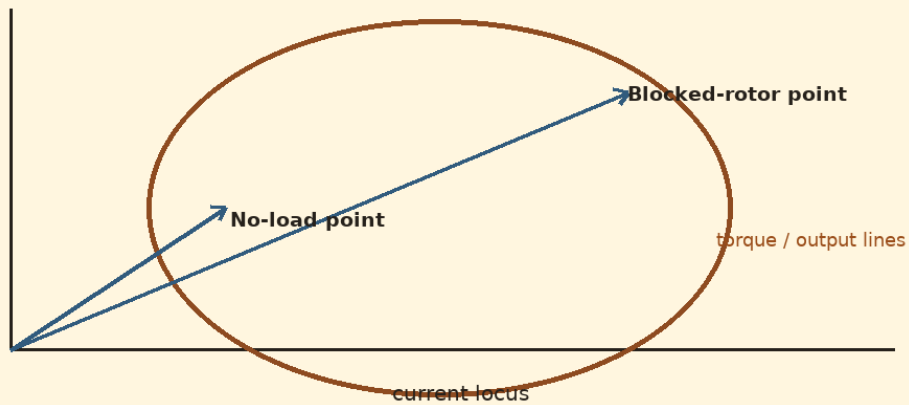
Step 4: Solving for rated current 30 A gives slip $s = 0.05435$.

Step 5: Operating power factor = 0.874 lagging; speed = 945.6 rpm.

Step 6: Input power = 18.16 kW; shaft-equivalent output = 14.03 kW; efficiency = 77.3%.

Final: The no-load readings give $R_c = 128 \, \Omega$, $X_m = 26.19 \, \Omega$ and constant loss is approximately 1.138 kW.; Solving for rated current 30 A gives slip $s = 0.05435$.; Operating power factor = 0.874 lagging; speed = 945.6 rpm.; Input power = 18.16 kW; shaft-equivalent output = 14.03 kW; efficiency = 77.3%.

Circle Diagram



Question 73

Original: 6B | EM-I ELE 205 Makeup, 08 January 2008

A 6 HP, 220 V, 50 Hz, 6 pole, 3 phase δ connected Induction motor gave the following test data No Load Test : 220 V (L-L) 6A 475 W Blocked Rotor Test : 110 V (L-L) 27 A 1930 W Calculate from circle diagram for full load condition the line current, PF, torque, Slip and efficiency. Stator copper loss at stand still is twice the rotor copper loss. (06)

Answer 73

Start from the induction-motor test or equivalent-circuit data and convert line quantities to per-phase quantities wherever required.

Step 1: Reducing the circle-diagram data to equivalent-circuit form gives $R_{01} = 2.647 \Omega$ and $X_{01} = 6.541 \Omega$.

Step 2: Since standstill stator Cu loss is twice rotor Cu loss, $R_1 = 2R_2'$, so $R_1 = 1.765 \Omega$ and $R_2' = 0.8825 \Omega$.

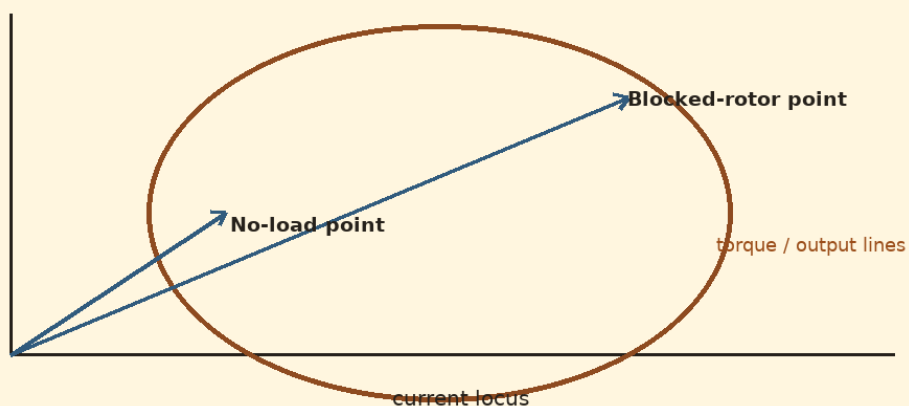
Step 3: at the rated shaft output 6 HP = 4.476 kW, solving the approximate circuit gives slip $s = 0.282$.

Step 4: the line current is 52.15 A; pf = 0.560 lagging; speed = 718 rpm.

Step 5: Developed torque is 65.0 N-m; shaft torque = 59.5 N-m; efficiency = 40.2%.

Final: Since standstill stator Cu loss is twice rotor Cu loss, $R_1 = 2R_2'$, so $R_1 = 1.765 \Omega$ and $R_2' = 0.8825 \Omega$; at the rated shaft output 6 HP = 4.476 kW, solving the approximate circuit gives slip $s = 0.282$; the line current is 52.15 A; pf = 0.560 lagging; speed = 718 rpm.; Developed torque is 65.0 N-m; shaft torque = 59.5 N-m; efficiency = 40.2%.

Circle Diagram



Question 74

Original: 4C | EM-I ELE 2103 End Sem. 27 November 2018

A 3-phase, 400 V, 50 Hz, star-connected induction motor gave the following test results (line values): No load test: 400 V, 10 A, 1.4 kW Blocked rotor test: 150 V, 40 A, 4.2 kW Construct the circle diagram to obtain the efficiency and speed of the motor when it draws 35 A. Scale: 1 cm=5 A. Assume rotor copper losses to be same as stator copper losses. (05)

Answer 74

Start from the induction-motor test or equivalent-circuit data and convert line quantities to per-phase quantities wherever required.

Step 1: the blocked-rotor readings give $R_{01} = 0.875 \Omega$ and $X_{01} = 1.980 \Omega$ per phase. With equal stator and rotor Cu losses, $R_1 = R_2' = 0.4375 \Omega$ and $X_1 = X_2' = 0.990 \Omega$.

Step 2: the no-load readings give $R_c = 114.3 \Omega$ and $X_m = 23.58 \Omega$ per phase.

Step 3: Solving the per-phase equivalent circuit at line current 35 A gives slip $s = 0.0678$.

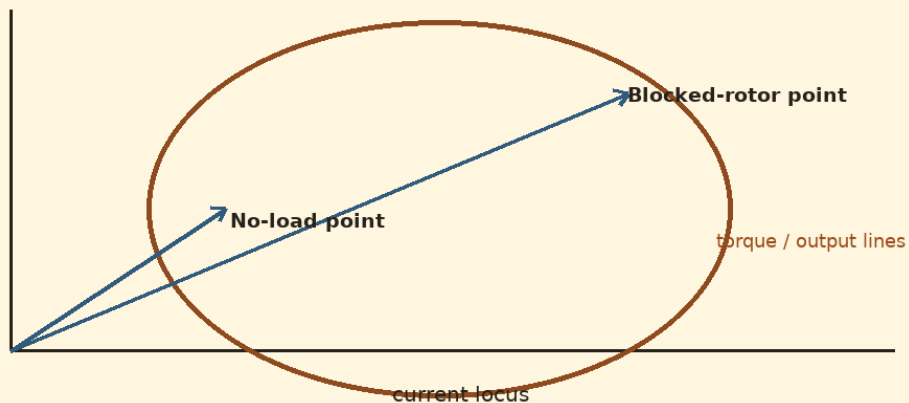
Step 4: the rotor speed is $(1-s)1500 = 1398$ rpm.

Step 5: the operating power factor is 0.876 lagging, input power = 21.25 kW, shaft-equivalent output = 17.31 kW.

Step 6: the efficiency is 81.5%. A hand-drawn circle diagram at scale 1 cm = 5 A should give close values.

Final: Solving the per-phase equivalent circuit at line current 35 A gives slip $s = 0.0678$.; the rotor speed is $(1-s)1500 = 1398$ rpm.; the operating power factor is 0.876 lagging, input power = 21.25 kW, shaft-equivalent output = 17.31 kW.; the efficiency is 81.5%. A hand-drawn circle diagram at scale 1 cm = 5 A should give close values.

Circle Diagram



Question 75

Original: 3C | EM-I ELE 2123 End Sem. 05 December 2023

A 4kW, 400V, 50Hz, 3 phase, 4 pole 6 connected induction motor has stator 4 resistance of 0.36Ω per phase and rotor resistance of 0.06Ω per phase. The no load and blocked rotor test data are as follows: No load test: 400 V, 3.3 A, p.f =0.174 Blocked rotor test: 210V, 16A, p.f =0.45 Estimate the single-phase equivalent circuit of the induction motor.

Answer 75

Start from the induction-motor test or equivalent-circuit data and convert line quantities to per-phase quantities wherever required.

Step 1: Δ motor: $V_{\text{phase}} = V_{\text{line}} = 400$ V and $I_{0\text{phase}} = I_{0\text{line}}/\sqrt{3} = 1.905$ A.

Step 2: No-load input = $\sqrt{3} V_L I_L \text{ pf} = 398$ W. Hence per phase $I_w = P_0/(3V_{\text{ph}}) = 0.331$ A, $I_m = 1.876$ A.

Step 3: $R_c = V_{ph}/I_w = 1208 \Omega$; $X_m = V_{ph}/I_m = 213 \Omega$.

Step 4: Blocked-rotor: $V_{ph} = 210 \text{ V}$, $I_{ph} = 16/\sqrt{3} = 9.238 \text{ A}$. $Z_{01} = 22.73 \Omega$, $R_{01} = Z_{01} \text{ pf} = 10.23 \Omega$, $X_{01} = 20.36 \Omega$.

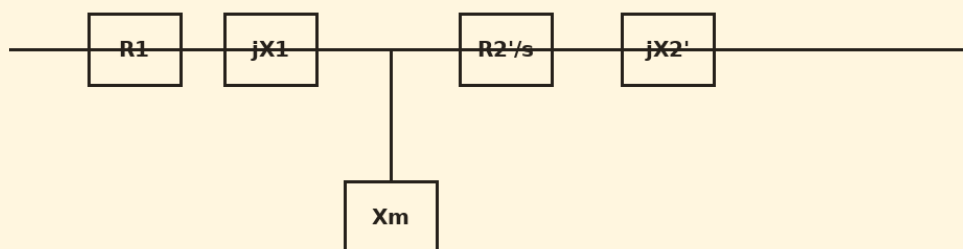
Step 5: using the test data, $R_2' = R_{01} - R_1 = 10.23 - 0.36 = 9.87 \Omega$ and $X_1 = X_2' = 10.18 \Omega$ if leakage is divided equally.

Step 6: Note: the stated rotor resistance 0.06Ω conflicts with the blocked-rotor resistance; the test-based referred value is used for the equivalent circuit.

Final: Blocked-rotor: $V_{ph} = 210 \text{ V}$, $I_{ph} = 16/\sqrt{3} = 9.238 \text{ A}$. $Z_{01} = 22.73 \Omega$, $R_{01} = Z_{01} \text{ pf} = 10.23 \Omega$, $X_{01} = 20.36 \Omega$; Note: the stated rotor resistance 0.06Ω conflicts with the blocked-rotor resistance; the test-based referred value is used for the equivalent circuit.

Equivalent Circuit

Approximate per-phase induction motor circuit



Question 76

Original: 3B | EM-I ELE 2123 End Sem. 25 November 2024

A 3-phase, 400 V, 3.7 kW, 7.5 A, 945 rpm, Δ connected squirrel cage induction 4 motor has per phase stator resistance of 5Ω . This motor is subjected to no-load and blocked-rotor tests. The test results are: No load test: 400 V, 3.25 A, 232 W Blocked rotor test: 92 V, 7.5 A, 650 W a) Determine the parameters of its approximate equivalent circuit. b) Draw and label the approximate equivalent circuit referred to stator.

Answer 76

Start from the induction-motor test or equivalent-circuit data and convert line quantities to per-phase quantities wherever required.

Step 1: Δ motor: $V_{ph} = 400 \text{ V}$, $I_{0ph} = 3.25/\sqrt{3} = 1.876 \text{ A}$.

Step 2: No-load branch: $R_c = 2069 \Omega$, $X_m = 214.4 \Omega$.

Step 3: Blocked-rotor: $V_{ph} = 92 \text{ V}$, $I_{ph} = 7.5/\sqrt{3} = 4.330 \text{ A}$, so $Z_{01} = 21.25 \Omega$, $R_{01} = 11.56 \Omega$ and $X_{01} = 17.82 \Omega$.

Step 4: Given $R_1 = 5 \Omega$, $R_2' = R_{01} - R_1 = 6.56 \Omega$.

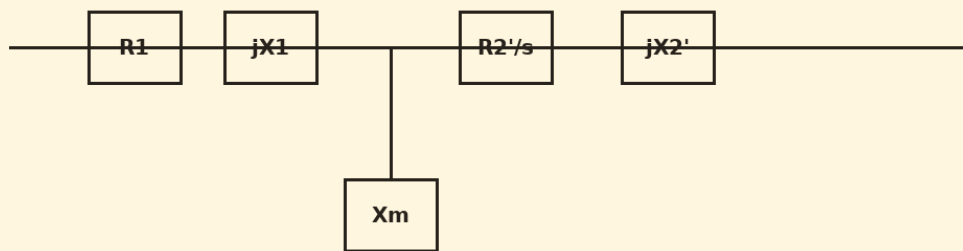
Step 5: Assuming equal leakage reactance: $X_1 = X_2' = 8.91 \Omega$.

Step 6: These parameters form the approximate equivalent circuit referred to the stator.

Final: No-load branch: $R_c = 2069 \Omega$, $X_m = 214.4 \Omega$; Blocked-rotor: $V_{ph} = 92 \text{ V}$, $I_{ph} = 7.5/\sqrt{3} = 4.330 \text{ A}$, so $Z_{01} = 21.25 \Omega$, $R_{01} = 11.56 \Omega$ and $X_{01} = 17.82 \Omega$; Given $R_1 = 5 \Omega$, $R_2' = R_{01} - R_1 = 6.56 \Omega$; Assuming equal leakage reactance: $X_1 = X_2' = 8.91 \Omega$.

Equivalent Circuit

Approximate per-phase induction motor circuit



Question 77

Original: 3B | EM-I ELE 2154 Makeup/GI, 28 July 2021

From the following test data given for a three phase 100HP, 3.3kV, 27A, 6pole 50Hz squirrel cage induction motor with star connected stator, develop the approximate equivalent circuit of the machine referred to stator side. No Load test (line values): 3.3kV, 6A, 24kW Blocked Rotor test (line values): 400V, 25A, 14kW DC resistance per phase: 3.75Ω. Also, Explain the reason why the rotor is blocked in the blocked-rotor test? (06)

Answer 77

No-load test, star connection: $V_{ph} = 3300/\sqrt{3} = 1905$ V, $I_0 = 6$ A, and $P_0/\text{phase} = 24000/3 = 8000$ W. The no-load power factor is $24000/(\sqrt{3} \times 3300 \times 6) = 0.700$. Therefore $I_w = 6 \times 0.700 = 4.20$ A and $I_m = \sqrt{6^2 - 4.20^2} = 4.29$ A.

The shunt branch parameters are $R_0 = V_{ph}/I_w = 1905/4.20 = 454$ Ω and $X_m = V_{ph}/I_m = 1905/4.29 = 445$ Ω.

Blocked-rotor test: $V_{ph} = 400/\sqrt{3} = 230.94$ V, $I_{br} = 25$ A, and $P_{br}/\text{phase} = 14000/3 = 4667$ W. Therefore $Z_{eq} = 230.94/25 = 9.24$ Ω and $R_{eq} = 4667/25^2 = 7.47$ Ω.

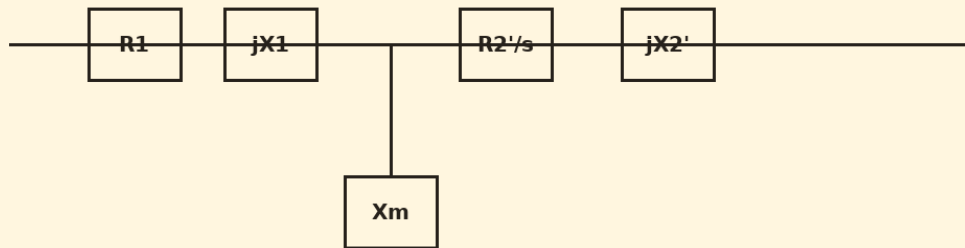
Given DC resistance per phase is 3.75 Ω, take $R_1 = 3.75$ Ω. Then $R_2' = R_{eq} - R_1 = 3.72$ Ω. The total leakage reactance is $X_{eq} = \sqrt{9.24^2 - 7.47^2} = 5.43$ Ω. If split equally, $X_1 = X_2' = 2.72$ Ω.

The rotor is blocked during the blocked-rotor test so that slip is unity. Under this condition the rotor frequency equals supply frequency, the rotor branch has standstill leakage reactance, and the series impedance can be measured at reduced voltage without accelerating the machine.

Final: Approximate equivalent-circuit parameters referred to stator are $R_1 = 3.75$ Ω, $R_2' = 3.72$ Ω, $X_1 = X_2' = 2.72$ Ω, $R_0 = 454$ Ω, $X_m = 445$ Ω.

Equivalent Circuit

Approximate per-phase induction motor circuit



Question 78

Original: 2A | EM-I ELE 2154 Online End Sem, 27 January 2022

The following test results were obtained on a 3-phase, 75kW, 3.3kV, 6-Pole, 50Hz, 6-connected three-phase squirrel cage induction motor. Determine the parameters of the approximated equivalent circuit. No-load Test: 3.3kV, 5A and 2500W Blocked rotor Test: 400V, 27A, and 15000W (04)

Answer 78

Start from the induction-motor test or equivalent-circuit data and convert line quantities to per-phase quantities wherever required.

Step 1: Δ motor: $V_{ph} = 3300 \text{ V}$, $I_{0ph} = 5/\sqrt{3} = 2.887 \text{ A}$.

Step 2: No-load branch: $R_c = 13.07 \text{ kohm}$ and $X_m = 1.147 \text{ kohm}$.

Step 3: Blocked-rotor: $V_{ph} = 400 \text{ V}$, $I_{ph} = 27/\sqrt{3} = 15.59 \text{ A}$.

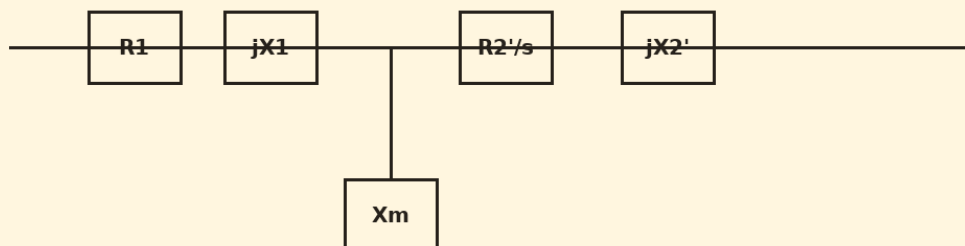
Step 4: $Z_{01} = 25.66 \Omega$, $R_{01} = P_{br}/(3I_{ph}^2) = 20.56 \Omega$, $X_{01} = 15.36 \Omega$.

Step 5: If no separate dc resistance is supplied, use the usual split: $R_1 = R_2' = 10.28 \Omega$, $X_1 = X_2' = 7.68 \Omega$.

Final: No-load branch: $R_c = 13.07 \text{ kohm}$ and $X_m = 1.147 \text{ kohm}$.; Blocked-rotor: $V_{ph} = 400 \text{ V}$, $I_{ph} = 27/\sqrt{3} = 15.59 \text{ A}$.; $Z_{01} = 25.66 \Omega$, $R_{01} = P_{br}/(3I_{ph}^2) = 20.56 \Omega$, $X_{01} = 15.36 \Omega$.; If no separate dc resistance is supplied, use the usual split: $R_1 = R_2' = 10.28 \Omega$, $X_1 = X_2' = 7.68 \Omega$.

Equivalent Circuit

Approximate per-phase induction motor circuit



6. Single-phase induction motors

Syllabus area: Week 10

Question 79

Original: 7B | EM-I ELE 205 End Sem. 04 December 2006

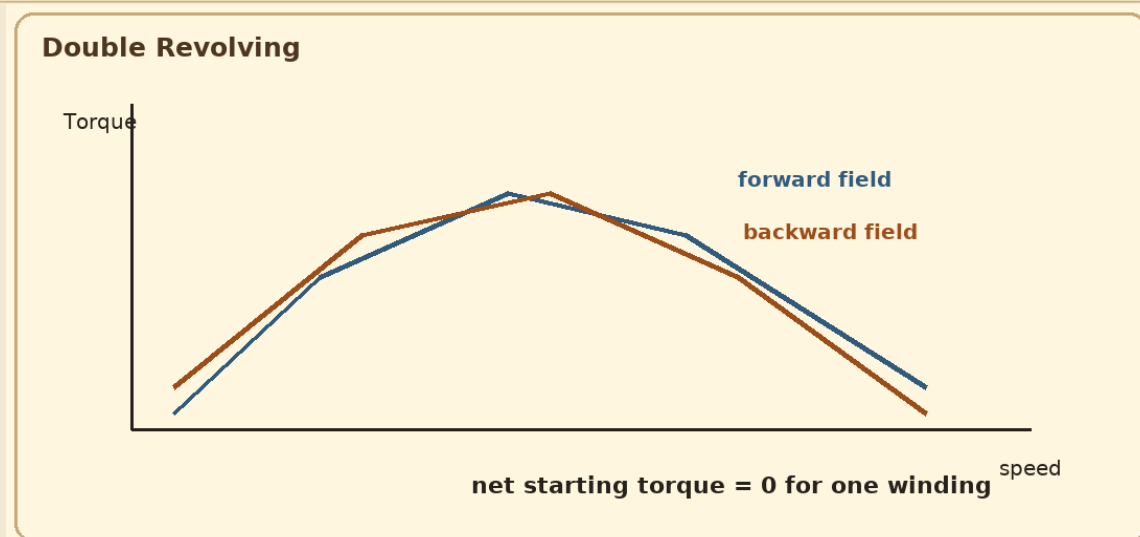
Show that a single phase current in a single phase winding produces only a pulsating magnetic field.

Answer 79

Use double-revolving-field theory or the single-phase equivalent circuit, depending on whether the question is conceptual or numerical.

Step 1: A single-phase current in a sinusoidally distributed stator winding produces a magnetic field fixed in space but alternating in magnitude with time. Single-phase winding produces a pulsating, not rotating, field $i = I_m \cos(\omega t)$ $F(\theta, t) = F_m \cos(\theta) \cos(\omega t)$ $F(\theta, t) = (F_m/2) \cos(\theta - \omega t) + (F_m/2) \cos(\theta + \omega t)$

Final: A single-phase current in a sinusoidally distributed stator winding produces a magnetic field fixed in space but alternating in magnitude with time. Single-phase winding produces a pulsating, not rotating, field $i = I_m \cos(\omega t)$ $F(\theta, t) = F_m \cos(\theta) \cos(\omega t)$ $F(\theta, t) = (F_m/2) \cos(\theta - \omega t) + (F_m/2) \cos(\theta + \omega t)$



Question 80

Original: 5C | EM-I ELE 205 End Sem. 30 November 2010

A 240V, 50Hz, 2 pole single phase induction motor has the following equivalent circuit impedances: $r_1=2.2\Omega$, $r_2'=3.8\Omega$, $x_1=3\Omega$, $x_2'=2.1\Omega$, $x_m=86\Omega$. Friction, windage and core losses=50W. Calculate input current, power factor, output power and efficiency at a full load speed of 2820RPM. (05)

Answer 80

Use double-revolving-field theory or the single-phase equivalent circuit, depending on whether the question is conceptual or numerical.

Step 1: Slip = $(3000-2820)/3000 = 0.06$.

Step 2: Use the double-revolving-field equivalent circuit. Forward branch: $(jX_m/2)$ parallel $[r_2'/(2s)+jx_2'/2]$. Backward branch: $(jX_m/2)$ parallel $[r_2'/(2(2-s))+jx_2'/2]$.

Step 3: Substituting $r_1=2.2$, $x_1=3$, $r_2'=3.8$, $x_2'=2.1$, $x_m=86$ gives input current = 7.98 A.

Step 4: Input power factor = 0.765 lagging; input power = 1.465 kW.

Step 5: Forward air-gap power = 1.265 kW; backward air-gap power = 0.059 kW.

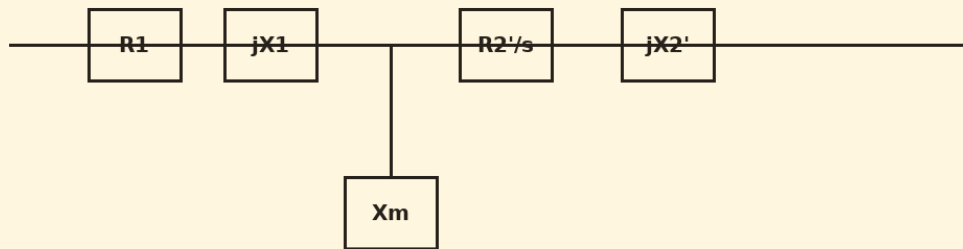
Step 6: Mechanical power developed = $(1-s)(P_{gf}-P_{gb}) = 1.134 \text{ kW}$.

Step 7: Output after 50 W losses = 1.084 kW; efficiency = 74.0%.

Final: Input power factor = 0.765 lagging; input power = 1.465 kW.; Forward air-gap power = 1.265 kW; backward air-gap power = 0.059 kW.; Mechanical power developed = $(1-s)(P_{gf}-P_{gb}) = 1.134 \text{ kW}$.; Output after 50 W losses = 1.084 kW; efficiency = 74.0%.

Equivalent Circuit

Approximate per-phase induction motor circuit



Question 81

Original: 6A | EM-I ELE 205 Makeup. 08 January 2008

With the help of double field revolving theory, show that single phase induction motor is not self starting. (04)

Answer 81

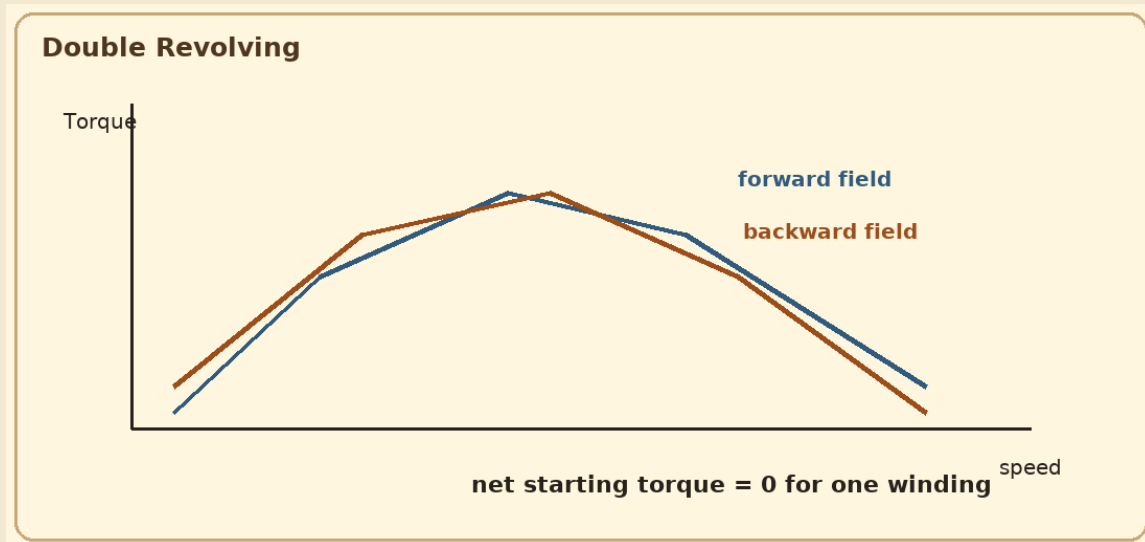
A single-phase stator winding supplied by sinusoidal current produces a pulsating magnetic field, not a rotating field. This pulsating field can be resolved into two rotating magnetic fields of equal magnitude rotating in opposite directions at synchronous speed.

At standstill, the forward and backward fields induce equal rotor currents and produce equal torques in opposite directions. Hence the net starting torque is zero. This proves that a single-phase induction motor with only one stator winding is not self-starting.

If the rotor is given an initial push in either direction, the slip with respect to the forward field becomes less than one, while the slip with respect to the backward field becomes greater than one. The forward torque then exceeds the backward torque, and the motor continues to accelerate in the direction of the initial rotation.

Therefore an auxiliary starting means is required. Common methods are split-phase winding, capacitor-start winding, shaded-pole construction, or repulsion-start arrangement.

Final: A single-phase induction motor is not self-starting because its forward and backward field torques cancel at standstill.



Question 82

Original: 4B | EM-I ELE 2103 End Sem, 27 November 2018

With necessary phasor diagram, explain how a capacitor can help in starting of a single-phase induction motor. (03)

Answer 82

A capacitor helps a single-phase induction motor start by creating a phase displacement between the current in the main winding and the current in the auxiliary winding. The two windings are placed physically apart in space, usually about 90 electrical degrees, and the capacitor makes their currents nearly time-displaced.

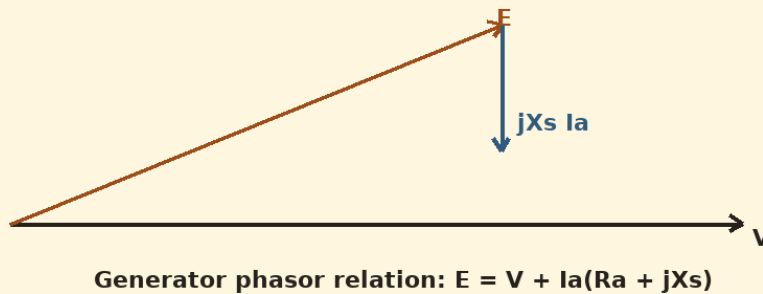
With both space displacement and time displacement, the two winding currents produce an approximate rotating magnetic field at starting. This rotating field gives a definite starting torque, unlike the pulsating field of a single winding.

In the phasor diagram, main-winding current I_m lags the supply voltage because the main winding is inductive. The auxiliary current I_a is made to lead or be less lagging by the series capacitor. The larger the phase angle between I_m and I_a , the larger the starting torque, with maximum torque ideally near 90° phase displacement.

After the motor accelerates, the starting capacitor and auxiliary winding may be disconnected by a centrifugal switch in a capacitor-start motor. In capacitor-run motors, a smaller capacitor remains in circuit to improve running power factor and smoothness.

Final: The capacitor creates a two-phase effect, producing starting torque by making the stator field approximately rotating.

Phasor



Question 83

Original: 3A | EM-I ELE 2123 End Sem, 05 December 2023

How is torque produced in a capacitor start single phase induction motor. Is there any need for this capacitor after starting? Explain.

Answer 83

Use double-revolving-field theory or the single-phase equivalent circuit, depending on whether the question is conceptual or numerical.

A single-phase induction motor is not self-starting because a single stator winding produces only a pulsating field. A capacitor-start motor uses an auxiliary winding displaced in space and a capacitor to create phase displacement in current.

Step 1: the main winding current and auxiliary winding current are separated in time phase.

Step 2: the windings are physically displaced by about 90 electrical degrees in space.

Step 3: the two currents produce a rotating field, giving non-zero starting torque.

Step 4: After the motor reaches about 70-80 percent of rated speed, the starting capacitor is usually disconnected by a centrifugal switch or relay.

Step 5: In capacitor-run motors, a smaller capacitor remains connected for better running power factor and smoother torque.

Final: After the motor reaches about 70-80 percent of rated speed, the starting capacitor is usually disconnected by a centrifugal switch or relay.; In capacitor-run motors, a smaller capacitor remains connected for better running power factor and smoother torque.

Question 84

Original: 3A | EM-I ELE 2123 End Sem, 25 November 2024

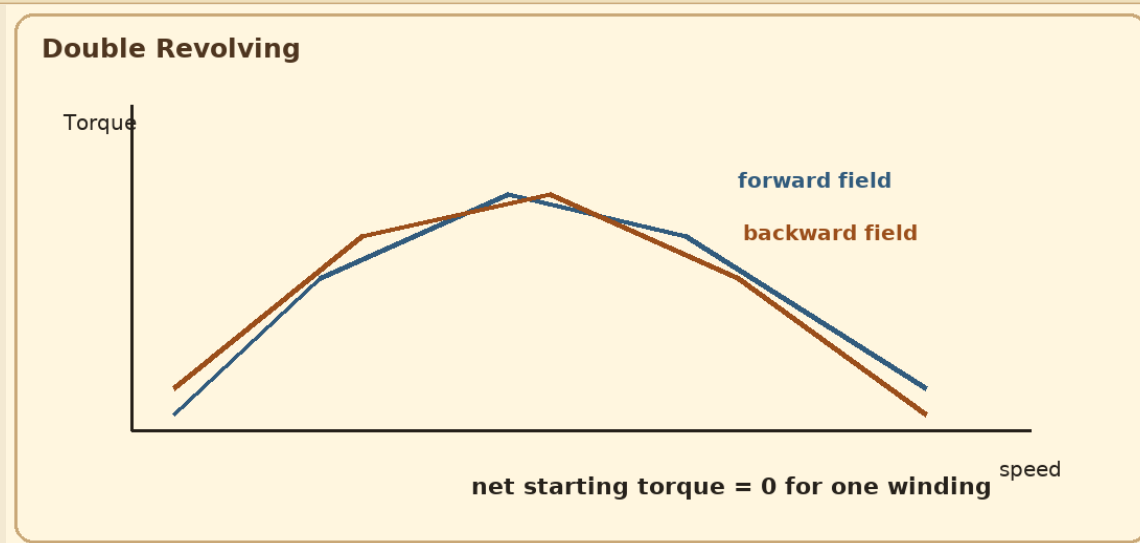
With the help of double field revolving theory, prove that a single-phase induction motor containing only one stator winding produces no starting torque. Justify your answer with suitable characteristics.

Answer 84

Use double-revolving-field theory or the single-phase equivalent circuit, depending on whether the question is conceptual or numerical.

Step 1: According to double revolving field theory, a single-phase pulsating magnetic field can be resolved into two rotating magnetic fields of equal magnitude rotating in opposite directions at synchronous speed. Double revolving field theory $F = F_m \cos(\theta) \cos(\omega t)$ $F = (F_m/2) \cos(\theta - \omega t) + (F_m/2) \cos(\theta + \omega t)$ - At standstill the rotor has unit slip with respect to both the forward and backward component fields.

Final: Double revolving field theory $F = F_m \cos(\theta) \cos(\omega t)$ $F = (F_m/2) \cos(\theta - \omega t) + (F_m/2) \cos(\theta + \omega t)$ - At standstill the rotor has unit slip with respect to both the forward and backward component fields.



Question 85

Original: 1C | EM-I ELE 2154 Online End Sem. 27 January 2022

Explain the operation of a single-phase induction motor using double field revolving theory. (02)

Answer 85

A single-phase stator winding produces a pulsating air-gap field. By double-field revolving theory, this pulsating field is resolved into two equal rotating fields:

$$F = F_m \cos(\theta) \cos(\omega t)$$

$$= (F_m/2) \cos(\theta - \omega t) + (F_m/2) \cos(\theta + \omega t).$$

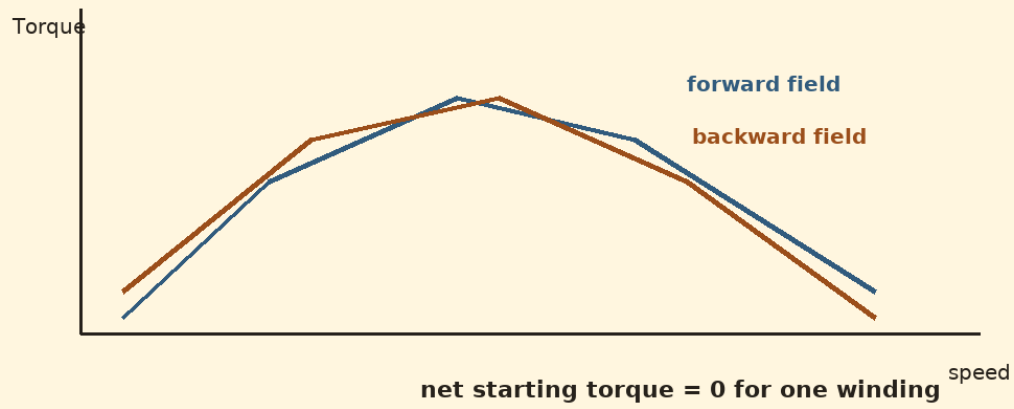
The first term is a forward rotating field and the second term is a backward rotating field. Both have half the peak magnitude of the original pulsating field and both rotate at synchronous speed in opposite directions.

At standstill, rotor slip with respect to the forward field is 1 and slip with respect to the backward field is also 1. Therefore the two fields induce equal rotor currents and develop equal torques in opposite directions. The net starting torque is zero, so a single-phase induction motor with only one stator winding is not self-starting.

If the rotor is first rotated in one direction, the forward slip becomes less than 1 and the backward slip becomes greater than 1. The forward torque then becomes larger than the backward torque and the motor accelerates in the direction of rotation.

Final: Double-revolving-field theory explains both facts: zero starting torque at standstill and continued running after the rotor is given an initial direction.

Double Revolving



7. Induction generator

Syllabus area: Week 10

Question 86

Original: 6C | EM-I ELE 205 End Sem. 08 December 2007

List out advantages, disadvantages and applications of induction generator. (03)

Answer 86

An induction generator is the same basic machine as an induction motor, but it is driven by a prime mover above synchronous speed. Then slip becomes negative and the electromagnetic torque opposes the prime mover, so mechanical power is converted into electrical power.

Advantages:

- It is rugged, cheap, and simple because there is no commutator and no separate DC field system.
- It can be connected easily to a strong grid; voltage and frequency are then fixed by the grid.
- It is suitable for variable-speed prime movers such as wind turbines and small hydro turbines.
- Maintenance is low compared with a DC generator or a separately excited synchronous generator.

Disadvantages:

- It needs magnetizing reactive power from the grid or from capacitors.
- Stand-alone voltage regulation is poor because terminal voltage depends on speed, capacitance, magnetization, and load.
- Frequency control is poor in isolated operation.
- Power factor is generally lagging unless reactive compensation is provided.

Applications include wind-energy conversion, micro/mini hydro plants, regenerative braking of induction-motor drives, energy recovery in cranes/elevators, and small grid-connected generation.

Final: Induction generators are simple and rugged, but their dependence on external reactive power and weak isolated voltage control limits their use for bulk standalone generation.

Question 87

Original: 3B | EM-I ELE 205 Makeup. 05 January 2016

Explain the working of Induction generator. Mention one application. 4M

Answer 87

An induction generator is an induction machine driven by a prime mover at a speed greater than synchronous speed. In this condition slip is negative: $s = (N_s - N_r)/N_s < 0$. The rotor cuts the rotating field in the opposite relative direction compared with motoring, so electromagnetic torque opposes the prime mover and electrical power is delivered to the supply.

A grid-connected induction generator takes magnetizing reactive power from the grid and supplies real power to the grid. Its frequency and voltage are practically fixed by the grid, making operation simple and rugged.

For stand-alone operation, capacitors must be connected across the terminals to provide magnetizing reactive power. Voltage buildup depends on residual magnetism, capacitance, speed, and load. This is why standalone voltage regulation is poor compared with a synchronous generator.

Applications include wind energy conversion, small hydro plants, regenerative braking, and energy recovery in industrial drives.

Final: An induction generator works above synchronous speed with negative slip; it is rugged and simple but needs external reactive power for excitation.

Question 88

Original: 3A | EM-I ELE 205 Makeup. 08 January 2008

What is an induction generator? Why it is not proffered for generation of bulk power. (02)

Answer 88

Treat the machine as an induction motor driven above synchronous speed, so the slip is negative and real power flow reverses.

An induction generator is an induction machine driven above synchronous speed by a prime mover. In this condition slip becomes negative and power flows from mechanical shaft to the electrical system.

Step 1: for motoring: $N_r < N_s$, slip is positive and the machine absorbs electrical power.

Step 2: for generating: $N_r > N_s$, slip is negative and electromagnetic torque opposes the prime mover while electrical power is delivered.

Final: For motoring: $N_r < N_s$, slip is positive and the machine absorbs electrical power.; for generating: $N_r > N_s$, slip is negative and electromagnetic torque opposes the prime mover while electrical power is delivered.

Question 89

Original: 3A | EM-I ELE 2154 Makeup/GI. 28 July 2021

Explain the reason why induction generators are not used as stand- alone generators. Explain this mode of operation of an induction machine. What are its applications? (04)

Answer 89

Treat the machine as an induction motor driven above synchronous speed, so the slip is negative and real power flow reverses.

An induction generator is an induction machine driven above synchronous speed by a prime mover. In this condition slip becomes negative and power flows from mechanical shaft to the electrical system.

Step 1: for motoring: $N_r < N_s$, slip is positive and the machine absorbs electrical power.

Step 2: for generating: $N_r > N_s$, slip is negative and electromagnetic torque opposes the prime mover while electrical power is delivered.

Step 3: the machine still requires magnetizing reactive power; when grid-connected it takes this from the grid, and in isolated operation capacitors are needed for self-excitation.

Step 4: Applications include wind turbines, small hydro, regenerative braking and energy recovery drives. It is generally not preferred as a stand-alone bulk generator because voltage and frequency regulation are poor without a stiff grid and because external reactive power is required.

Final: Applications include wind turbines, small hydro, regenerative braking and energy recovery drives. It is generally not preferred as a stand-alone bulk generator because voltage and frequency regulation are poor without a stiff grid and because external reactive power is required.

8. Synchronous machine construction and alternator fundamentals

Syllabus area: Week 11

Question 90

Original: 2C | EM-II ELE 204 Makeup, 08 July 2014

A 1500 kVA, 6.6 kV, 3-phase star connected wound rotor alternator with a resistance of 0.4Ω and reactance of 6Ω per phase, delivers full load current at 0.8 power factor lagging and normal terminal voltage. Estimate the excitation emf required and respective load angle. (04)

Answer 90

Use the synchronous-machine phasor equation and keep the generator/motor sign convention explicit before substituting values.

Step 1: the full-load current is 131.22 A; $V_{ph} = 3,810.5 \text{ V}$.

Step 2: using the generator phasor convention, $E = V + I_a(R_a + jX_s)$. At 0.8 lagging, I_a lags V by 36.87° .

Step 3: $E = 4,366.1 \text{ V/phase}$ at $\delta = 7.8768^\circ$.

Step 4: the percentage regulation = 14.580 %.

Final: The full-load current is 131.22 A; $V_{ph} = 3,810.5 \text{ V}$; $E = 4,366.1 \text{ V/phase}$ at $\delta = 7.8768^\circ$; the percentage regulation = 14.580 %.

Question 91

Original: 4B | EM-II ELE 204 Makeup, 09 July 2015

Two alternators of identical rating are required to share a load of 2.5 MW at 0.8 pf lagging. Alternator A has a frequency drop of 1Hz per MW. Similar data for Alternator B is 1.5 Hz per MW. No-load frequencies are respectively 51.5 Hz and 51 Hz. At what frequency will the system operate and how much load is supplied by each machine. (03)

Answer 91

Use the synchronous-machine phasor equation and keep the generator/motor sign convention explicit before substituting values.

Step 1: for alternators operating in parallel, machines settle at a common frequency. For a governor droop characteristic, the load supplied by each machine is found from its frequency-load line. For A: $f = 51.5 - 1(PA)$ For B: $f = 51.0 - 1.5(PB)$ $PA + PB = 2.5 \text{ MW}$ Equating frequencies: $51.5 - PA = 51.0 - 1.5PB$. With $PA = 2.5 - PB$: $51.5 - (2.5 - PB) = 51.0 - 1.5PB$ $49.0 + PB = 51.0 - 1.5PB$ $2.5PB = 2.0 \Rightarrow PB = 0.8 \text{ MW}$ $PA = 1.7 \text{ MW}$ $f = 51.5 - 1.7 = 49.8 \text{ Hz}$ Thus machine A supplies 1.7 MW, machine B supplies 0.8 MW, and the common operating frequency is 49.8 Hz.

Final: With $PA = 2.5 - PB$: $51.5 - (2.5 - PB) = 51.0 - 1.5PB$ $49.0 + PB = 51.0 - 1.5PB$ $2.5PB = 2.0 \Rightarrow PB = 0.8 \text{ MW}$ $PA = 1.7 \text{ MW}$ $f = 51.5 - 1.7 = 49.8 \text{ Hz}$ Thus machine A supplies 1.7 MW, machine B supplies 0.8 MW, and the common operating frequency is 49.8 Hz.

Question 92

Original: 2C | EM-II ELE 2202 Makeup, 16 June 2017

What are the advantages of placing the field system of a large alternator on its rotor and the 3-phase windings on its stator? (02)

Answer 92

Use the synchronous-machine phasor equation and keep the generator/motor sign convention explicit before substituting values.

In large alternators the DC field winding is placed on the rotor and the 3-phase armature winding is placed on the stator. 3-phase armature winding on stator; field system on rotor - High-voltage armature conductors are stationary, so insulation, bracing and cooling are easier.

Step 1: Only low-voltage DC excitation must be fed to the rotating rotor through slip rings/brushless exciter.

Step 2: the stator can be made large with better ventilation ducts and mechanical support.

Final: Field system on rotor - High-voltage armature conductors are stationary, so insulation, bracing and cooling are easier.; Only low-voltage DC excitation must be fed to the rotating rotor through slip rings/brushless exciter.

Question 93

Original: 4A | EM-II ELE 2202 Makeup, 19 June 2018

With neat sketch, explain how an alternator can be synchronized to the grid using 'Bright lamp method'. What are the conditions to be met to synchronize two 3-phase alternators? (05)

Answer 93

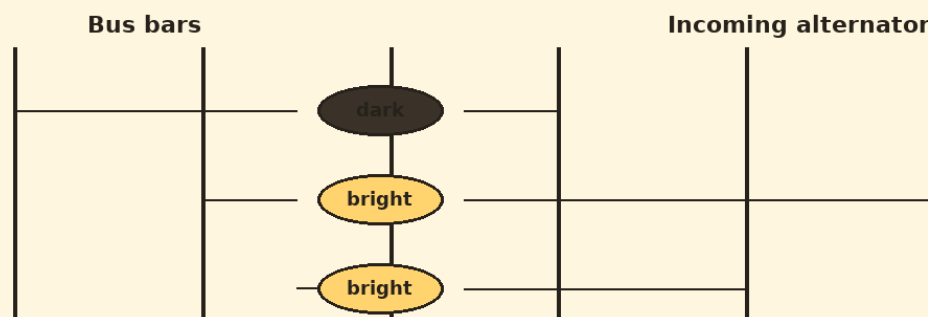
Use the synchronous-machine phasor equation and keep the generator/motor sign convention explicit before substituting values.

In the two-bright one-dark lamp method, three lamps are connected between corresponding and cross-connected phases of the bus bars and incoming alternator. Two-bright one-dark lamp synchronization Correct instant: one lamp dark, two equally bright, and lamp flicker is slow.

Step 1: Adjust prime mover speed until the lamp flicker is slow; this means frequencies are nearly equal.

Final: Adjust prime mover speed until the lamp flicker is slow; this means frequencies are nearly equal.

Lamp Method



Question 94

Original: 1A | EM-II ELE 2225 End Sem, 09 May 2024

'The inherent nature of synchronous machines is to rotate in synchronism with the supply frequency while that of induction motors is to rotate with a slip'. Differentiate the above two machines. Use necessary schematic diagrams to justify your answer.

Answer 94

A synchronous machine runs at exactly synchronous speed after it is locked to the rotating magnetic field. Its speed is $N_s = 120f/P$ and is fixed by supply frequency and number of poles. In steady state there is no slip between rotor field and stator rotating field.

An induction motor must run with slip. If the rotor reached synchronous speed, relative speed between rotor conductors and the stator field would be zero, rotor emf would be zero, rotor current would be zero, and torque would vanish. Therefore an induction motor develops torque only when N_r is less than N_s in motoring operation.

In a synchronous machine, rotor excitation is supplied by DC field current or permanent magnets. Real power is controlled mainly by load angle, and reactive power can be controlled by excitation. In an induction motor, excitation is drawn from the AC supply as magnetizing current, so the power factor is normally lagging.

Thus synchronous machines are constant-speed machines tied to supply frequency, while induction motors are asynchronous machines whose speed falls slightly with load.

Final: The synchronous machine operates at zero slip in synchronism; the induction motor requires nonzero slip for torque production.

Question 95

Original: 1C | EM-II ELE 2225 End Sem. 09 May 2024

‘Unlike asynchronous machines, Synchronous machines can be operated at different power factors’. Justify this statement with the help of necessary characteristics. 3

Answer 95

Use the synchronous-machine phasor equation and keep the generator/motor sign convention explicit before substituting values.

In a synchronous machine connected to a constant-frequency, constant-voltage bus, the rotor speed is fixed at synchronous speed. The active power is mainly controlled by mechanical input/load angle, while the reactive power and stator current power factor are controlled by field excitation.

Step 1: Under-excited synchronous motor: draws lagging current and absorbs reactive power, behaving like an inductive load.

Step 2: Normally excited: armature current is minimum and power factor is approximately unity.

Step 3: Over-excited: draws leading current and supplies reactive power, so it can act as a synchronous condenser.

Step 4: This is unlike an induction motor, whose magnetizing current is always lagging and whose power factor is mainly load-dependent. V-curves of synchronous motor Hence synchronous machines can be deliberately operated at lagging, unity, or leading power factor by varying field current.

Final: The active power is mainly controlled by mechanical input/load angle, while the reactive power and stator current power factor are controlled by field excitation.; Over-excited: draws leading current and supplies reactive power, so it can act as a synchronous condenser.

9. Alternator armature reaction, voltage regulation, OCC/SCC and phasor diagrams

Syllabus area: Weeks 11-12

Question 96

Original: 2B | EM-II ELE 204 End Sem. 13 May 2014

The following data were obtained for the OCC of a 10MVA, 13 kV, 3 phase, 50Hz, star connected synchronous generator: If (A) 50 75 100 125 150 162.5 200 250 300 VOC (line) (kV) 6.2 8.7 10.5 11.8 12.8 13.2 14.2 15.2 15.9 An excitation of 100A causes the full load current to flow during the short-circuit test. An excitation current of 290 A is required to deliver full load current at rated terminal voltage during zero power factor test. The effective armature resistance is 1.2Ω per phase. Draw OCC and Potier triangle to calculate (i) Synchronous reactance by EMF method and (ii) Leakage reactance by Z.P.F. method. (06)

Answer 96

Rated line voltage is 13 kV, so rated phase voltage is $V_{ph} = 13/\sqrt{3} = 7.505$ kV. The rated current is $I = 10 \text{ MVA}/(\sqrt{3} \times 13 \text{ kV}) = 444.1$ A.

For the EMF method, use the OCC and SCC at the same field current. The short-circuit test states that $I_f = 100$ A gives full-load current. From the OCC at $I_f = 100$ A, open-circuit line voltage is 10.5 kV, so open-circuit phase emf is $E_{oc} = 10.5/\sqrt{3} = 6.062$ kV.

Thus synchronous impedance is $Z_s = E_{oc}/I = 6062/444.1 = 13.65 \Omega$ per phase. With $R_a = 1.2 \Omega$ per phase: $X_s = \sqrt{(Z_s^2 - R_a^2)} = \sqrt{(13.65^2 - 1.2^2)} = 13.60 \Omega$.

For the ZPF/Potier part, plot the OCC and mark the rated zero-power-factor point P at $I_f = 290$ A and $V = 13$ kV line. The full-load short-circuit field current is $I_{sc} = 100$ A, so draw a horizontal line 100 A to the left from P, giving point D at $I_f = 190$ A and $V = 13$ kV line.

Draw the air-gap line from the initial straight part of the OCC. Using the first OCC point, its slope is about $6.2/50 = 0.124$ kV/A. Through D draw a line parallel to this air-gap line to meet the OCC. The intersection is near $I_f = 199$ A and $V = 14.15$ kV line. Hence the vertical side of the Potier triangle is approximately $14.15 - 13.0 = 1.15$ kV line.

Per phase leakage-reactance drop $= 1.15/\sqrt{3} = 0.664$ kV. Therefore Potier/leakage reactance is $X_p = 664/444.1 = 1.50 \Omega$ per phase. A plotted graph may give a slightly different value depending on scale, but it should be about $1.5 \Omega/\text{phase}$.

Final: EMF method gives $X_s = 13.6 \Omega/\text{phase}$; ZPF/Potier construction gives leakage reactance X_p approximately $1.5 \Omega/\text{phase}$.

Question 97

Original: 1B | EM-II ELE 204 Makeup. 08 July 2014

A 6.6 kV, 3 phase, 50 Hz, star connected alternator gave the following data for open circuit and short circuit tests. If the per phase effective armature resistance is 0.2Ω , determine the voltage regulation at full load current of 500 A at 0.8 power factor lagging by MMF method. Field Current in Amp 3.2 5.0 7.5 10.0 14.0 Open circuit line voltage in kV 3.1 4.9 6.6 7.5 8.24 Short circuit current in Amp 500 -- 1170 -- -- (06)

Answer 97

Use the synchronous-machine phasor equation and keep the generator/motor sign convention explicit before substituting values.

Step 1: MMF method: first correct terminal voltage for armature resistance drop, then combine OC field current and SC field current as MMFs.

Step 2: V' line $= 6,739.4$ V; corresponding I_{f1} from OCC $= 7.8871$ A.

Step 3: Field current for full-load short-circuit current 500 A is $I_{f2} = 3.2000$ A.

Step 4: for 0.8 lagging, resultant $I_f = \sqrt{(I_{f1}^2 + I_{f2}^2 + 2I_{f1}I_{f2} \sin \phi)} = 10.136$ A.

Step 5: OCC voltage corresponding to resultant field current = 7,525.1 V line.

Step 6: the percentage regulation = $(E - V) / V = 14.017\%$.

Final: Field current for full-load short-circuit current 500 A is $I_{f2} = 3.2000$ A.; for 0.8 lagging, resultant $I_f = \sqrt{(I_{f1}^2 + I_{f2}^2 + 2I_{f1}I_{f2} \sin \phi)} = 10.136$ A.; OCC voltage corresponding to resultant field current = 7,525.1 V line.; the percentage regulation = $(E - V) / V = 14.017\%$.

Question 98

Original: 3A | EM-II ELE 204 Makeup, 08 July 2014

A 5 MVA, 11 kV, 50 Hz, 4 pole, star connected alternator with synchronous reactance of 15Ω is connected to an infinite bus. Determine the synchronizing power and torque per mechanical degree of phase displacement at (a) no-load (b) full load, 0.8 power factor lagging. (04)

Answer 98

Use the synchronous-machine phasor equation and keep the generator/motor sign convention explicit before substituting values.

Step 1: No-load: $E = V_{ph} = 11 \text{ kV}/\sqrt{3}$. P_{syn} per mechanical degree = $(3EV/X) \times (P/2) \times \pi/180$.

Step 2: No-load result = 281.6 kW/mechanical degree and synchronizing torque = 1792.6 N·m/mechanical degree.

Step 3: at full-load 0.8 lagging: $I_a = 262.43$ A and $E = V + j15I_a = 9.264$ kV/phase at $\delta = 19.87^\circ$.

Step 4: $P_{syn} = (3EV/X \cos \delta)(P/2)(\pi/180) = 386.3$ kW/mechanical degree.

Step 5: Synchronizing torque = $P_{syn}/\omega_s = 2459$ N·m/mechanical degree.

Final: No-load result = 281.6 kW/mechanical degree and synchronizing torque = 1792.6 N·m/mechanical degree.; at full-load 0.8 lagging: $I_a = 262.43$ A and $E = V + j15I_a = 9.264$ kV/phase at $\delta = 19.87^\circ$.; $P_{syn} = (3EV/X \cos \delta)(P/2)(\pi/180) = 386.3$ kW/mechanical degree.; Synchronizing torque = $P_{syn}/\omega_s = 2459$ N·m/mechanical degree.

Question 99

Original: 3B | EM-II ELE 204 Makeup, 08 July 2014

A 3 phase, 6600 V, 50 Hz, 1000 rpm, star connected alternator is delivering a constant power of 4 MW to an infinite bus system. Its synchronous impedance per phase is $(0+j0.75) \Omega$. Determine the excitation emf required if the power factor is to be adjusted 0.7 lagging. Also obtain the current and power factor when the excitation is set to its minimum value required to deliver the same power. (06)

Answer 99

Use the synchronous-machine phasor equation and keep the generator/motor sign convention explicit before substituting values.

Step 1: at 0.7 lagging, $I = P/(\sqrt{3} V \text{ pf}) = 499.9$ A.

Step 2: $E = V + j0.75I = 4.087$ kV/phase at $\delta = 3.68^\circ$.

Step 3: Minimum excitation for the same power occurs at $\delta = 90^\circ$: $E_{min} = P X/(3V) = 262.4$ V/phase.

Step 4: at this excitation, $I = (E \text{ at } 90^\circ - V)/(jX) = 5093$ A.

Step 5: the operating power factor is 0.0687 leading. The current is very large because the given synchronous reactance is very small.

Final: $E = V + j0.75I = 4.087$ kV/phase at $\delta = 3.68^\circ$.; Minimum excitation for the same power occurs at $\delta = 90^\circ$: $E_{min} = P X/(3V) = 262.4$ V/phase.; at this excitation, $I = (E \text{ at } 90^\circ - V)/(jX) = 5093$ A.; the operating power factor is 0.0687 leading. The current is very large because the

given synchronous reactance is very small.

Question 100

Original: 5A | EM-II ELE 204 Makeup, 08 July 2014

A 3 phase, 50 Hz, 440 V, synchronous motor has a synchronous impedance of $(0.5+j4)$ Ω /phase. Draw a set of excitation circles with excitation emfs of 50%, 90% and 125% of its terminal voltage. For an armature current of 35 A, graphically determine the load angle and power factor for each of the above excitations. (04)

Answer 100

Use the synchronous-machine phasor equation and keep the generator/motor sign convention explicit before substituting values.

Step 1: Take $V_{ph} = 440/\sqrt{3} = 254$ V and $Z_s = 0.5+j4$ Ω . For a synchronous motor, $I = (V - E \text{ at } -\delta)/Z_s$.

Step 2: for $|I| = 35$ A, solve $|V - E| = |Z|I$ for each excitation circle.

Step 3: $E = 0.5V$: $\delta = 19.69^\circ$, pf = 0.419 lagging.

Step 4: $E = 0.9V$: $\delta = 33.47^\circ$, pf = 0.942 lagging.

Step 5: $E = 1.25V$: $\delta = 25.63^\circ$, pf = 0.938 leading.

Step 6: These three points lie on the corresponding excitation circles in the current plane.

Final: For $|I| = 35$ A, solve $|V - E| = |Z|I$ for each excitation circle.; $E = 0.5V$: $\delta = 19.69^\circ$, pf = 0.419 lagging.; $E = 0.9V$: $\delta = 33.47^\circ$, pf = 0.942 lagging.; $E = 1.25V$: $\delta = 25.63^\circ$, pf = 0.938 leading.

Question 101

Original: 2A | EM-II ELE 204 Makeup, 09 July 2015

Draw the EMF and MMF diagrams when a pure inductive load is connected to a 3 phase wound rotor synchronous generator with negligible armature resistance. Hence discuss the armature reaction effect. (03)

Answer 101

Use the synchronous-machine phasor equation and keep the generator/motor sign convention explicit before substituting values.

Armature reaction is the effect of armature current mmf on the main field flux of an alternator. Its nature depends strongly on load power factor. Unity pf cross-magnetising Lagging pf demagnetising + cross Leading pf magnetising + cross - Unity power factor: armature mmf is approximately perpendicular to field mmf, so the effect is mainly cross-magnetising.

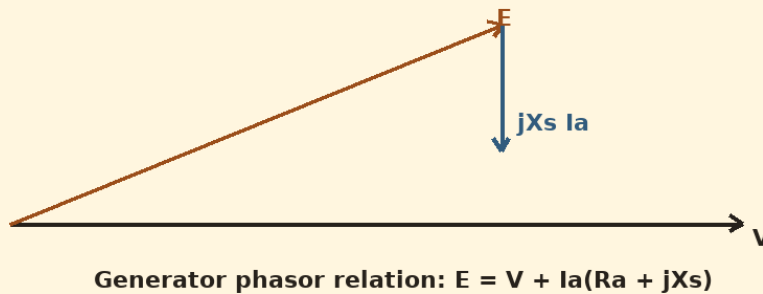
Step 1: Lagging power factor/inductive load: armature mmf has a demagnetising component, weakening the main field and causing voltage drop.

Step 2: Leading power factor/capacitive load: armature mmf has a magnetising component, strengthening the field and possibly causing voltage rise.

Step 3: for intermediate lagging or leading loads, armature reaction has both cross-magnetising and magnetising/demagnetising components.

Final: Lagging power factor/inductive load: armature mmf has a demagnetising component, weakening the main field and causing voltage drop.; Leading power factor/capacitive load: armature mmf has a magnetising component, strengthening the field and possibly causing voltage rise.

Phasor



Question 102

Original: 2B | EM-II ELE 204 Makeup, 09 July 2015

A 440 V, 3phase, 50 Hz, star connected alternator gave the following test data: Field current (A) 2 4 6 7 8 10 12 14 16 Open Circuit voltage per phase (V) 90 166 229 254 274 306 328 342 350 ZPF voltage per phase (V) 0 46 119 181 230 266 The effective armature resistance is 0.15Ω per phase, Calculate the voltage regulation at full load current of 40 A at 0.8 pf lagging by ZPF method. (07)

Answer 102

Rated phase voltage is $V = 440/\sqrt{3} = 254 \text{ V}$ and full-load current is 40 A. The resistance drop is $I_a R_a = 40 \times 0.15 = 6 \text{ V}$ per phase. The power-factor angle for 0.8 lagging is 36.87° .

For the Potier construction, plot OCC as phase voltage versus field current and plot the ZPF curve on the same axes. The ZPF data correspond to the listed field-current readings up to 12 A. At rated voltage 254 V on the ZPF curve, interpolation gives $I_{f,zpf} \approx 11.33 \text{ A}$. On the OCC, 254 V corresponds to $I_{f,oc} = 7 \text{ A}$.

The vertical separation between OCC and ZPF near this point gives an approximate Potier leakage drop. OCC voltage at $I_f = 11.33 \text{ A}$ is about 317 V per phase, so $I_a X_p \approx 317 - 254 = 63 \text{ V}$. Thus $X_p \approx 63/40 = 1.58 \Omega$.

The horizontal separation gives the approximate armature-reaction field current: $I_{ar} \approx 11.33 - 7 = 4.33 \text{ A}$. Now form the phasor air-gap voltage for 0.8 lagging:

$E_g = V + I_a(R_a + jX_p)$, with I_a lagging by 36.87° .

This gives $E_g \approx 300 \text{ V}$ per phase. From the OCC, 300 V corresponds to about 9.65 A field current. Add armature-reaction field current: $I_{f,total} \approx 9.65 + 4.33 = 13.98 \text{ A}$. From OCC, this field current gives open-circuit emf $E_o \approx 342 \text{ V}$ per phase.

Voltage regulation = $(E_o - V)/V \times 100 = (342 - 254)/254 \times 100 \approx 34.6 \text{ percent}$.

Final: By approximate ZPF/Potier interpolation, voltage regulation is about 34.6 percent lagging.

Question 103

Original: 3B | EM-II ELE 204 Makeup, 09 July 2015

A 3 phase, star connected cylindrical rotor alternator with a synchronous reactance of 5 per phase with negligible armature resistance is supplying 250 A at 0.8 power factor lagging to a 11 kV infinite bus. Determine (a) excitation emf and load angle (b) If excitation is increased by 15 % without changing its driving torque, determine the new values of load angle, armature current and power factor (03)

Answer 103

Use the synchronous-machine phasor equation and keep the generator/motor sign convention explicit before substituting values.

Step 1: Initial current = 250 A at 0.8 lagging. $E = V + j5I = 7.171 \text{ kV/phase}$ at $\delta = 8.02^\circ$.

Step 2: Real power supplied = $\sqrt{3} \times 11 \text{ kV} \times 250 \times 0.8 = 3.811 \text{ MW}$.

Step 3: After 15% excitation increase, $E' = 8.247 \text{ kV / phase}$. With driving torque unchanged, P remains constant.

Step 4: New load angle from $P = 3E'V / X \sin \delta$ is $\delta = 6.965^\circ$.

Step 5: New current = $(E' \text{ at } \delta - V) / (|X|) = 417.9 \text{ A}$.

Step 6: the operating power factor is 0.479 lagging.

Final: After 15% excitation increase, $E' = 8.247 \text{ kV / phase}$. With driving torque unchanged, P remains constant.; New load angle from $P = 3E'V / X \sin \delta$ is $\delta = 6.965^\circ$.; New current = $(E' \text{ at } \delta - V) / (|X|) = 417.9 \text{ A}$.; the operating power factor is 0.479 lagging.

Question 104

Original: 2B | EM-II ELE 2202 End Sem, 10 May 2016

A 6.6 kV, 3 phase, 50Hz, star connected alternator gave the following data for open circuit and short circuit tests. If the per phase effective armature resistance is 0.2Ω , determine the voltage regulation at full load current of 500A at 0.75 pf lagging by MMF method. Open circuit line voltage (kV) 3.1 4.9 6.6 7.5 8.2 Field current (A) 3.2 5.0 7.5 10.0 14.0 Short circuit current (A) 500 -- 1170 -- -- (06)

Answer 104

Use the synchronous-machine phasor equation and keep the generator/motor sign convention explicit before substituting values.

Step 1: V' line after resistance correction = 6,730.9 V.

Step 2: OCC field current $I_{f1} = 7.8636 \text{ A}$; SCC field current $I_{f2} = 3.2000 \text{ A}$ for 500 A.

Step 3: for 0.75 lagging, resultant $I_f = 10.265 \text{ A}$.

Step 4: Corresponding OCC line emf = 7,546.3 V. The percentage regulation = 14.338 %.

Final: V' line after resistance correction = 6,730.9 V.; OCC field current $I_{f1} = 7.8636 \text{ A}$; SCC field current $I_{f2} = 3.2000 \text{ A}$ for 500 A.; for 0.75 lagging, resultant $I_f = 10.265 \text{ A}$.; Corresponding OCC line emf = 7,546.3 V. The percentage regulation = 14.338 %.

Question 105

Original: 5A | EM-II ELE 2202 End Sem, 10 May 2016

A 15 kW, 400 V, 50 Hz, 3 phase, star connected synchronous motor has its synchronous impedance of $(1+j5) \Omega$ per phase. If the excitation is maintained constant at 277 V per phase, determine the maximum load the synchronous motor can drive and corresponding current and power factor. (05)

Answer 105

Use the synchronous-machine phasor equation and keep the generator/motor sign convention explicit before substituting values.

Step 1: $V_{ph} = 400/\sqrt{3} = 230.94$ V, $E = 277$ V/phase, $Z_s = 1 + j5 \Omega$.

Step 2: for each load angle, $I = (V - E \text{ at } -\delta)/Z_s$. Gross mechanical power = $P_{in} - 3I^2R$.

Step 3: Maximum gross mechanical power occurs at $\delta = 78.69^\circ$.

Step 4: Maximum load = 28.78 kW.

Step 5: Corresponding current = 63.54 A; current angle = -21.72° , so pf = 0.929 lagging.

Final: $V_{ph} = 400/\sqrt{3} = 230.94$ V, $E = 277$ V/phase, $Z_s = 1 + j5 \Omega$.; Maximum gross mechanical power occurs at $\delta = 78.69^\circ$.; Maximum load = 28.78 kW.; Corresponding current = 63.54 A; current angle = -21.72° , so pf = 0.929 lagging.

Question 106

Original: 3B | EM-II ELE 2202 End Sem. 23 April 2018

The open circuit and short circuit test readings for a 3 phase, star connected 1,000 kVA, 2,000 V, 50 Hz synchronous generator are: Field current (A) 10 20 30 40 50 60 OC voltage (L-L) (V) 800 1,500 2,000 2,350 2,600 2,750 SC current (A) - 200 300 - - - The effective armature resistance is 0.15Ω per phase. Determine the full load percentage regulation at 0.8 p.f. lagging by MMF method. (06)

Answer 106

Use the synchronous-machine phasor equation and keep the generator/motor sign convention explicit before substituting values.

Step 1: the rated current is obtained from 288.68 A. V' line after R_a drop = 2,060.5 V.

Step 2: from OCC $I_{f1} = 31.728$ A; from SCC $I_{f2} = 28.868$ A.

Step 3: Resultant field current for 0.8 lagging = 54.214 A.

Step 4: Corresponding OCC line emf = 2,663.2 V. Regulation = 33.160 %.

Final: The rated current is obtained from 288.68 A. V' line after R_a drop = 2,060.5 V.; from OCC $I_{f1} = 31.728$ A; from SCC $I_{f2} = 28.868$ A.; Resultant field current for 0.8 lagging = 54.214 A.; Corresponding OCC line emf = 2,663.2 V. Regulation = 33.160 %.

Question 107

Original: 2C | EM-II ELE 2202 End Sem. 24 April 2017

Using relevant phasor diagram, discuss the behaviour of a cylindrical rotor synchronous motor supplying a constant load but operating under varying excitation conditions. (03)

Answer 107

For a cylindrical-rotor synchronous motor at constant mechanical load and terminal voltage, the real power input is approximately constant. Therefore the in-phase component of armature current remains nearly fixed, while the reactive component changes with field excitation.

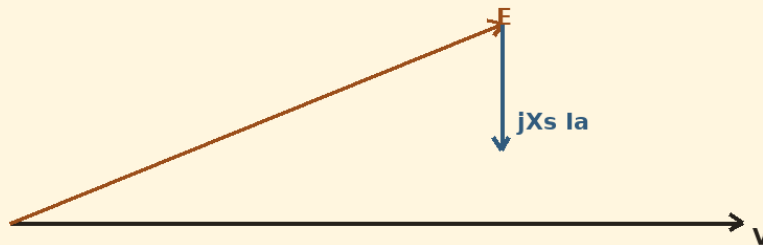
At under-excitation, the induced emf is small. The motor draws a lagging current from the supply and absorbs reactive power, behaving like an inductive load. At normal excitation, armature current is minimum and the power factor is approximately unity. At over-excitation, the induced emf is large and the motor draws leading current, supplying reactive power to the system.

The phasor relation for a motor is $E = V - I_a(R_a + jX_s)$. Neglecting resistance, the tip of the current phasor moves as excitation changes while the real-power component remains set by the load. This produces the familiar V-curve of armature current versus field current and inverted-V curve of power factor versus field current.

This behavior is important because a synchronous motor can be used as a synchronous condenser for power-factor correction.

Final: Varying excitation changes reactive power and power factor from lagging to unity to leading, while real power is set mainly by shaft load.

Phasor



$$\text{Generator phasor relation: } E = V + I_a(R_a + jX_s)$$

Question 108

Original: 3B | EM-II ELE 2202 End Sem, 24 April 2017

Along with necessary waveforms, discuss the behaviour of a three-phase alternator subjected to a symmetrical 3-phase short circuit. (03)

Answer 108

When a three-phase alternator is subjected to a sudden symmetrical short circuit, the armature current does not immediately settle to its steady-state value. The current waveform contains an AC component whose envelope decays in stages, and it may also contain a DC offset depending on the instant of fault.

Immediately after the fault, flux linkages in the machine cannot change instantly. The current is limited by subtransient reactance X_d'' , so the initial current is very high. This is the subtransient period and lasts only a few cycles.

After damper-winding currents decay, the current is governed by transient reactance X_d' . This gives the transient short-circuit current, lower than the subtransient current but still higher than steady state.

Finally, after field transients decay, the current is limited by synchronous reactance X_d . The steady-state short-circuit current is the smallest of the three. Thus the reactance sequence is $X_d'' < X_d' < X_d$ and the current sequence is $I_{\text{subtransient}} > I_{\text{transient}} > I_{\text{steady}}$.

Final: A sudden three-phase short circuit produces subtransient, transient, and steady-state currents, with decaying offset and progressively increasing effective reactance.

Question 109

Original: 5C | EM-II ELE 2202 End Sem, 24 April 2017

A three-phase, 11 kV, 800 kW, 6 pole, 50 Hz star connected synchronous motor has synchronous reactance of 4Ω / phase. It operates at 0.6 lagging power factor. Neglecting all losses, find synchronising power and torque developed per mechanical degree of displacement. (04)

Answer 109

Use the synchronous-machine phasor equation and keep the generator/motor sign convention explicit before substituting values.

Step 1: Input current = $800 \text{ kW} / (\sqrt{3} \times 11 \text{ kV} \times 0.6) = 69.98 \text{ A}$.

Step 2: using the motor phasor convention, $E = V - jX_s I$. With I lagging by 53.13° , $E = 6.129 \text{ kV/phase}$ and load angle = 1.57° .

Step 3: Synchronizing power per mechanical degree = $(3EV/X \cos \delta)(P/2)(\pi/180)$.

Step 4: for 6 poles, $P_{syn} = 1.528 \text{ MW/mechanical degree}$.

Step 5: Synchronizing torque = $P_{syn}/\omega_s = 14.59 \text{ kN}\cdot\text{m per mechanical degree}$.

Final: Input current = $800 \text{ kW}/(\sqrt{3} \times 11 \text{ kV} \times 0.6) = 69.98 \text{ A}$.; Synchronizing power per mechanical degree = $(3EV/X \cos \delta)(P/2)(\pi/180)$.; for 6 poles, $P_{syn} = 1.528 \text{ MW/mechanical degree}$.; Synchronizing torque = $P_{syn}/\omega_s = 14.59 \text{ kN}\cdot\text{m per mechanical degree}$.

Question 110

Original: 3A | EM-II ELE 2202 End Sem. 29 April 2019

Describe the operation of alternator with constant excitation and variable load with suitable phasor diagrams. What is the significance of the condition with minimum excitation? Analyze the relation between power factor and excitation with the help of suitable curve. (05)

Answer 110

In an alternator operating at constant excitation, the magnitude of internal generated emf E is approximately fixed. When load is varied, armature current I_a and load angle δ change so that the electrical output power follows the mechanical input and load demand.

For lagging loads, armature current lags terminal voltage. The armature reaction is mainly demagnetizing, and the synchronous-reactance drop $jX_s I_a$ causes a larger difference between E and V . Voltage regulation is positive and terminal voltage tends to fall.

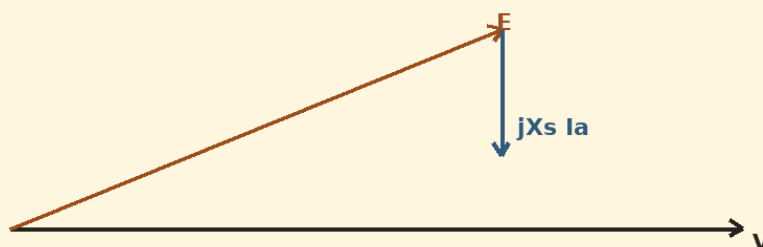
For unity power factor load, armature reaction is mainly cross-magnetizing. The voltage drop is moderate and regulation is smaller than for lagging load.

For leading loads, armature reaction is magnetizing. The terminal voltage can rise, and voltage regulation may become negative. This is why capacitive loads can cause voltage rise in alternators.

Minimum excitation for a given real power corresponds to operation near the stability limit of the power-angle curve. Below that excitation, the required real power cannot be transmitted without losing synchronism.

Final: With constant excitation, increasing load mainly increases I_a and δ ; the terminal-voltage behavior depends strongly on load power factor.

Phasor



$$\text{Generator phasor relation: } E = V + I_a(R_a + jX_s)$$

Question 111

Original: 4A | EM-II ELE 2202 End Sem. 29 April 2019

A 10 kVA, 440 V, 50 Hz, 3-phase star-connected alternator has the following open circuit characteristics. With zero pf full-load applied, an excitation of 14 A produced a terminal line voltage of 500 V and an excitation of 6 A produced a voltage of 200 V. On short circuit, 4 A excitation was required to circulate the full load current. Using the Potier triangle method, determine the voltage regulation at full-load 0.6 pf lagging. Assume armature resistance of 2 Ω per phase. Field current (A) 1 3 5 8 11 15 Terminal Line Voltage (V) 100 300 440 550 600 635 (05)

Answer 111

Rated current is $I = 10 \text{ kVA} / (\sqrt{3} \times 440 \text{ V}) = 13.12 \text{ A}$. Rated phase voltage is $V = 440 / \sqrt{3} = 254 \text{ V}$. Resistance drop at full load is $I_a R_a = 13.12 \times 2 = 26.2 \text{ V}$ per phase. The 0.6 lagging power-factor angle is 53.13° .

From the ZPF data, line voltage changes from 200 V at $I_f = 6 \text{ A}$ to 500 V at $I_f = 14 \text{ A}$. Interpolating for rated line voltage 440 V gives $I_{f,zpf} = 12.4 \text{ A}$.

The SCC field current for full-load current is 4 A, so take the armature-reaction field component $F_{ar} = 4 \text{ A}$. Move left from the rated ZPF point by 4 A to $I_f = 8.4 \text{ A}$. From OCC, $I_f = 8.4 \text{ A}$ gives about 556.7 V line, while the rated point is 440 V line. Therefore Potier leakage drop is approximately $116.7 \text{ V line} = 67.4 \text{ V}$ per phase.

Hence $X_p = 67.4 / 13.12 = 5.14 \Omega$. Now compute the air-gap voltage for 0.6 lagging:

$E_g = V + I_a(R_a + jX_p)$, with I_a lagging by 53.13° .

This gives $|E_g| \approx 324.3 \text{ V}$ per phase, or 561.7 V line. From OCC, 561.7 V line corresponds to field current about 8.70 A. Add armature reaction field current: $I_{f,\text{total}} = 8.70 + 4 = 12.70 \text{ A}$.

OCC voltage at $I_f = 12.70 \text{ A}$ is about 615 V line. Therefore voltage regulation = $(615 - 440) / 440 \times 100 \approx 39.7 \text{ percent}$.

Final: By approximate Potier-triangle interpolation, voltage regulation at full-load 0.6 pf lagging is about 39.7 percent.

Question 112

Original: 4B | EM-II ELE 2202 End Sem. 29 April 2019

A factory has an average load of 1000kW at a power factor of 0.6 lag. A synchronous motor of 86% efficiency is used later to supply an additional mechanical load of 65kW and also to improve the overall power factor to 0.92 lag. Determine the power factor at which the synchronous motor operates. Also comment on the type of excitation required for the synchronous motor for this application and draw the corresponding phasor diagram relating terminal voltage and excitation emf. (05)

Answer 112

Use the synchronous-machine phasor equation and keep the generator/motor sign convention explicit before substituting values.

Step 1: Initial reactive power = 1,333.3 kVar lagging.

Step 2: Motor electrical input for 65 kW mechanical load = $65 / 0.86 = 75.581 \text{ kW}$.

Step 3: Total active power after adding motor = 1,075.6 kW.

Step 4: Required total reactive power at 0.92 lagging = 458.20 kVar.

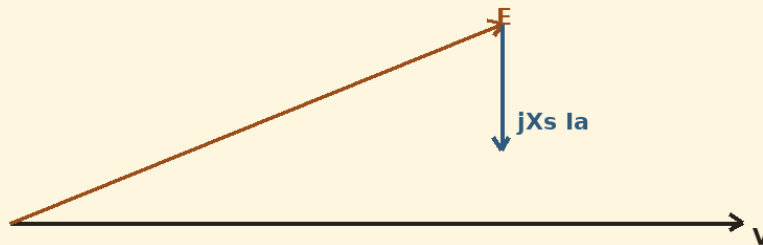
Step 5: Motor reactive power = -875.14 kVar (negative means leading).

Step 6: the required synchronous-motor rating is 878.40 kVA; motor pf = 0.08604 leading. The motor must be over-excited.

Final: Total active power after adding motor = 1,075.6 kW.; Required total reactive power at 0.92 lagging = 458.20 kVar.; Motor reactive power = -875.14 kVar (negative means leading).;

the required synchronous-motor rating is 878.40 kVA; motor pf = 0.08604 leading. The motor must be over-excited.

Phasor



$$\text{Generator phasor relation: } E = V + I_a(R_a + jX_s)$$

Question 113

Original: 5C | EM-II ELE 2202 End Sem, 29 April 2019

A 10MVA, 6-pole star-connected alternator is synchronised to 13kV bus bar supplies half load at 0.84 pf lag. Find the excitation emf and air-gap voltage. Also calculate the synchronising power developed with a displacement of 2.5 mechanical degrees. Synchronous reactance is 7Ω and leakage reactance is 1.7Ω . Neglect the armature winding resistance. (04)

Answer 113

Use the synchronous-machine phasor equation and keep the generator/motor sign convention explicit before substituting values.

Step 1: Half load = 5 MVA at 0.84 lagging, so $I = 222.1$ A approximately.

Step 2: Excitation emf: $E = V + jX_s I = 8.450$ kV/phase at $\delta = 8.89^\circ$.

Step 3: Air-gap voltage using leakage reactance only: $E_g = V + jX_{II} I = 7.717$ kV/phase.

Step 4: A displacement of 2.5 mechanical degrees in a 6-pole machine equals 7.5 electrical degrees = 0.1309 rad.

Step 5: Synchronizing power developed = $(3EV/X_s \cos \delta) \times \Delta\delta = 3.52$ MW.

Final: Excitation emf: $E = V + jX_s I = 8.450$ kV/phase at $\delta = 8.89^\circ$.; Air-gap voltage using leakage reactance only: $E_g = V + jX_{II} I = 7.717$ kV/phase.; A displacement of 2.5 mechanical degrees in a 6-pole machine equals 7.5 electrical degrees = 0.1309 rad.; Synchronizing power developed = $(3EV/X_s \cos \delta) \times \Delta\delta = 3.52$ MW.

Question 114

Original: 3A | EM-II ELE 2202 Makeup, 02 July 2016

A 6.6 kV, 3 phase, 50Hz, star connected alternator gave the following data for open circuit test. Field current (A) 1.6 3.2 5 7.5 10 14 Open circuit phase voltage (kV) 0.9 1.8 2.8 3.8 4.4 4.8 A field current of 6 A circulates full load current of 500 A. The per phase effective armature resistance is 0.5Ω . Determine the voltage regulation at full load current at 0.8 pf lagging by EMF method. 6M

Answer 114

Use the synchronous-machine phasor equation and keep the generator/motor sign convention explicit before substituting values.

Step 1: Reading the OCC at field current $I_f = 6$ A, interpolate OCC phase voltage between $I_f = 5$ A and 7.5 A: $E_{oc} = 3.2$ kV/phase.

Step 2: the same field gives full-load short-circuit current 500 A, so $Z_s = 3200/500 = 6.4 \Omega$.

Step 3: $X_s = \sqrt{6.4^2 - 0.5^2} = 6.38 \Omega$.

Step 4: $V_{ph} = 6.6 \text{ kV}/\sqrt{3} = 3.811 \text{ kV}$; $I_a = 500$ A at 0.8 lag.

Step 5: $E = V + I_a(0.5 + j6.38) = 6.393 \text{ kV/phase}$.

Step 6: the percentage regulation = 67.8%.

Final: $X_s = \sqrt{6.4^2 - 0.5^2} = 6.38 \Omega$; $V_{ph} = 6.6 \text{ kV}/\sqrt{3} = 3.811 \text{ kV}$; $I_a = 500$ A at 0.8 lag; $E = V + I_a(0.5 + j6.38) = 6.393 \text{ kV/phase}$; the percentage regulation = 67.8%.

Question 115

Original: 5A | EM-II ELE 2202 Makeup, 02 July 2016

A 3.3 kV, 3 phase, 50 Hz, 4 pole star connected synchronous motor has a synchronous impedance of $0.2 + j3 \Omega$ per phase. For an input current of 200A at 0.9 p.f lag, determine the maximum gross power and torque developed. 5M

Answer 115

Use the synchronous-machine phasor equation and keep the generator/motor sign convention explicit before substituting values.

Step 1: Given $I = 200$ A at 0.9 lagging, motor internal emf is $E = V - I(0.2 + j3)$.

Step 2: This gives $E = 1690.5$ V/phase and initial load angle about 18.0° .

Step 3: Keeping excitation constant, maximize gross power $P_g = P_{in} - 3I^2R$ over load angle.

Step 4: Maximum gross power = 3.024 MW at δ approx 86.19° .

Step 5: for 4 poles, $N_s = 1500$ rpm and $\omega_s = 157.08$ rad/s.

Step 6: Maximum gross torque = $3.024 \text{ MW}/157.08 = 19.25 \text{ kN}\cdot\text{m}$.

Final: Keeping excitation constant, maximize gross power $P_g = P_{in} - 3I^2R$ over load angle.; Maximum gross power = 3.024 MW at δ approx 86.19° ; for 4 poles, $N_s = 1500$ rpm and $\omega_s = 157.08$ rad/s.; Maximum gross torque = $3.024 \text{ MW}/157.08 = 19.25 \text{ kN}\cdot\text{m}$.

Question 116

Original: 2C | EM-II ELE 2202 Makeup, 13 June 2019

A 3-phase, star connected alternator is rated 1,600 kVA, 13.5 kV. Its per-phase effective armature resistance & synchronous reactance are 1 & 40 respectively. a) Calculate the percentage voltage regulation for a load of 1,250 kW at 0.8 pf lagging. b) Draw the phasor diagram for the given load. (03)

Answer 116

Use the synchronous-machine phasor equation and keep the generator/motor sign convention explicit before substituting values.

Step 1: Load kVA = $1250/0.8 = 1562.5$ kVA. Current = $1562.5 \text{ kVA}/(\sqrt{3} \times 13.5 \text{ kV}) = 66.82$ A.

Step 2: using the generator phasor convention, $E = V + I_a(1 + j40)$.

Step 3: $V_{ph} = 13.5 \text{ kV}/\sqrt{3} = 7794$ V; I_a lags by 36.87° .

Step 4: $E_{ph} = 9681.5$ V at $\delta = 12.52^\circ$.

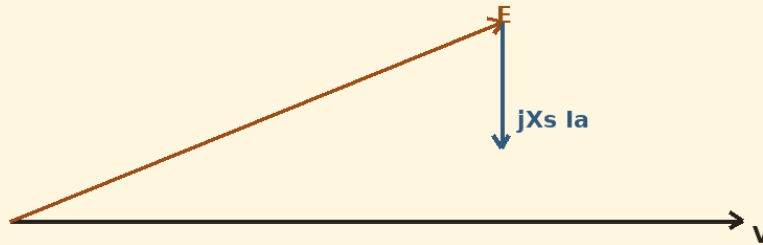
Step 5: the percentage regulation = $(9681.5 - 7794)/7794 \times 100 = 24.21\%$.

Step 6: Phasor diagram: take V as reference, draw I_a lagging by 36.87° , then add $I_a R_a$ in phase with I_a and $jI_a X_s$ leading I_a by 90° to reach E .

Final: $V_{ph} = 13.5 \text{ kV}/\sqrt{3} = 7794$ V; I_a lags by 36.87° ; $E_{ph} = 9681.5$ V at $\delta = 12.52^\circ$; the percentage regulation = $(9681.5 - 7794)/7794 \times 100 = 24.21\%$; Phasor diagram: take V as

reference, draw I_a lagging by 36.87° , then add $I_a R_a$ in phase with I_a and $jI_a X_s$ leading I_a by 90° to reach E.

Phasor



$$\text{Generator phasor relation: } E = V + I_a(R_a + jX_s)$$

Question 117

Original: 3B | EM-II ELE 2202 Makeup, 13 June 2019

Explain the effect of load power factor on armature reaction in alternators. (03)

Answer 117

Use the synchronous-machine phasor equation and keep the generator/motor sign convention explicit before substituting values.

Armature reaction is the effect of armature current mmf on the main field flux of an alternator. Its nature depends strongly on load power factor. Unity pf cross-magnetising Lagging pf demagnetising + cross Leading pf magnetising + cross - Unity power factor: armature mmf is approximately perpendicular to field mmf, so the effect is mainly cross-magnetising.

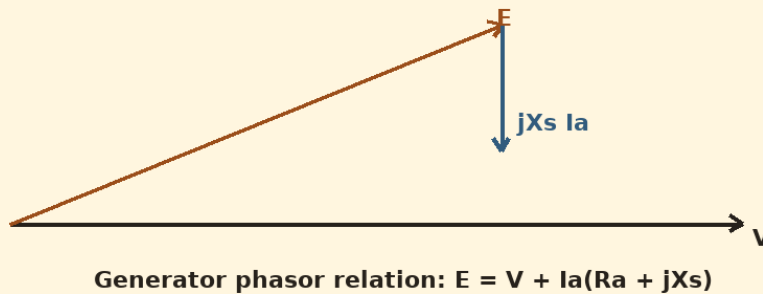
Step 1: Lagging power factor/inductive load: armature mmf has a demagnetising component, weakening the main field and causing voltage drop.

Step 2: Leading power factor/capacitive load: armature mmf has a magnetising component, strengthening the field and possibly causing voltage rise.

Step 3: for intermediate lagging or leading loads, armature reaction has both cross-magnetising and magnetising/demagnetising components.

Final: Lagging power factor/inductive load: armature mmf has a demagnetising component, weakening the main field and causing voltage drop.; Leading power factor/capacitive load: armature mmf has a magnetising component, strengthening the field and possibly causing voltage rise.

Phasor



Question 118

Original: 4B | EM-II ELE 2202 Makeup, 13 June 2019

A 1000 kW, 3.3 kV, 24 pole, 50Hz, 3-phase star connected synchronous motor has synchronous reactance of 3.4Ω per phase and the resistance is negligible. The motor is fed from infinite bus bar at 3.3kV. Its field excitation is adjusted to result in upf operation at rated load. Compute the maximum power and torque that the motor can deliver with its excitation remains constant at this value. (05)

Answer 118

Use the synchronous-machine phasor equation and keep the generator/motor sign convention explicit before substituting values.

Step 1: at rated load and upf: $I = 1000 \text{ kW}/(\sqrt{3} \times 3.3 \text{ kV}) = 174.95 \text{ A}$.

Step 2: using the motor phasor convention with negligible resistance, $E = V - jX_s I$.

Step 3: $V_{ph} = 1905.3 \text{ V}$; $E = 1996.0 \text{ V/phase}$ at load angle 17.34° .

Step 4: with excitation constant, maximum power = $3EV/X_s = 3.355 \text{ MW}$.

Step 5: $N_s = 120f/P = 250 \text{ rpm}$, $\omega_s = 26.18 \text{ rad/s}$.

Step 6: Maximum torque = $P_{max}/\omega_s = 128.17 \text{ kN}\cdot\text{m}$.

Final: $V_{ph} = 1905.3 \text{ V}$; $E = 1996.0 \text{ V/phase}$ at load angle 17.34° .; with excitation constant, maximum power = $3EV/X_s = 3.355 \text{ MW}$.; $N_s = 120f/P = 250 \text{ rpm}$, $\omega_s = 26.18 \text{ rad/s}$.; Maximum torque = $P_{max}/\omega_s = 128.17 \text{ kN}\cdot\text{m}$.

Question 119

Original: 3C | EM-II ELE 2202 Makeup, 16 June 2017

Explain the effect of load power factor on armature reaction in alternators. (03)

Answer 119

Use the synchronous-machine phasor equation and keep the generator/motor sign convention explicit before substituting values.

Armature reaction is the effect of armature current mmf on the main field flux of an alternator. Its nature depends strongly on load power factor. Unity pf cross-magnetising Lagging pf demagnetising + cross Leading pf magnetising + cross - Unity power factor: armature mmf is approximately perpendicular to field mmf, so the effect is mainly cross-magnetising.

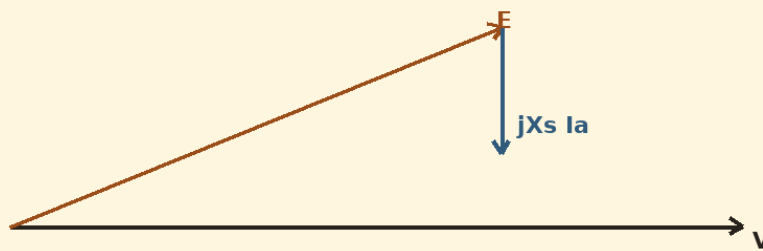
Step 1: Lagging power factor/inductive load: armature mmf has a demagnetising component, weakening the main field and causing voltage drop.

Step 2: Leading power factor/capacitive load: armature mmf has a magnetising component, strengthening the field and possibly causing voltage rise.

Step 3: for intermediate lagging or leading loads, armature reaction has both cross-magnetising and magnetising/demagnetising components.

Final: Lagging power factor/inductive load: armature mmf has a demagnetising component, weakening the main field and causing voltage drop.; Leading power factor/capacitive load: armature mmf has a magnetising component, strengthening the field and possibly causing voltage rise.

Phasor



$$\text{Generator phasor relation: } E = V + I_a(R_a + jX_s)$$

Question 120

Original: 4A | EM-II ELE 2202 Makeup, 16 June 2017

A 30kVA, 440V 3-phase 50Hz star connected alternator gave the following test data. If the per phase resistance is 0.3Ω , calculate the voltage regulation at full load 0.8 pf lagging by EMF method. If (A) 2 4 6 8 10 OC Voltage(VLine) 155 287 395 475 530 S.C Current (A) 11 22 34 46 57 (04)

Answer 120

Use the synchronous-machine phasor equation and keep the generator/motor sign convention explicit before substituting values.

Step 1: the rated current is obtained from $30 \text{ kVA}/(\sqrt{3} \times 440) = 39.36 \text{ A}$.

Step 2: from OCC, field current for 440 V line is 7.125 A. From SCC, current at this field is 40.75 A.

Step 3: $Z_s = (440/\sqrt{3})/40.75 = 6.234 \Omega$; $X_s = \sqrt{(Z_s^2 - 0.3^2)} = 6.227 \Omega$.

Step 4: for 0.8 lagging load, $E = V + I_a(0.3 + j6.227) = 451.97 \text{ V/phase}$.

Step 5: the percentage regulation = 77.92%.

Final: From OCC, field current for 440 V line is 7.125 A. From SCC, current at this field is 40.75 A.; $Z_s = (440/\sqrt{3})/40.75 = 6.234 \Omega$; $X_s = \sqrt{(Z_s^2 - 0.3^2)} = 6.227 \Omega$.; for 0.8 lagging load, $E = V + I_a(0.3 + j6.227) = 451.97 \text{ V/phase}$.; the percentage regulation = 77.92%.

Question 121

Original: 5B | EM-II ELE 2202 Makeup, 16 June 2017

A 1000 kW, 3.3 kV, 24 pole, 50Hz, 3-phase star connected synchronous motor has synchronous reactance of 3.4Ω per phase and the resistance is negligible. The motor is fed from infinite bus bar at 3.3kV. Its field excitation is adjusted to result in upf operation at rated load. Compute the maximum power and torque that the motor can deliver with its excitation remains constant at this value. (05)

Answer 121

Use the synchronous-machine phasor equation and keep the generator/motor sign convention explicit before substituting values.

Step 1: at rated load and upf: $I = 1000 \text{ kW}/(\sqrt{3} \times 3.3 \text{ kV}) = 174.95 \text{ A}$.

Step 2: using the motor phasor convention with negligible resistance, $E = V - jX_s I$.

Step 3: $V_{ph} = 1905.3 \text{ V}$; $E = 1996.0 \text{ V/phase}$ at load angle 17.34° .

Step 4: with excitation constant, maximum power = $3EV/X_s = 3.355 \text{ MW}$.

Step 5: $N_s = 120f/P = 250 \text{ rpm}$, $\omega_s = 26.18 \text{ rad/s}$.

Step 6: Maximum torque = $P_{max}/\omega_s = 128.17 \text{ kN}\cdot\text{m}$.

Final: $V_{ph} = 1905.3 \text{ V}$; $E = 1996.0 \text{ V/phase}$ at load angle 17.34° ; with excitation constant, maximum power = $3EV/X_s = 3.355 \text{ MW}$; $N_s = 120f/P = 250 \text{ rpm}$, $\omega_s = 26.18 \text{ rad/s}$; Maximum torque = $P_{max}/\omega_s = 128.17 \text{ kN}\cdot\text{m}$.

Question 122

Original: 3B | EM-II ELE 2202 Makeup, 19 June 2018

The O.C. and S.C. tests conducted on a 1,200 kVA, 2,200 V, 50 Hz, 3-phase, star connected alternator gave the following data: Field current (A) 10 20 30 40 50 OC voltage (V/phase) 850 1,600 2,200 2,450 2,600 SC current (A) -- 220 330 -- -- The effective armature resistance is 0.25Ω / phase. Plot the OCC & SCC and find the voltage regulation at full-load, 0.8 pf lagging using EMF method. (06)

Answer 122

Use the synchronous-machine phasor equation and keep the generator/motor sign convention explicit before substituting values.

Step 1: the rated current is obtained from $1200 \text{ kVA}/(\sqrt{3} \times 2.2 \text{ kV}) = 314.9 \text{ A}$.

Step 2: using rated phase voltage $2200/\sqrt{3} = 1270 \text{ V}$, the OCC field current is 15.60 A.

Step 3: from SCC, 20 A gives 220 A and 30 A gives 330 A, so I_{sc} at 15.60 A is about 171.6 A.

Step 4: $Z_s = E_{oc}/I_{sc} = 1270/171.6 = 7.401 \Omega$; $X_s = \sqrt{(Z_s^2 - R_a^2)} = 7.397 \Omega$.

Step 5: at 0.8 lagging, $E = V + I_a(0.25 + j7.397) = 3279.6 \text{ V/phase}$.

Step 6: Regulation = 158.2%. Note: if the table voltage is interpreted as rated phase voltage, the regulation changes; the star-rated line-voltage interpretation is used here.

Final: At 0.8 lagging, $E = V + I_a(0.25 + j7.397) = 3279.6 \text{ V/phase}$; Regulation = 158.2%. Note: if the table voltage is interpreted as rated phase voltage, the regulation changes; the star-rated line-voltage interpretation is used here.

Question 123

Original: 2A | EM-II ELE 2225 End Sem. 09 May 2024

A 3-phase, star-connected alternator is rated at 1,600 kVA, 13.5 kV. Its per-phase effective armature resistance & synchronous reactance are 1Ω & 40Ω respectively. a) Calculate the percentage voltage regulation for a load of 1,250 kW at 0.8 pf lagging. b) Draw the phasor diagram for the given load condition. c) Suggest a method to improve the voltage regulation, without any active power loading. 4

Answer 123

Load kVA = $1250/0.8 = 1562.5$ kVA. Line current is $I = 1562.5 \text{ kVA}/(\sqrt{3} \times 13.5 \text{ kV}) = 66.82$ A. Per-phase rated voltage is $V = 13.5 \text{ kV}/\sqrt{3} = 7794$ V.

For a generator, $E = V + I_a(R_a + jX_s)$. At 0.8 lagging power factor, current lags V by 36.87° . Substitution with $R_a = 1\Omega$ and $X_s = 40\Omega$ gives $|E| = 9681.5$ V per phase.

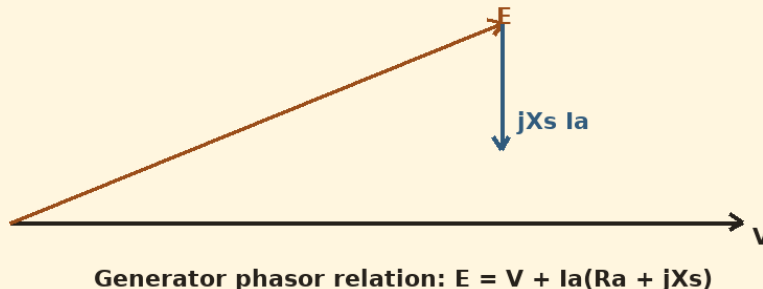
Voltage regulation = $(E - V)/V \times 100 = (9681.5 - 7794)/7794 \times 100 = 24.2$ percent.

Phasor diagram: take V as horizontal reference, draw I_a lagging by 36.87° , draw $I_a R_a$ in phase with I_a , draw $jI_a X_s$ leading I_a by 90° , and add these drops to V to reach E .

Voltage regulation can be improved without increasing active power by supplying leading reactive power, for example by over-exciting a synchronous motor connected in parallel or by using capacitor compensation.

Final: Voltage regulation = 24.2 percent positive; leading VAR compensation improves regulation.

Phasor



Question 124

Original: 1A | EM-II ELE 2225 Makeup. 26 June 2024

With the aid of EMF-MMF diagram, explain the effect of armature reaction when an alternator is connected to (a) capacitive load (b) Inductive load (c) Lagging load 3

Answer 124

Use the synchronous-machine phasor equation and keep the generator/motor sign convention explicit before substituting values.

Armature reaction is the effect of armature current mmf on the main field flux of an alternator. Its nature depends strongly on load power factor. Unity pf cross-magnetising Lagging pf demagnetising + cross Leading pf magnetising + cross - Unity power factor: armature mmf is approximately perpendicular to field mmf, so the effect is mainly cross-magnetising.

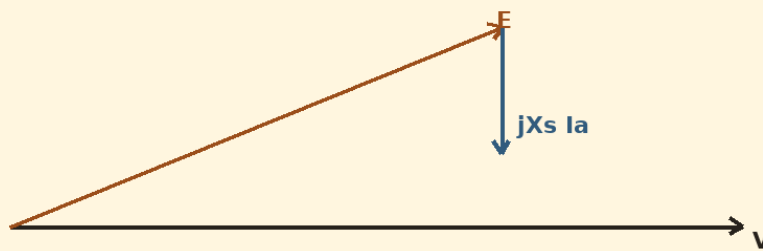
Step 1: Lagging power factor/inductive load: armature mmf has a demagnetising component, weakening the main field and causing voltage drop.

Step 2: Leading power factor/capacitive load: armature mmf has a magnetising component, strengthening the field and possibly causing voltage rise.

Step 3: for intermediate lagging or leading loads, armature reaction has both cross-magnetising and magnetising/demagnetising components.

Final: Lagging power factor/inductive load: armature mmf has a demagnetising component, weakening the main field and causing voltage drop.; Leading power factor/capacitive load: armature mmf has a magnetising component, strengthening the field and possibly causing voltage rise.

Phasor



$$\text{Generator phasor relation: } E = V + I_a(R_a + jX_s)$$

Question 125

Original: 2A | EM-II ELE 2251 End Sem. 31 May 2023

Assume a purely resistive load connected across the terminals of an alternator. Will the voltage regulation of the alternator be zero? Justify your answer. (02)

Answer 125

Use the synchronous-machine phasor equation and keep the generator/motor sign convention explicit before substituting values.

No. A purely resistive load does not generally give zero voltage regulation in an alternator.

Step 1: at unity power factor, armature reaction is mainly cross-magnetising, not zero.

Step 2: the synchronous reactance drop $jX_s I_a$ is still present and is in quadrature with I_a .

Step 3: Armature resistance drop $I_a R_a$ is also present in a practical machine.

Therefore internal emf E is generally greater than terminal voltage V even at unity power factor. $E = V + I_a(R_a + jX_s)$ The percentage regulation = $(|E| - |V|) / |V| \times 100$ percent Zero regulation occurs only at a particular leading power factor where the voltage rise due to leading current balances the internal drops.

Final: Therefore internal emf E is generally greater than terminal voltage V even at unity power factor. $E = V + I_a(R_a + jX_s)$ The percentage regulation = $(|E| - |V|) / |V| \times 100$ percent Zero regulation occurs only at a particular leading power factor where the voltage rise due to leading current balances the internal drops.

Question 126

Original: 3C | EM-II ELE 2251 End Sem. 31 May 2023

A 3-phase cylindrical rotor alternator operating in a captive plant has a rating of 80 kVA, 440 V. This alternator is supplying power to a drive system at a rated current and 0.75 lagging power factor. Consider an armature resistance of $0.25\ \Omega$ per phase and a synchronous reactance of $3.2\ \Omega$ per phase. Determine the voltage regulation. (03)

Answer 126

Use the synchronous-machine phasor equation and keep the generator/motor sign convention explicit before substituting values.

Step 1: the rated current is obtained from $80\text{ kVA}/(\sqrt{3} \times 440\text{ V}) = 104.97\text{ A}$.

Step 2: at 0.75 lagging, current angle = -41.41° .

Step 3: for cylindrical rotor alternator: $E = V + I_a(R_a + jX_s) = V + I_a(0.25 + j3.2)$.

Step 4: $E_{ph} = 548.6\text{ V}$; $V_{ph} = 254.0\text{ V}$.

Step 5: the percentage regulation = $(548.6 - 254.0)/254.0 \times 100 = 115.95\%$.

Final: At 0.75 lagging, current angle = -41.41° .; for cylindrical rotor alternator: $E = V + I_a(R_a + jX_s) = V + I_a(0.25 + j3.2)$.; $E_{ph} = 548.6\text{ V}$; $V_{ph} = 254.0\text{ V}$.; the percentage regulation = $(548.6 - 254.0)/254.0 \times 100 = 115.95\%$.

Question 127

Original: 4A | EM-II ELE 2251 End Sem. 31 May 2023

"V" and "inverted V" curves have significant importance in analyzing the behavior of synchronous machines. Justify this statement suitably. Describe the operation of alternator with constant excitation and variable load with suitable phasor diagrams. What is the significance of the condition with minimum excitation? Analyze the relation between power factor and excitation with the help of a suitable curve. (04)

Answer 127

V-curves and inverted V-curves show the effect of excitation on synchronous-machine current and power factor. For a synchronous motor at constant voltage and mechanical load, increasing field current first reduces armature current to a minimum at unity power factor and then increases it again at leading power factor. Hence the I_a versus I_f curve is V-shaped. The power factor versus I_f curve is inverted-V shaped: lagging at under-excitation, unity near normal excitation, and leading at over-excitation.

For an alternator operating with constant excitation and variable load, the internal emf E is fixed in magnitude. As load increases, armature current increases and the load angle δ changes so that the electrical power output balances the mechanical input. The terminal voltage relation is $E = V + I_a(R_a + jX_s)$. The phasor diagram changes with load power factor because $jX_s I_a$ is perpendicular to I_a and armature reaction may be magnetizing, demagnetizing, or cross-magnetizing.

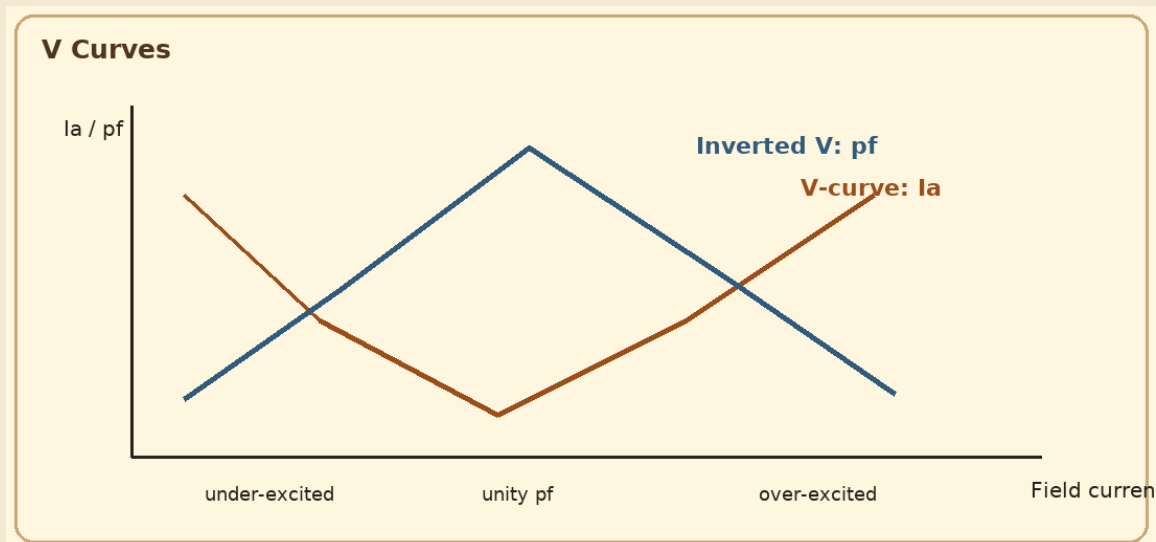
Minimum excitation for a specified real power occurs at the stability boundary where the power-angle equation is just able to transmit the required power. Below this excitation the machine cannot carry the load without losing synchronism. Practically, operation is kept with a margin below the maximum power angle.

As excitation is varied at constant load, the power factor shifts from lagging to unity to leading.

Under-excited synchronous motors absorb reactive power; over-excited motors supply reactive power.

This is why synchronous machines are used for power-factor correction.

Final: V-curves locate minimum armature current and power-factor control; constant-excitation alternator phasors show how load angle and armature reaction change with load.



Question 128

Original: 4A | EM-II ELE 2251 Grade Improvement, 11 August 2021

A 60 kVA, 381.05 V, 50 Hz, Y - connected alternator has an effective resistance of 0.016Ω and armature related self-inductance of 0.23 mH . With the help of accurate phasor diagrams and related analysis, determine the induced voltage in the armature delivers rated current at a load power factor of 0.7 leading. (05)

Answer 128

Use the synchronous-machine phasor equation and keep the generator/motor sign convention explicit before substituting values.

Step 1: the rated current is obtained from $60 \text{ kVA}/(\sqrt{3} \times 381.05) = 90.91 \text{ A}$.

Step 2: $X_s = 2 \pi f L = 2 \pi \times 50 \times 0.23 \text{ mH} = 0.0723 \Omega$.

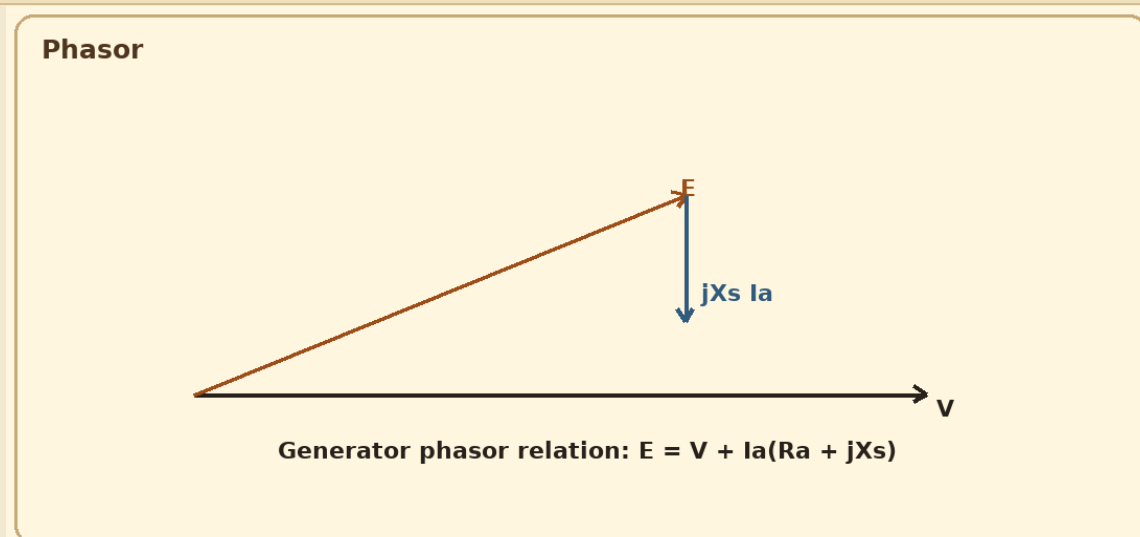
Step 3: for 0.7 leading pf, I_a leads V by 45.57° .

Step 4: Generator equation: $E = V + I_a(0.016 + j0.0723)$.

Step 5: $E_{ph} = 216.40 \text{ V}$ at 1.49° . $V_{ph} = 381.05/\sqrt{3} = 220.0 \text{ V}$.

Step 6: Regulation = -1.64% because leading current produces a voltage-rise effect.

Final: For 0.7 leading pf, I_a leads V by 45.57° .; Generator equation: $E = V + I_a(0.016 + j0.0723)$.; $E_{ph} = 216.40 \text{ V}$ at 1.49° . $V_{ph} = 381.05/\sqrt{3} = 220.0 \text{ V}$.; Regulation = -1.64% because leading current produces a voltage-rise effect.



Question 129

Original: 5A | EM-II ELE 2251 Grade Improvement, 11 August 2021

3-phase, star connected alternator is rated 1,600 kVA, 13.5 kV. Its per-phase effective armature resistance & synchronous reactance are 1 & 40 respectively. a) Calculate the percentage voltage regulation for a load of 1,250 kW at 0.8 pf lagging. b) Draw the phasor diagram for the given load. (05)

Answer 129

Use the synchronous-machine phasor equation and keep the generator/motor sign convention explicit before substituting values.

Step 1: Load kVA = $1250/0.8 = 1562.5$ kVA. Current = $1562.5 \text{ kVA}/(\sqrt{3} \times 13.5 \text{ kV}) = 66.82$ A.

Step 2: using the generator phasor convention, $E = V + I_a(1+j40)$.

Step 3: $V_{ph} = 13.5 \text{ kV}/\sqrt{3} = 7794$ V; I_a lags by 36.87° .

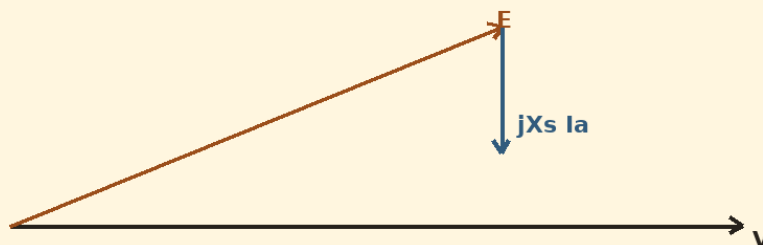
Step 4: $E_{ph} = 9681.5$ V at $\delta = 12.52^\circ$.

Step 5: the percentage regulation = $(9681.5 - 7794)/7794 \times 100 = 24.21\%$.

Step 6: Phasor diagram: take V as reference, draw I_a lagging by 36.87° , then add $I_a R_a$ in phase with I_a and $jI_a X_s$ leading I_a by 90° to reach E .

Final: $V_{ph} = 13.5 \text{ kV}/\sqrt{3} = 7794$ V; I_a lags by 36.87° ; $E_{ph} = 9681.5$ V at $\delta = 12.52^\circ$; the percentage regulation = $(9681.5 - 7794)/7794 \times 100 = 24.21\%$; Phasor diagram: take V as reference, draw I_a lagging by 36.87° , then add $I_a R_a$ in phase with I_a and $jI_a X_s$ leading I_a by 90° to reach E .

Phasor



Question 130

Original: 5B | EM-II ELE 2251 Grade Improvement, 11 August 2021

What is the effect of load power factor on the armature reaction in an alternator? A 1500 kVA, 6600 V, 3 - phase, Y - connected alternator with a resistance of 0.4Ω /phase and synchronous reactance of 6Ω /phase delivers a full - load current at a power factor 0.8 (lagging) and normal rated voltage. Estimate the terminal voltage (line - line) for the same excitation and load current at 0.8 p.f. (leading). (05)

Answer 130

At full load, $I_a = 1500 \text{ kVA}/(\sqrt{3} \times 6.6 \text{ kV}) = 131.2$ A. Rated phase voltage is $V = 6600/\sqrt{3} = 3810.5$ V. For the initial 0.8 lagging load, I lags V by 36.87° and the generator phasor equation is $E = V + I_a(R_a + jX_s)$.

Substitution gives $E = 3810.5 + 131.2 \angle -36.87^\circ (0.4 + j6) = 4324.9 + j598.3$ V, so $|E| = 4366$ V per phase.

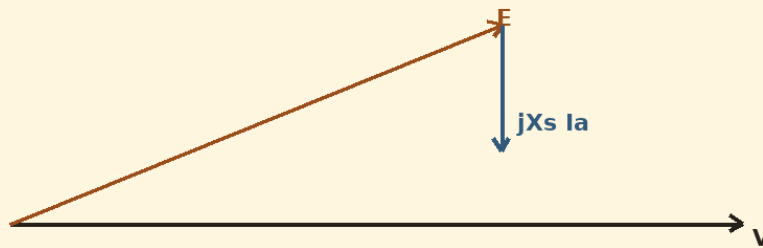
For the same excitation and same current magnitude at 0.8 leading, I leads V by 36.87° . The impedance drop is $I_a \angle +36.87^\circ (0.4 + j6) = -430.4 + j661.3$ V. With $|E|$ unchanged, $|V + \text{drop}| = 4366$ V.

Therefore $(V - 430.4)^2 + 661.3^2 = 4366^2$, giving $V = 4746$ V per phase. The corresponding terminal line voltage is $\sqrt{3} \times 4746 = 8.22$ kV.

For armature reaction: lagging load is mainly demagnetizing, unity pf is mainly cross-magnetizing, and leading load is magnetizing. Hence a leading load can raise terminal voltage for the same excitation.

Final: Terminal voltage for the same excitation and 0.8 leading load is approximately 8.22 kV line-to-line.

Phasor



Question 131

Original: 6B | EM-II ELE 2251 Grade Improvement, 11 August 2021

Using relevant phasor diagram, discuss the behaviour of a cylindrical rotor synchronous motor supplying a constant load but operating under varying excitation conditions. (05)

Answer 131

For a cylindrical-rotor synchronous motor at constant mechanical load and terminal voltage, the real power input is approximately constant. Therefore the in-phase component of armature current remains nearly fixed, while the reactive component changes with field excitation.

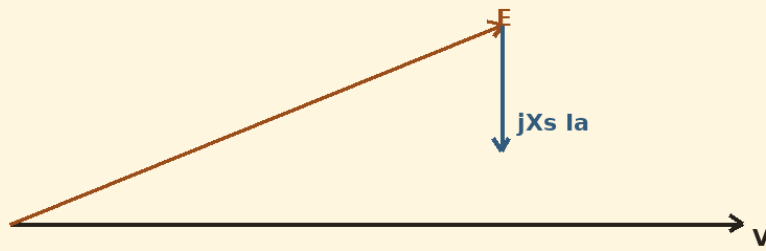
At under-excitation, the induced emf is small. The motor draws a lagging current from the supply and absorbs reactive power, behaving like an inductive load. At normal excitation, armature current is minimum and the power factor is approximately unity. At over-excitation, the induced emf is large and the motor draws leading current, supplying reactive power to the system.

The phasor relation for a motor is $E = V - I_a(R_a + jX_s)$. Neglecting resistance, the tip of the current phasor moves as excitation changes while the real-power component remains set by the load. This produces the familiar V-curve of armature current versus field current and inverted-V curve of power factor versus field current.

This behavior is important because a synchronous motor can be used as a synchronous condenser for power-factor correction.

Final: Varying excitation changes reactive power and power factor from lagging to unity to leading, while real power is set mainly by shaft load.

Phasor



Generator phasor relation: $E = V + I_a(R_a + jX_s)$

10. Synchronization and parallel operation of alternators

Syllabus area: Week 11

Question 132

Original: 1C | EM-II ELE 204 End Sem. 13 May 2014

List the necessary conditions to be satisfied while synchronising 3 phase alternators. With neat connection diagram, explain “Two Bright One Dark Lamp method” of synchronisation. (04)

Answer 132

For successful synchronization of a three-phase alternator with bus bars, the incoming alternator must have: same phase sequence, equal line voltage, equal frequency, and zero phase-angle difference at the instant of closing. The waveform should also be sinusoidal and the machine must be running in the correct direction.

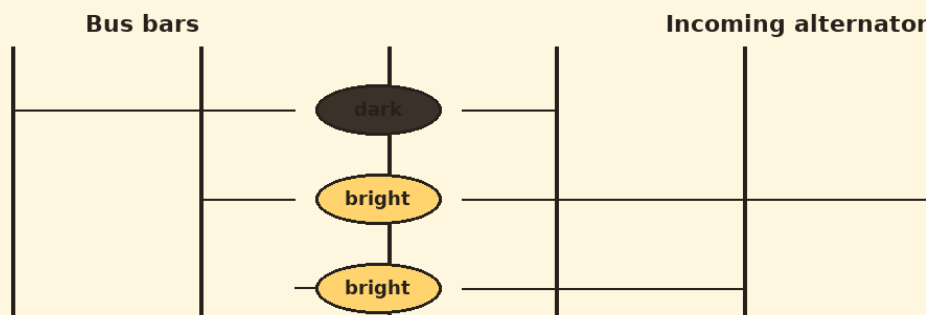
In the two-bright one-dark lamp method, three lamps are connected between corresponding and cross phases of the incoming alternator and bus bars. If the phase sequence is correct, the lamps brighten and darken cyclically. At the correct synchronizing instant, one lamp is dark while the other two are equally bright.

The dark lamp indicates that the voltage across that pair of terminals is zero, i.e. those two phase voltages are in phase. The two bright lamps confirm that the other phase voltages are correctly displaced by 120° and that the phase sequence is correct.

The breaker should be closed at the middle of the dark period when the flicker frequency is very low. Slow flicker means the incoming frequency is almost equal to bus-bar frequency.

Final: Synchronize only when voltage, frequency, phase sequence, and phase angle are correct; in the two-bright one-dark method, close the breaker when one lamp is dark and the other two are equally bright.

Lamp Method



Question 133

Original: 2A | EM-II ELE 204 End Sem. 13 May 2014

Two identical 3 phase star connected generators are supplying equal active power when operating in parallel to supply a total load of 1 MW at 6 kV and 0.8 pf lagging. The synchronous impedance of each machine is $(2+j40) \Omega$ per phase. The field of the first generator is so excited that its armature current is 55 A lagging. Find (i) armature current of the second machine, (ii) power factor of each machine (iii) open circuit emf of each machine (04)

Answer 133

Use the synchronous-machine phasor equation and keep the generator/motor sign convention explicit before substituting values.

Step 1: the total load is 1000 kW at 0.8 lagging, so total $Q = 750$ kVar. Equal active-power sharing gives $P_1 = P_2 = 500$ kW.

Step 2: for generator 1, $I_1 = 55$ A. $\text{pf}_1 = 500 \text{ kW}/(\sqrt{3} \times 6000 \times 55) = 0.875$ lagging.

Step 3: $Q_1 = 276.95$ kVar, so $Q_2 = 750 - 276.95 = 473.05$ kVar.

Final: The total load is 1000 kW at 0.8 lagging, so total $Q = 750$ kVar. Equal active-power sharing gives $P_1 = P_2 = 500$ kW.; for generator 1, $I_1 = 55$ A. $\text{pf}_1 = 500 \text{ kW}/(\sqrt{3} \times 6000 \times 55) = 0.875$ lagging.; $Q_1 = 276.95$ kVar, so $Q_2 = 750 - 276.95 = 473.05$ kVar.

Question 134

Original: 3B | EM-II ELE 204 End Sem. 13 May 2014

Two identical 2000 KW alternators operate in parallel. The governor of the first machine is such that the frequency drops uniformly from 50 Hz on no-load to 48 Hz on full load. The corresponding uniform speed drop of the second machine is 50 Hz to 47.5 Hz. (i) How will the two machines share a load of 3000 KW? (ii) What is the maximum load at upf that can be delivered without overloading either machine? (04)

Answer 134

Let P_1 and P_2 be the kW loads on the two alternators. For alternator 1, frequency falls from 50 Hz at no-load to 48 Hz at 2000 kW, so droop slope is $2/2000 = 0.001$ Hz/kW. For alternator 2, slope is $2.5/2000 = 0.00125$ Hz/kW.

At common operating frequency, $50 - 0.001P_1 = 50 - 0.00125P_2$, so $0.001P_1 = 0.00125P_2$. Also $P_1 + P_2 = 3000$ kW. Solving gives $P_1 = 1667$ kW and $P_2 = 1333$ kW. The common frequency is $50 - 0.001 \times 1667 = 48.33$ Hz.

For maximum total load without overloading either machine, alternator 1 reaches its 2000 kW limit first at $f = 48$ Hz. At this frequency, alternator 2 carries $(50 - 48)/0.00125 = 1600$ kW.

Final: For 3000 kW, sharing is 1667 kW and 1333 kW at 48.33 Hz. Maximum total load without overload = 3600 kW.

Question 135

Original: 3C | EM-II ELE 204 End Sem. 13 May 2014

Two 3 phase alternators having synchronous impedance of $(0.1+j2)$ /phase and $(0.2+j3)$ /phase respectively are connected in parallel to supply a common load of impedance $(2+j1)$ /phase. If the induced emf of each machine is 1100 V/phase where the induced emf of second machine is lagging that of first machine by 10° electrical. Find (i) common voltage (ii) armature current and power output of each machine (04)

Answer 135

Use the synchronous-machine phasor equation and keep the generator/motor sign convention explicit before substituting values.

Step 1: Let V be the common terminal phase voltage. Applying KCL at the bus gives $(E_1 - V)/Z_1 + (E_2 - V)/Z_2 = V/Z_L$.

Step 2: Take $E_1 = 1100$ at 0° V/phase, $E_2 = 1100$ at -10° V/phase, $Z_1 = 0.1+j2$, $Z_2 = 0.2+j3$, $Z_L = 2+j1 \Omega$.

Step 3: Solving the complex node equation gives common terminal voltage $V = 812.2$ at -24.20° V/phase.

Step 4: $I_1 = (E_1 - V)/Z_1 = 244.6$ at -44.31° A; $I_2 = (E_2 - V)/Z_2 = 123.3$ at -63.67° A.

Step 5: the corresponding three-phase real powers are $P_1 = 3 \text{ Re}(V I_1 \times) = 559.7$ kW; $P_2 = 3 \text{ Re}(V I_2 \times) = 231.9$ kW.

Final: $I_1 = (E_1 - V)/Z_1 = 244.6$ at -44.31° A; $I_2 = (E_2 - V)/Z_2 = 123.3$ at -63.67° A.; the corresponding three-phase real powers are $P_1 = 3 \operatorname{Re}(V I_1^*) = 559.7$ kW; $P_2 = 3 \operatorname{Re}(V I_2^*) = 231.9$ kW.

Question 136

Original: 4B | EM-II ELE 204 Makeup, 08 July 2014

Two station generators A and B operate in parallel. Respective rated capacities are 50 MW and 25 MW. Full load speed regulation of A is 3% and that of B is 3.5%. The no load frequency of both A and B is 50 Hz. Calculate (i) the common operating frequency and load sharing for a connected load of 50 MW (ii) Maximum power shared by both machines without over loading either of machine. (03)

Answer 136

For generator A, full-load frequency drop is 3 percent of 50 Hz = 1.5 Hz at 50 MW, so slope = 0.03 Hz/MW. For generator B, full-load drop is 3.5 percent of 50 Hz = 1.75 Hz at 25 MW, so slope = 0.07 Hz/MW.

At a common frequency, $50 - 0.03P_A = 50 - 0.07P_B$, hence $0.03P_A = 0.07P_B$. With $P_A + P_B = 50$ MW, solving gives $P_A = 35$ MW and $P_B = 15$ MW.

The common frequency is $f = 50 - 0.03 \times 35 = 48.95$ Hz.

For maximum total load without overload, A reaches 50 MW at $f = 50 - 0.03 \times 50 = 48.5$ Hz. At this frequency, B supplies $(50 - 48.5)/0.07 = 21.43$ MW, which is below its 25 MW rating.

Final: At 50 MW load, A supplies 35 MW and B supplies 15 MW at 48.95 Hz. Maximum non-overload load = 71.43 MW.

Question 137

Original: 3A | EM-II ELE 204 Makeup, 09 July 2015

A three phase load of 8 MW at 0.8 pf lagging is to be supplied at a line voltage of 11 kV by running two alternators in parallel with equal excitation emfs. The synchronous impedances per phase of the alternators are $(0.75 + j10)$ and $(1.2 + j8) \Omega$ respectively. For each machine determine (i) the currents supplied and power factor (ii) The excitation emf and load angle. (04)

Answer 137

Use the synchronous-machine phasor equation and keep the generator/motor sign convention explicit before substituting values.

Step 1: Assuming equal excitation emfs means the two internal emf phasors are equal. Let E be the common internal emf.

Step 2: $I_1 = (E - V)/Z_1$ and $I_2 = (E - V)/Z_2$, with $I_1 + I_2 = I_{\text{load}}$.

Step 3: Load current = $10 \text{ MVA}/(\sqrt{3} \times 11 \text{ kV})$ at 0.8 lag = 524.86 at -36.87° A.

Step 4: Solving gives $E = 8.150$ kV/phase at $\delta = 12.08^\circ$.

Step 5: Machine 1 current = 234.5 A at pf 0.775 lagging; power = 3.462 MW.

Step 6: Machine 2 current = 290.7 A at pf 0.819 lagging; power = 4.538 MW.

Final: Load current = $10 \text{ MVA}/(\sqrt{3} \times 11 \text{ kV})$ at 0.8 lag = 524.86 at -36.87° A.; Solving gives $E = 8.150$ kV/phase at $\delta = 12.08^\circ$.; Machine 1 current = 234.5 A at pf 0.775 lagging; power = 3.462 MW.; Machine 2 current = 290.7 A at pf 0.819 lagging; power = 4.538 MW.

Question 138

Original: 3C | EM-II ELE 204 Makeup, 09 July 2015

Mention the essential conditions to synchronise an incoming alternator with infinite bus system. (03)

Answer 138

Use the synchronous-machine phasor equation and keep the generator/motor sign convention explicit before substituting values.

Before an incoming alternator is connected to live bus bars, the following conditions must be satisfied:

Step 1: Terminal voltage magnitude of incoming alternator equals bus-bar voltage.

Step 2: Frequency of incoming alternator equals bus-bar frequency.

Step 3: Phase sequence is the same.

Step 4: Phase angle at the instant of closing is zero or very small.

Step 5: Waveform should be similar and the machine must be running at correct speed. These conditions are checked by lamps, synchroscope, voltmeters, frequency meters and phase-sequence indicators.

Final: Waveform should be similar and the machine must be running at correct speed. These conditions are checked by lamps, synchroscope, voltmeters, frequency meters and phase-sequence indicators.

Question 139

Original: 5B | EM-II ELE 204 Makeup, 09 July 2015

Two station generators A and B operate in parallel. Respective rated capacities are 50 MW and 25 MW. The no load frequency of both generators is 50 Hz. The drooping characteristics are respectively 0.06 Hz/MW and 0.07 Hz/MW. Calculate (i) the common operating frequency and load sharing for a connected load of 50 MW, (ii) Maximum power shared by both machines without over loading either of machine. (04)

Answer 139

Use the droop equations $f = 50 - 0.06P_A$ and $f = 50 - 0.07P_B$, with $P_A + P_B = 50$ MW.

Equating frequency gives $0.06P_A = 0.07P_B$. With $P_B = 50 - P_A$, $0.06P_A = 0.07(50 - P_A)$, so $P_A = 26.92$ MW and $P_B = 23.08$ MW.

The common frequency is $f = 50 - 0.06 \times 26.92 = 48.38$ Hz.

For maximum loading, generator B reaches its 25 MW rating first at $f = 50 - 0.07 \times 25 = 48.25$ Hz. At this frequency A supplies $(50 - 48.25)/0.06 = 29.17$ MW. Therefore total maximum load is $29.17 + 25 = 54.17$ MW.

Final: At 50 MW load, A = 26.92 MW, B = 23.08 MW, $f = 48.38$ Hz. Maximum total without overload = 54.17 MW.

Question 140

Original: 3A | EM-II ELE 2202 End Sem, 10 May 2016

Two station generators A and B operate in parallel. Frequency drop of A is 0.025 Hz/MW and that of B is 0.07 Hz/MW. If no load frequency of both A and B is 50 Hz, calculate the common operating frequency and load sharing for a connected load of 65 MW. (04)

Answer 140

Let P_A and P_B be the MW supplied by A and B. Their frequency-load equations are $f = 50 - 0.025P_A$ and $f = 50 - 0.07P_B$.

Since the machines operate in parallel, frequency is common: $0.025P_A = 0.07P_B$. Also $P_A + P_B = 65$ MW.

Substitute $P_B = 65 - P_A$: $0.025P_A = 0.07(65 - P_A)$. Thus $0.095P_A = 4.55$ and $P_A = 47.89$ MW. Hence $P_B = 17.11$ MW.

Common frequency is $f = 50 - 0.025 \times 47.89 = 48.80$ Hz.

Final: Alternator A supplies 47.9 MW, alternator B supplies 17.1 MW, and common operating frequency is 48.80 Hz.

Question 141

Original: 4B | EM-II ELE 2202 End Sem. 23 April 2018

A 10 MVA, 11 kV, 3 phase, 50 Hz, 2 pole, star connected alternator operates in parallel with other synchronous generators. Its synchronous reactance is 1.2Ω per phase. The armature resistance may be neglected. At full load, 0.8 lagging power factor, a) Determine the excitation emf. b) If the excitation is suddenly increased by 10 % without changing the load, determine the synchronizing power and torque per mechanical degree of phase displacement. (06)

Answer 141

Rated current is $I = 10 \text{ MVA}/(\sqrt{3} \times 11 \text{ kV}) = 524.9$ A. Per-phase voltage is $V = 11 \text{ kV}/\sqrt{3} = 6350.9$ V. At full load 0.8 lagging, I lags by 36.87° .

Generated emf is $E = V + jX_s I = 6350.9 + j1.2(524.9 \angle -36.87^\circ) = 6728.8 + j503.9$ V. Therefore $|E| = 6747.6$ V per phase and $\delta = 4.28^\circ$.

When excitation is increased by 10 percent, $E_2 = 1.1E = 7422.4$ V per phase. Real power remains $P = 10 \times 0.8 = 8$ MW, so $P = 3EV \sin(\delta)/X_s$ gives $\delta_2 = \sin^{-1}(PX_s/(3E_2V)) = 3.89^\circ$.

Synchronizing power coefficient per electrical radian is $P_{\text{syn}} = 3EV \cos(\delta)/X_s = 117.6$ MW/electrical radian. Since it is a 2-pole alternator, one mechanical degree equals one electrical degree. Therefore synchronizing power per mechanical degree = $117.6 \times \pi/180 = 2.05$ MW/degree.

At 3000 rpm, $\omega_s = 314.16$ rad/s. Synchronizing torque per mechanical degree = $2.052 \text{ MW}/314.16 = 6.53$ kN m/degree.

Final: $E = 11.69$ kV line initially; after 10 percent excitation rise, $\delta = 3.89^\circ$, synchronizing power = 2.05 MW/mechanical degree, synchronizing torque = 6.53 kN m/mechanical degree.

Question 142

Original: 3A | EM-II ELE 2202 End Sem. 24 April 2017

Two station alternators A & B operate in parallel. Respective rated capacities are 50 MW and 25 MW. The no-load frequency of both generators is 50 Hz. The drooping characteristics are 0.06 Hz/MW and 0.07 Hz/MW respectively. Calculate: (a) The common operating frequency & load sharing for a connected load of 50 MW. (b) Maximum power shared by both machines without overloading either of them. (04)

Answer 142

The droop equations are $f = 50 - 0.06P_A$ and $f = 50 - 0.07P_B$. For a 50 MW total load, $P_A + P_B = 50$ MW.

Equating frequency gives $0.06P_A = 0.07P_B$. Solving with $P_A + P_B = 50$ gives $P_A = 26.92$ MW and $P_B = 23.08$ MW.

The common operating frequency is $f = 50 - 0.06 \times 26.92 = 48.38$ Hz.

For maximum load without overloading, B reaches 25 MW first at $f = 50 - 0.07 \times 25 = 48.25$ Hz. A then supplies $(50 - 48.25)/0.06 = 29.17$ MW. Total load = 54.17 MW.

Final: A supplies 26.92 MW, B supplies 23.08 MW at 48.38 Hz; maximum safe total load = 54.17 MW.

Question 143

Original: 2C | EM-II ELE 2202 End Sem. 29 April 2019

Two alternators X and Y with capacities 80MW and 30MW respectively operate in parallel in a power station. Full load speed regulation of X is 2.5% and that of Y is 3.2%. If no load frequency of both the machines are 50Hz, calculate the common operating frequency with a connected load of 60MW. (04)

Answer 143

For alternator X, full-load drop = 2.5 percent of 50 = 1.25 Hz at 80 MW, so slope = $1.25/80 = 0.015625$ Hz/MW. For alternator Y, full-load drop = 3.2 percent of 50 = 1.6 Hz at 30 MW, so slope = $1.6/30 = 0.05333$ Hz/MW.

Let loads be P_X and P_Y . At common frequency, $0.015625P_X = 0.05333P_Y$. Also $P_X + P_Y = 60$ MW.

Solving gives $P_X = 46.40$ MW and $P_Y = 13.60$ MW. Common frequency is $f = 50 - 0.015625 \times 46.40 = 49.27$ Hz.

Final: Alternator X supplies 46.4 MW, alternator Y supplies 13.6 MW, and the common frequency is 49.27 Hz.

Question 144

Original: 3B | EM-II ELE 2202 Makeup. 02 July 2016

With neat diagram explain the working principle of Synchroscope. Also explain how Synchroscope is used to synchronise two alternators? 4M

Answer 144

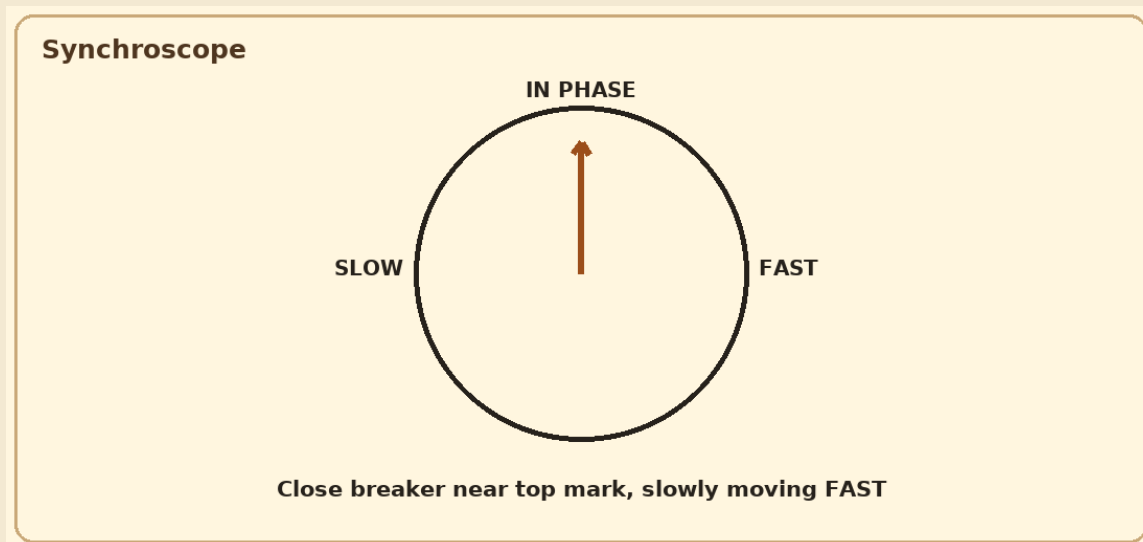
A synchroscope indicates whether the incoming alternator voltage is leading or lagging the bus-bar voltage and how fast the phase angle is changing. It is connected between the incoming alternator and the live bus bars through potential transformers.

The pointer rotates when there is a small frequency difference between the incoming alternator and bus bars. If the incoming frequency is higher, the pointer rotates in the FAST direction; if lower, it rotates in the SLOW direction. The speed of rotation is proportional to the frequency difference.

Before closing the breaker, adjust excitation until voltages are equal, adjust prime mover speed until the synchroscope pointer moves slowly, and verify correct phase sequence. The breaker is closed when the pointer is at the 12 o'clock in-phase mark and moving slowly in the FAST direction.

The pointer should not be allowed to stand still away from the top mark because that means equal frequency but wrong phase angle.

Final: A synchroscope is used to close the alternator breaker at the instant when incoming voltage and bus voltage are in phase, with equal voltage, equal frequency, and correct phase sequence.



Question 145

Original: 3C | EM-II ELE 2202 Makeup, 13 June 2019

Explain the conditions for synchronization of an incoming 3-phase alternator with live bus bars. (02)

Answer 145

Use the synchronous-machine phasor equation and keep the generator/motor sign convention explicit before substituting values.

Before an incoming alternator is connected to live bus bars, the following conditions must be satisfied:

Step 1: Terminal voltage magnitude of incoming alternator equals bus-bar voltage.

Step 2: Frequency of incoming alternator equals bus-bar frequency.

Step 3: Phase sequence is the same.

Step 4: Phase angle at the instant of closing is zero or very small.

Step 5: Waveform should be similar and the machine must be running at correct speed. These conditions are checked by lamps, synchroscope, voltmeters, frequency meters and phase-sequence indicators.

Final: Waveform should be similar and the machine must be running at correct speed. These conditions are checked by lamps, synchroscope, voltmeters, frequency meters and phase-sequence indicators.

Question 146

Original: 5C | EM-II ELE 2202 Makeup, 13 June 2019

Two station alternators A & B operate in parallel. Respective rated capacities are 50 MW and 25 MW. The no-load frequency of both generators is 50 Hz. The drooping characteristics are 0.06 Hz/MW and 0.07 Hz/MW respectively. Calculate: (a) The common operating frequency & load sharing for a connected load of 50 MW. (b) Maximum power shared by both machines without overloading either of them. (04)

Answer 146

Use $f = 50 - 0.06P_A$ for alternator A and $f = 50 - 0.07P_B$ for alternator B. At 50 MW total load, $P_A + P_B = 50$.

Equating common frequency: $0.06P_A = 0.07P_B$. Hence $P_A = 26.92$ MW and $P_B = 23.08$ MW. The frequency is $f = 50 - 0.06 \times 26.92 = 48.38$ Hz.

For maximum total load without overload, alternator B reaches its 25 MW rating first. At $P_B = 25$ MW, $f = 50 - 0.07 \times 25 = 48.25$ Hz. At this frequency $P_A = (50 - 48.25)/0.06 = 29.17$ MW.

Final: For 50 MW, load sharing is A = 26.92 MW and B = 23.08 MW at 48.38 Hz. Maximum load without overloading either machine = 54.17 MW.

Question 147

Original: 2B | EM-II ELE 2202 Makeup, 16 June 2017

Two star-connected synchronous generators are connected in parallel to supply a balanced load of 1,500 kVA at 11 kV and 0.8 lagging power factor. The per-phase synchronous reactances of the two generators are 35 Ω and 40 Ω respectively. The prime movers of the 2 generators are adjusted so that they share the total active power equally. The phase current in one generator is 43 A at lagging power factor. Calculate: a) Phase current of the second generator. b) Power factors at which each generator operate. c) Induced emf of each machine. (05)

Answer 147

Use the synchronous-machine phasor equation and keep the generator/motor sign convention explicit before substituting values.

Step 1: the total load is 1500 kVA at 0.8 lag $\rightarrow P = 1200$ kW, $Q = 900$ kVAr.

Step 2: Equal prime-mover setting gives $P_1 = P_2 = 600$ kW.

Step 3: for the first generator, $\cos \phi_1 = P_1/(\sqrt{3} V_L I_1) = 0.73237$, so $\text{pf}_1 = 0.73237$ lagging and $Q_1 = 557.84$ kVAr.

Step 4: $Q_2 = 900 - Q_1 = 342.16$ kVAr. Hence $I_2 = 36.253$ A and $\text{pf}_2 = 0.86868$ lagging.

Step 5: using $E = V + jX_s I$ per phase:

Step 6: $E_1 = 7,457.5$ V/phase at 8.4994° ; $E_2 = 7,180.6$ V/phase at 10.104° .

Final: Equal prime-mover setting gives $P_1 = P_2 = 600$ kW.; for the first generator, $\cos \phi_1 = P_1/(\sqrt{3} V_L I_1) = 0.73237$, so $\text{pf}_1 = 0.73237$ lagging and $Q_1 = 557.84$ kVAr.; $Q_2 = 900 - Q_1 = 342.16$ kVAr. Hence $I_2 = 36.253$ A and $\text{pf}_2 = 0.86868$ lagging.; $E_1 = 7,457.5$ V/phase at 8.4994° ; $E_2 = 7,180.6$ V/phase at 10.104° .

Question 148

Original: 3A | EM-II ELE 2202 Makeup, 16 June 2017

A 3-phase, 3,000 kVA, 6.6 kV, 50 Hz, 750 rpm, star connected synchronous generator has a synchronous reactance of 2.9 Ω per phase and a negligible resistance. It is running in parallel with infinite bus. Calculate synchronizing power and synchronizing torque per mechanical degree of displacement at full-load, 0.8 pf leading. (03)

Answer 148

Rated current is $I = 3000 \text{ kVA}/(\sqrt{3} \times 6.6 \text{ kV}) = 262.4$ A. For 750 rpm at 50 Hz, the number of poles is $P = 120f/N_s = 8$. Phase voltage is $V = 6.6 \text{ kV}/\sqrt{3} = 3810.5$ V.

At 0.8 leading power factor, current leads V by 36.87° . With negligible resistance, generator emf is $E = V + jX_s I$. Substitution gives $|E| = 3408.7$ V per phase and load angle $\delta = 10.29^\circ$.

Synchronizing power coefficient per electrical radian is $P_{\text{syn},e} = 3EV \cos(\delta)/X_s = 13.22$ MW/electrical radian.

Electrical angle changes $P/2 = 4$ times as fast as mechanical angle. Therefore synchronizing power per mechanical degree is $P_{\text{syn},e} \times (P/2) \times \pi/180 = 0.923$ MW/mechanical degree.

At 750 rpm, $\omega_s = 78.54$ rad/s. Synchronizing torque per mechanical degree = $0.923 \text{ MW}/78.54 = 11.75$ kN m/mechanical degree.

Final: $E = 5.90$ kV line, synchronizing power = 0.923 MW/mechanical degree, synchronizing torque = 11.75 kN m/mechanical degree.

Question 149

Original: 4B | EM-II ELE 2202 Makeup, 19 June 2018

Two identical 3-phase star-connected generators having synchronous impedance of $(2 + j40) \Omega$ /phase supplying equal power, operate in parallel. They supply a total load of 1,000 kW at 6,000 V and at 0.8 power factor lagging. The field of the first generator is so excited that its armature current is 55 A lagging. Find (i) The open circuit emf of each machine (ii) The armature current and power factor of the other machine. (05)

Answer 149

Use the synchronous-machine phasor equation and keep the generator/motor sign convention explicit before substituting values.

Step 1: the total load is 1000 kW at 0.8 lagging, so total $Q = 750$ kVAr. Equal active-power sharing gives $P_1 = P_2 = 500$ kW.

Step 2: for generator 1, $I_1 = 55$ A. $\text{pf}_1 = 500 \text{ kW} / (\sqrt{3} \times 6000 \times 55) = 0.875$ lagging.

Step 3: $Q_1 = 276.95$ kVAr, so $Q_2 = 750 - 276.95 = 473.05$ kVAr.

Step 4: $I_2 = \sqrt{(P_2^2 + Q_2^2)} / (\sqrt{3}V) = 66.23$ A; $\text{pf}_2 = 0.726$ lagging.

Step 5: using $E = V + (2 + j40)I$: $E_1 = 4.990$ kV/phase and $E_2 = 5.685$ kV/phase.

Final: The total load is 1000 kW at 0.8 lagging, so total $Q = 750$ kVAr. Equal active-power sharing gives $P_1 = P_2 = 500$ kW.; for generator 1, $I_1 = 55$ A. $\text{pf}_1 = 500 \text{ kW} / (\sqrt{3} \times 6000 \times 55) = 0.875$ lagging.; $Q_1 = 276.95$ kVAr, so $Q_2 = 750 - 276.95 = 473.05$ kVAr.; $I_2 = \sqrt{(P_2^2 + Q_2^2)} / (\sqrt{3}V) = 66.23$ A; $\text{pf}_2 = 0.726$ lagging.

Question 150

Original: 3A | EM-II ELE 2225 Makeup, 26 June 2024

Two identical 2000 kW alternators operate in parallel. The governor of the first machine is such that the frequency drops uniformly from 50 Hz on no- load to 48 Hz on full load. The corresponding uniform speed drop of the second machine is 50 Hz to 47.5 Hz. (a) How will the two machines share a load of 3000 kW? (b) What is the maximum load at upf that can be delivered without overloading either machine? 3

Answer 150

The droop slopes are $2/2000 = 0.001$ Hz/kW for alternator 1 and $2.5/2000 = 0.00125$ Hz/kW for alternator 2.

For a 3000 kW total load, common frequency gives $0.001P_1 = 0.00125P_2$ and $P_1 + P_2 = 3000$. Hence $P_1 = 1667$ kW and $P_2 = 1333$ kW.

Common frequency is $f = 50 - 0.001 \times 1667 = 48.33$ Hz.

Maximum total load occurs when alternator 1 reaches 2000 kW. At $f = 48$ Hz, alternator 2 carries $(50 - 48)/0.00125 = 1600$ kW. Total = 3600 kW.

Final: For 3000 kW, the machines share 1667 kW and 1333 kW at 48.33 Hz. Maximum load without overload = 3600 kW.

Question 151

Original: 3B | EM-II ELE 2225 Makeup, 26 June 2024

List the necessary conditions to be satisfied while synchronizing 3 phase alternators. With neat connection diagram, explain “Two Bright One Dark Lamp method” of synchronization. 3

Answer 151

Conditions for synchronizing are: equal rms line voltage, same frequency, same phase sequence, and zero phase-angle difference at the instant of closing. In practice voltage is checked by voltmeters, frequency by

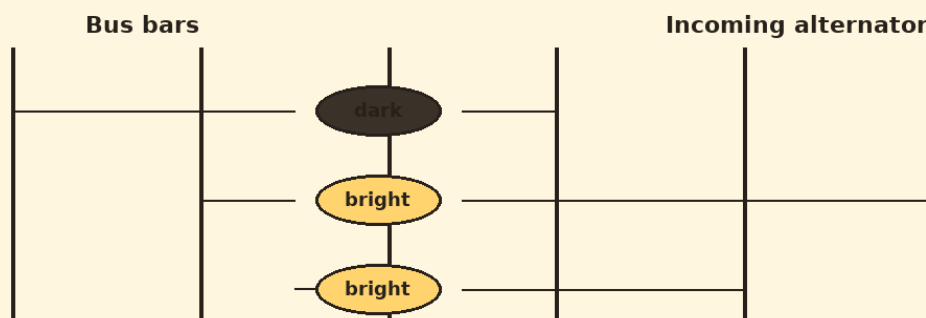
frequency meters or lamp flicker, phase sequence by lamp behavior/phase-sequence indicator, and phase coincidence by lamps or synchroscope.

In the two-bright one-dark method, one lamp is connected across corresponding phases and the other two are cross-connected. When the phase sequence is correct, the lamps brighten and darken in a regular pattern. The synchronizing instant is indicated when one lamp is dark and the other two are equally bright.

The breaker is closed at the middle of the dark interval, preferably when the flicker is slow. Slow flicker means the incoming alternator frequency is nearly equal to the bus-bar frequency.

Final: Close the incoming alternator only when one lamp is dark, two are equally bright, and the flicker is slow, after matching voltage and phase sequence.

Lamp Method



Question 152

Original: 4B-i | EM-II ELE 2251 Grade Improvement. 11 August 2021

State the conditions to be satisfied for successful synchronisation of an alternator with infinite busbars. Also mention the techniques through which these conditions are ensured to have been met. (part i)

Answer 152

For successful synchronization with infinite bus bars, the incoming alternator must satisfy four conditions: terminal voltage equal to bus-bar voltage, frequency equal to bus-bar frequency, same phase sequence, and phase opposition/zero phase difference across the breaker contacts at closing instant.

These are ensured as follows. Voltage is matched by adjusting field excitation. Frequency is matched by adjusting prime-mover speed. Phase sequence is checked by a phase-sequence indicator or by the lamp method. Phase coincidence is checked by a synchroscope or synchronizing lamps.

Common techniques are the three-dark-lamp method, two-bright one-dark lamp method, and synchroscope method. For large alternators, synchroscope with check synchronizing relay is preferred because it is safer and more precise.

Final: Voltage, frequency, phase sequence, and phase angle must all match; they are verified by voltmeters, frequency meters, phase-sequence checks, synchronizing lamps, and synchroscope/check relay.

11. Salient-pole machines, Blondel two-reaction theory and slip test

Syllabus area: Week 12

Question 153

Original: 4B | EM-II ELE 204 End Sem. 13 May 2014

A 3.5 MVA, 6.6 kV, 32 pole, 50 Hz, 3 phase synchronous generator has the direct and quadrature axis synchronous reactance as measured by the slip test of 9.6Ω and 6Ω per phase respectively with negligible armature resistance. When supplying rated load 0.8 pf lag, determine (i) the excitation emf needed to maintain the rated terminal voltage (ii) synchronising power per mechanical degree of rotor displacement. (04)

Answer 153

Use Blondel two-reaction theory: resolve armature current into direct-axis and quadrature-axis components and use X_d and X_q separately.

Step 1: the rated current is obtained from 306.17 A; $\delta = 16.654^\circ$.

Step 2: $I_d = 246.19$ A, $I_q = 182.01$ A; $E = 6,014.1$ V/phase.

Step 3: Synchronizing power per electrical degree = 159,460 W/electrical degree.

Step 4: for 32 poles, per mechanical degree = 2,551,359 W/mechanical degree.

Step 5: Synchronizing torque per mechanical degree = 129,940 N m/mechanical degree.

Final: $I_d = 246.19$ A, $I_q = 182.01$ A; $E = 6,014.1$ V/phase.; Synchronizing power per electrical degree = 159,460 W/electrical degree.; for 32 poles, per mechanical degree = 2,551,359 W/mechanical degree.; Synchronizing torque per mechanical degree = 129,940 N m/mechanical degree.

Slip Test



$I_{min} \rightarrow X_d = V/I_{min}$; $I_{max} \rightarrow X_q = V/I_{max}$

Question 154

Original: 4C | EM-II ELE 204 End Sem. 13 May 2014

Draw a neat connection diagram for measurement of direct and quadrature axis reactance by slip test. (02)

Answer 154

In the slip test, the stator winding of the salient-pole alternator is supplied from a low-voltage three-phase source while the field winding is kept open. The rotor is driven slowly at a speed slightly different from synchronous speed. A voltmeter, ammeter, and wattmeter are connected on the stator side.

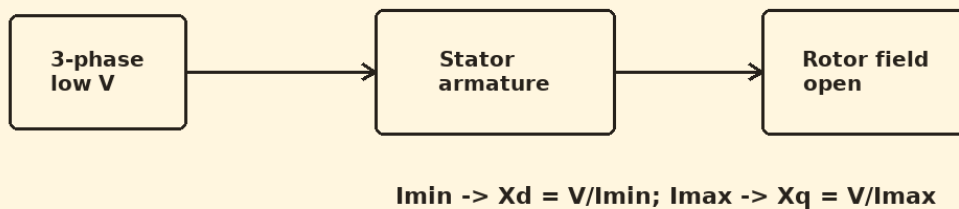
As the rotor direct axis and quadrature axis successively align with the stator mmf, the stator current oscillates between minimum and maximum values. When the stator mmf is along the direct axis, reactance is high and current is minimum. When it is along the quadrature axis, reactance is lower and current is maximum.

Therefore $X_d = V_{ph}/I_{min}$ and $X_q = V_{ph}/I_{max}$, using corresponding maximum/minimum voltage readings if the applied voltage is not perfectly constant.

Connection diagram description: connect a three-phase variac/low-voltage supply to the armature through ammeter and wattmeter; keep the DC field winding open; mechanically drive the rotor slowly; measure line voltage and line current variations.

Final: Slip test gives X_d from minimum armature current and X_q from maximum armature current under low-voltage open-field operation.

Slip Test



Question 155

Original: 4A | EM-II ELE 204 Makeup, 08 July 2014

With the aid of phasor diagram based on Blondel's two reaction theory, derive an expression for the active power output of a 3 phase Salient pole alternator in terms of excitation emf, terminal voltage, direct and quadrature axis reactance. Neglect armature resistance. (04)

Answer 155

Blondel two-reaction theory is used for salient-pole alternators because the air gap is non-uniform. The direct-axis reluctance is lower than the quadrature-axis reluctance, so X_d is greater than X_q . Hence armature reaction must be resolved into direct-axis and quadrature-axis components.

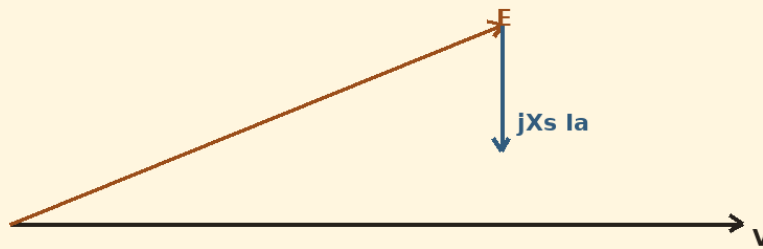
Take terminal voltage V as reference and draw armature current I_a at the load power-factor angle. The excitation emf E lies along the direct axis. Resolve I_a into I_d along the direct axis and I_q along the quadrature axis.

Neglecting armature resistance, the phasor equation is $E = V + jX_d I_d + jX_q I_q$. The q-axis current component produces a voltage drop $jX_q I_q$, and the d-axis component produces $jX_d I_d$. Adding these components vectorially to V gives the excitation emf E .

For a lagging load, the current lags V and the direct-axis component is demagnetizing. For a leading load, the direct-axis component may be magnetizing. The phasor diagram therefore explains both voltage regulation and the extra reluctance power term in salient-pole machines.

Final: The salient-pole phasor diagram differs from the cylindrical-rotor diagram by resolving I_a into I_d and I_q and using two reactances, X_d and X_q .

Phasor



$$\text{Generator phasor relation: } E = V + I_a(R_a + jX_s)$$

Question 156

Original: 4A | EM-II ELE 204 Makeup, 09 July 2015

A 5 MVA, 6.6 KV, 3- phase, 6.6 KV 50 Hz, star connected salient pole alternator is connected to an infinite bus. The direct axis reactance is 12Ω while that of quadrature axis is 9.5Ω per phase. The armature resistance is 1.5Ω per phase. When the generator is operating at rated MVA at 0.9 pf lagging, calculate the electromagnetic power developed and reluctance power. (05)

Answer 156

Use Blondel two-reaction theory: resolve armature current into direct-axis and quadrature-axis components and use X_d and X_q separately.

Step 1: using the standard salient-pole power-angle model; armature resistance is included approximately as copper loss in electromagnetic power.

Step 2: the rated current is obtained from 437.39 A; terminal active output = 4.5 MW; stator copper loss = 860.88 kW.

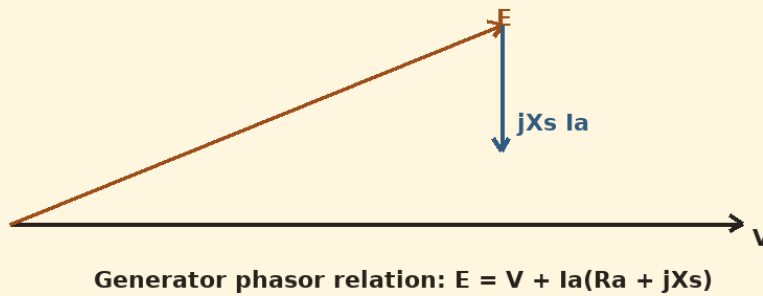
Step 3: from the lossless d-q phasor approximation: $\delta = 33.633^\circ$, $E = 7,693.9$ V/phase.

Step 4: Electromagnetic power developed = terminal output + copper loss = 5.3609 MW.

Step 5: Reluctance power component from the salient-pole power equation = 0.44052 MW.

Final: The rated current is obtained from 437.39 A; terminal active output = 4.5 MW; stator copper loss = 860.88 kW.; from the lossless d-q phasor approximation: $\delta = 33.633^\circ$, $E = 7,693.9$ V/phase.; Electromagnetic power developed = terminal output + copper loss = 5.3609 MW.; Reluctance power component from the salient-pole power equation = 0.44052 MW.

Phasor



Question 157

Original: 4A | EM-II ELE 2202 End Sem, 10 May 2016

Explain Blondel's two reaction theory for Salient Pole Alternator. Derive the circuit model and sketch the phasor diagram showing the relationship between terminal voltage and internal voltage. (05)

Answer 157

Use Blondel two-reaction theory: resolve armature current into direct-axis and quadrature-axis components and use X_d and X_q separately.

Blondel two-reaction theory resolves armature reaction in a salient-pole machine into two perpendicular components: direct-axis component and quadrature-axis component. Salient-pole generator: I_a is resolved into direct-axis I_d and quadrature-axis I_q components.

Step 1: the direct axis is along the rotor field pole axis and has reactance X_d .

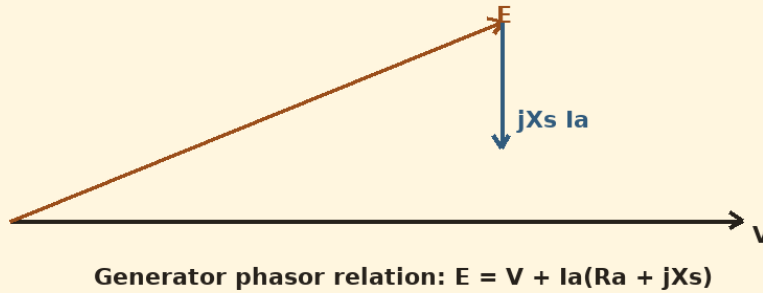
Step 2: the quadrature axis is 90 electrical degrees away and has reactance X_q .

Step 3: Armature current I_a is resolved into I_d and I_q .

Step 4: using the generator phasor convention neglecting R_a , the phasor relation is $E_f = V + jX_d I_d + jX_q I_q$.

Final: Blondel two-reaction theory resolves armature reaction in a salient-pole machine into two perpendicular components: direct-axis component and quadrature-axis component. Salient-pole generator: I_a is resolved into direct-axis I_d and quadrature-axis I_q components.; the quadrature axis is 90 electrical degrees away and has reactance X_q .

Phasor



Question 158

Original: 4B | EM-II ELE 2202 End Sem, 10 May 2016

A 5 MVA, 6.6 kV, 3- phase, 50 Hz, star connected salient pole alternator with per phase direct and quadrature axes reactances of 12Ω and 10Ω respectively is connected to an infinite bus. When the generator is operating at its rated MVA and power factor of 0.9 lagging, calculate the excitation emf required, synchronizing power per electrical degree of displacement. (05)

Answer 158

Rated current is $I = 5 \text{ MVA}/(\sqrt{3} \times 6.6 \text{ kV}) = 437.4 \text{ A}$. Per-phase terminal voltage is $V = 6.6 \text{ kV}/\sqrt{3} = 3810.5 \text{ V}$. At 0.9 lagging power factor, current lags terminal voltage by $\phi = \cos^{-1}(0.9) = 25.84^\circ$.

For a salient-pole alternator neglecting resistance, resolve armature current into direct and quadrature components with respect to the rotor axes. The phasor relation is $E = V + jX_d I_d + jX_q I_q$. Solving the two-reaction phasor diagram for the given load gives load angle $\delta = 34.55^\circ$ and generated emf $E = 7702 \text{ V}$ per phase.

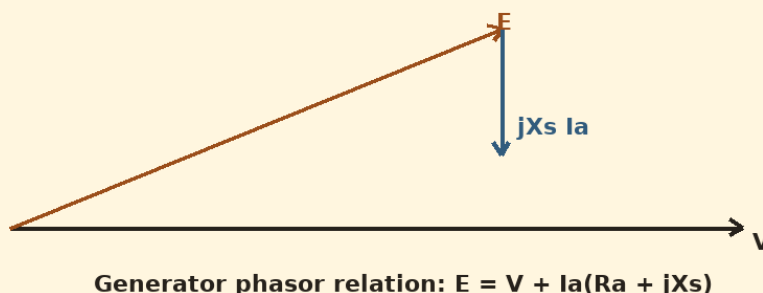
The power-angle equation is $P = (3EV/X_d) \sin \delta + (3V^2/2)(1/X_q - 1/X_d) \sin 2\delta$.

Synchronizing power coefficient is $dP/d\delta = (3EV/X_d) \cos \delta + 3V^2(1/X_q - 1/X_d) \cos 2\delta$. Substituting the above operating point gives $dP/d\delta = 6.30 \text{ MW}$ per electrical radian.

Per electrical degree, synchronizing power = $6.30 \times 10^6 \times \pi/180 = 0.110 \text{ MW/electrical degree}$.

Final: Excitation emf = 7.70 kV per phase; synchronizing power = 6.30 MW/electrical radian = 0.110 MW/electrical degree.

Phasor



Question 159

Original: 5B | EM-II ELE 2202 End Sem. 10 May 2016

Explain the phenomenon of hunting in synchronous motors. Find the natural frequency of oscillations of a 500 kVA, 11 kV, 1500 rpm, 50 Hz, non-salient pole alternator with synchronous reactance of 12Ω . The alternator is operating at full load 0.8 p.f lag. Consider the moment of inertia of the rotating part equal to $50 \times 10^3 \text{ kg-m}^2$. (05)

Answer 159

Use Blondel two-reaction theory: resolve armature current into direct-axis and quadrature-axis components and use X_d and X_q separately.

Step 1: Hunting is the oscillation of a synchronous rotor about its final equilibrium/load angle after a disturbance. Damper bars and excitation control help damp it.

Step 2: for the given alternator, rated current = $500 \text{ kVA}/(\sqrt{3} \times 11 \text{ kV}) = 26.24 \text{ A}$.

Step 3: at 0.8 lagging, $E = V + j12I = 6.545 \text{ kV/phase}$ and $\delta = 2.206^\circ$.

Step 4: Synchronizing power coefficient per electrical radian = $3EV/X \cos(\delta) = 10.383 \text{ MW/rad}$.

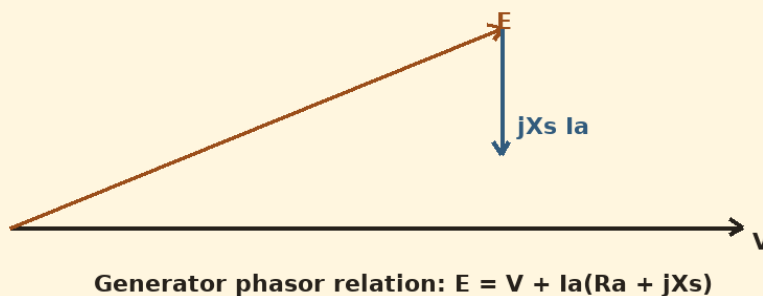
Step 5: for a 1500 rpm, 50 Hz alternator, $P = 4$ poles. Mechanical-radian coefficient = electrical coefficient $\times P/2 = 20.767 \text{ MW/rad mechanical}$.

Step 6: Synchronizing torque coefficient = $P_s/\omega_s = 132205 \text{ N-m/rad}$.

Step 7: Natural frequency = $(1/(2\pi))\sqrt{(T_s/J)} = 0.259 \text{ Hz}$.

Final: Synchronizing power coefficient per electrical radian = $3EV/X \cos(\delta) = 10.383 \text{ MW/rad}$;; for a 1500 rpm, 50 Hz alternator, $P = 4$ poles. Mechanical-radian coefficient = electrical coefficient $\times P/2 = 20.767 \text{ MW/rad mechanical}$;; Synchronizing torque coefficient = $P_s/\omega_s = 132205 \text{ N-m/rad}$;; Natural frequency = $(1/(2\pi))\sqrt{(T_s/J)} = 0.259 \text{ Hz}$.

Phasor



Question 160

Original: 4A | EM-II ELE 2202 End Sem. 23 April 2018

With a neat diagram, explain an experimental procedure to determine the X_d and X_q of a salient pole alternator. (04)

Answer 160

The slip test is used to determine the direct-axis and quadrature-axis synchronous reactances of a salient-pole alternator. The field winding is kept open and a low, balanced three-phase voltage is applied to the stator winding through meters. The rotor is driven slowly at a speed close to synchronous speed, so the stator mmf slowly slips past the rotor poles.

Because the rotor is salient, the reluctance seen by the armature mmf changes with rotor position. When the armature mmf is along the direct axis, the air-gap reluctance is minimum, reactance is maximum, and armature current is minimum. When the mmf is along the quadrature axis, reactance is minimum and armature current is maximum.

Record the maximum and minimum line current and the corresponding line voltage. For a star-connected stator, convert line voltage to phase voltage. Then:

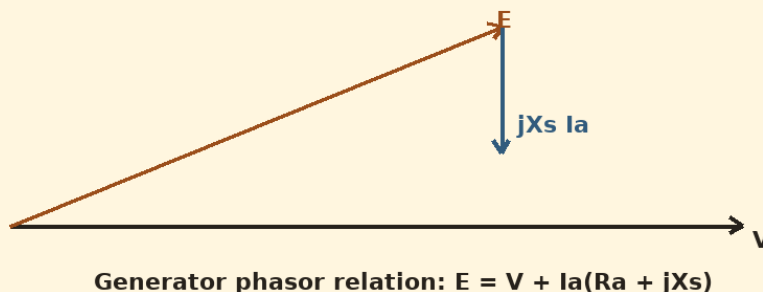
$$X_d = V_{\text{max,phase}} / I_{\text{min}}$$

$$X_q = V_{\text{min,phase}} / I_{\text{max}}$$

The applied voltage must be low to avoid excessive current, and the field winding is left open so that no DC excitation fixes the rotor field.

Final: Slip test gives X_d from the high-reactance/minimum-current position and X_q from the low-reactance/maximum-current position.

Phasor



Question 161

Original: 3C | EM-II ELE 2202 End Sem. 24 April 2017

Draw the connection diagram of slip test and briefly explain its significance. (03)

Answer 161

The slip-test connection has the stator connected to a low-voltage three-phase AC supply through meters, while the rotor field circuit is left open. The rotor is driven slowly near synchronous speed by an auxiliary motor or prime mover.

Because the rotor is salient, the magnetic reluctance seen by the armature mmf changes as the rotor position changes. When the armature mmf is along the direct axis, the air gap is effectively smaller, reactance is high, and current is minimum. When the mmf is along the quadrature axis, reactance is lower and current is maximum.

The direct-axis and quadrature-axis reactances are found from $X_d = V_{\text{ph}}/I_{\text{min}}$ and $X_q = V_{\text{ph}}/I_{\text{max}}$. If voltage varies during the test, use the simultaneous voltage and current readings.

The significance is that X_d and X_q are needed for Blondel two-reaction analysis, salient-pole phasor diagrams, power-angle characteristics, voltage regulation, and synchronizing power calculations.

Final: Slip test experimentally separates X_d and X_q by using the natural current pulsation caused by saliency.

Slip Test



$$I_{\min} \rightarrow X_d = V/I_{\min}; I_{\max} \rightarrow X_q = V/I_{\max}$$

Question 162

Original: 4A | EM-II ELE 2202 End Sem, 24 April 2017

A three-phase, 20 MVA, 11 kV, 50 Hz star-connected alternator has $X_d = 4 \Omega$ and $X_q = 3 \Omega$. Armature resistance is negligibly small. At full load, 0.8 lagging power factor, determine: a) Direct and quadrature axes components of the armature current. b) Excitation emf. c) Voltage regulation. d) Electromagnetic power. e) Reluctance power. (07)

Answer 162

Use Blondel two-reaction theory: resolve armature current into direct-axis and quadrature-axis components and use X_d and X_q separately.

Step 1: the rated current is obtained from 1,049.7 A, $V_{ph} = 6,350.9$ V, $\varphi = 36.87^\circ$.

Step 2: $\tan \delta = I_a X_q \cos \varphi / (V + I_a X_q \sin \varphi)$, so $\delta = 17.000^\circ$.

Step 3: $I_d = I_a \sin(\varphi + \delta) = 847.85$ A; $I_q = I_a \cos(\varphi + \delta) = 618.94$ A.

Step 4: Excitation emf $E = V \cos \delta + X_d I_d = 9,464.7$ V/phase.

Step 5: the percentage regulation = 49.031 %.

Step 6: Electromagnetic power = 16.000 MW; reluctance power = 2.8193 MW.

Final: $I_d = I_a \sin(\varphi + \delta) = 847.85$ A; $I_q = I_a \cos(\varphi + \delta) = 618.94$ A.; Excitation emf $E = V \cos \delta + X_d I_d = 9,464.7$ V/phase.; the percentage regulation = 49.031 %.; Electromagnetic power = 16.000 MW; reluctance power = 2.8193 MW.

Question 163

Original: 5A | EM-II ELE 2202 End Sem, 29 April 2019

What is the significance of reluctance power in salient pole synchronous machines? Discuss with power-angle characteristics. (03)

Answer 163

Reluctance power is the additional power component developed in a salient-pole synchronous machine because the direct-axis and quadrature-axis reluctances are unequal. Since X_d and X_q are different, the rotor tends to align itself in the position of minimum reluctance even apart from the excitation-emf torque component.

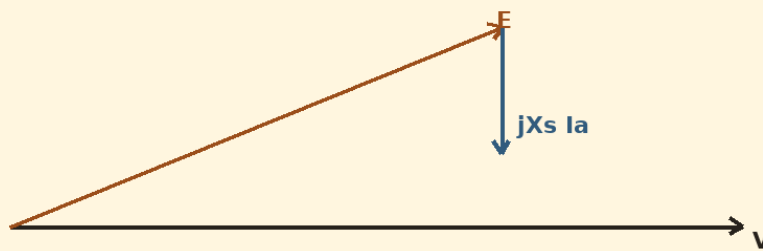
For a salient-pole generator neglecting resistance, the power-angle equation is $P = (3EV/X_d) \sin \delta + (3V^2/2)(1/X_q - 1/X_d) \sin 2\delta$.

The first term is the excitation power, also present in cylindrical-rotor machines. The second term is reluctance power. It depends on saliency and exists even when field excitation is reduced, provided X_d and X_q are unequal.

In the power-angle curve, reluctance power adds a second-harmonic component proportional to $\sin 2\delta$. It changes the maximum power point and improves the steady-state power capability of salient-pole machines.

Final: Reluctance power is the saliency-produced component of power; it is zero for cylindrical rotors where $X_d = X_q$.

Phasor



Generator phasor relation: $E = V + I_a(R_a + jX_s)$

Question 164

Original: 4A | EM-II ELE 2202 Makeup, 02 July 2016

A 10 kVA, 3 phase, 400 V, 50 Hz, star connected salient pole alternator has direct and quadrature axes reactance of 5 Ω and 2 Ω respectively. When it is delivering full load at 0.8 p.f lag to an infinite bus, determine the electromagnetic power and reluctance power.
5M

Answer 164

Use Blondel two-reaction theory: resolve armature current into direct-axis and quadrature-axis components and use X_d and X_q separately.

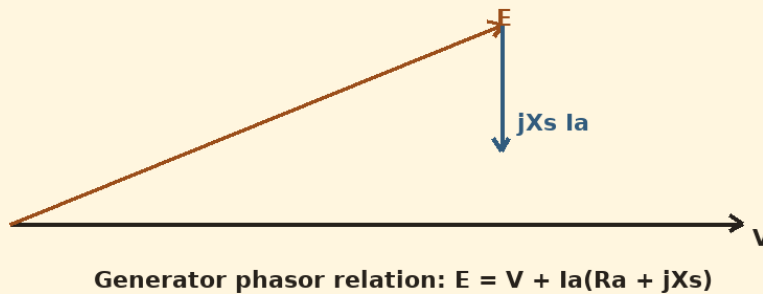
Step 1: the rated current is obtained from 14.434 A; $V_{ph} = 230.94$ V; $\delta = 5.3145^\circ$.

Step 2: $I_d = 9.6925$ A, $I_q = 10.695$ A; $E = 278.41$ V/phase.

Step 3: Electromagnetic power = 8.0000 kW; reluctance power = 4.4268 kW.

Final: The rated current is obtained from 14.434 A; $V_{ph} = 230.94$ V; $\delta = 5.3145^\circ$; $I_d = 9.6925$ A, $I_q = 10.695$ A; $E = 278.41$ V/phase.; Electromagnetic power = 8.0000 kW; reluctance power = 4.4268 kW.

Phasor



Question 165

Original: 4A | EM-II ELE 2202 Makeup, 13 June 2019

Draw and explain the phasor diagram of salient pole alternator based on Blondel's two reaction theory. (05)

Answer 165

Blondel two-reaction theory is used for salient-pole alternators because the air gap is non-uniform. The direct-axis reluctance is lower than the quadrature-axis reluctance, so X_d is greater than X_q . Hence armature reaction must be resolved into direct-axis and quadrature-axis components.

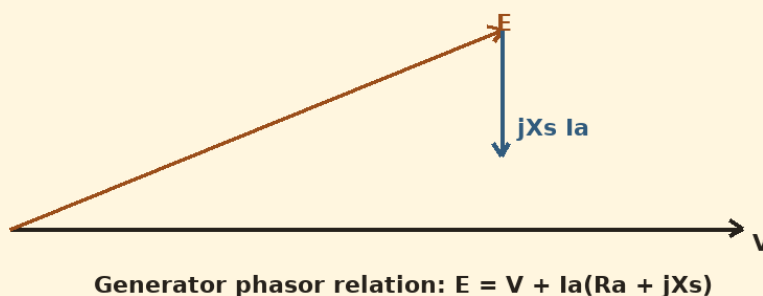
Take terminal voltage V as reference and draw armature current I_a at the load power-factor angle. The excitation emf E lies along the direct axis. Resolve I_a into I_d along the direct axis and I_q along the quadrature axis.

Neglecting armature resistance, the phasor equation is $E = V + jX_d I_d + jX_q I_q$. The q-axis current component produces a voltage drop $jX_q I_q$, and the d-axis component produces $jX_d I_d$. Adding these components vectorially to V gives the excitation emf E .

For a lagging load, the current lags V and the direct-axis component is demagnetizing. For a leading load, the direct-axis component may be magnetizing. The phasor diagram therefore explains both voltage regulation and the extra reluctance power term in salient-pole machines.

Final: The salient-pole phasor diagram differs from the cylindrical-rotor diagram by resolving I_a into I_d and I_q and using two reactances, X_d and X_q .

Phasor



Question 166

Original: 5A | EM-II ELE 2202 Makeup, 13 June 2019

What is meant by reluctance power in salient pole synchronous machines? Discuss with power-angle characteristics. (03)

Answer 166

Reluctance power is the additional power component developed in a salient-pole synchronous machine because the direct-axis and quadrature-axis reluctances are unequal. Since X_d and X_q are different, the rotor tends to align itself in the position of minimum reluctance even apart from the excitation-emf torque component.

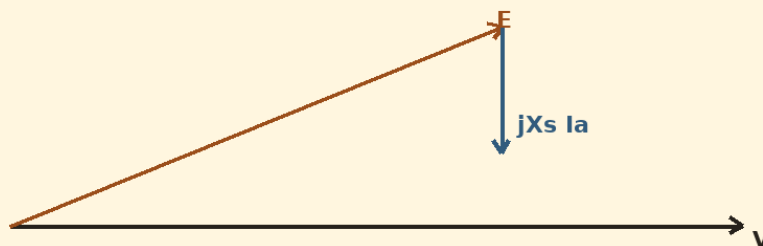
For a salient-pole generator neglecting resistance, the power-angle equation is $P = (3EV/X_d) \sin \delta + (3V^2/2)(1/X_q - 1/X_d) \sin 2\delta$.

The first term is the excitation power, also present in cylindrical-rotor machines. The second term is reluctance power. It depends on saliency and exists even when field excitation is reduced, provided X_d and X_q are unequal.

In the power-angle curve, reluctance power adds a second-harmonic component proportional to $\sin 2\delta$. It changes the maximum power point and improves the steady-state power capability of salient-pole machines.

Final: Reluctance power is the saliency-produced component of power; it is zero for cylindrical rotors where $X_d = X_q$.

Phasor



$$\text{Generator phasor relation: } E = V + I_a(R_a + jX_s)$$

Question 167

Original: 4B | EM-II ELE 2202 Makeup, 16 June 2017

Draw and explain the phasor diagram of salient pole alternator based on Blondel's two reaction theory. (03)

Answer 167

Blondel two-reaction theory is used for salient-pole alternators because the air gap is non-uniform. The direct-axis reluctance is lower than the quadrature-axis reluctance, so X_d is greater than X_q . Hence armature reaction must be resolved into direct-axis and quadrature-axis components.

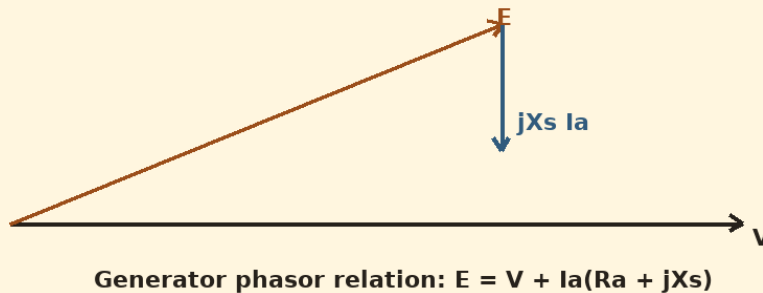
Take terminal voltage V as reference and draw armature current I_a at the load power-factor angle. The excitation emf E lies along the direct axis. Resolve I_a into I_d along the direct axis and I_q along the quadrature axis.

Neglecting armature resistance, the phasor equation is $E = V + jX_d I_d + jX_q I_q$. The q-axis current component produces a voltage drop $jX_q I_q$, and the d-axis component produces $jX_d I_d$. Adding these components vectorially to V gives the excitation emf E .

For a lagging load, the current lags V and the direct-axis component is demagnetizing. For a leading load, the direct-axis component may be magnetizing. The phasor diagram therefore explains both voltage regulation and the extra reluctance power term in salient-pole machines.

Final: The salient-pole phasor diagram differs from the cylindrical-rotor diagram by resolving I_a into I_d and I_q and using two reactances, X_d and X_q .

Phasor



Question 168

Original: 5A | EM-II ELE 2202 Makeup, 19 June 2018

Based on Blondel's two reaction theory, develop the phasor diagram of a salient pole synchronous generator. (04)

Answer 168

Use Blondel two-reaction theory: resolve armature current into direct-axis and quadrature-axis components and use X_d and X_q separately.

Blondel two-reaction theory resolves armature reaction in a salient-pole machine into two perpendicular components: direct-axis component and quadrature-axis component. Salient-pole generator: I_a is resolved into direct-axis I_d and quadrature-axis I_q components.

Step 1: the direct axis is along the rotor field pole axis and has reactance X_d .

Step 2: the quadrature axis is 90 electrical degrees away and has reactance X_q .

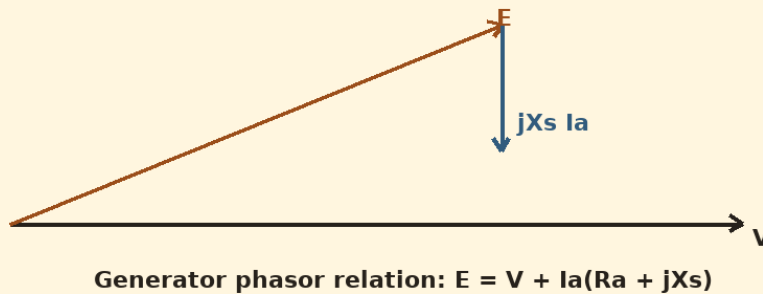
Step 3: Armature current I_a is resolved into I_d and I_q .

Step 4: using the generator phasor convention neglecting R_a , the phasor relation is $E_f = V + jX_d I_d + jX_q I_q$.

Step 5: This theory gives more accurate analysis than using a single synchronous reactance for salient-pole machines.

Final: The quadrature axis is 90 electrical degrees away and has reactance X_q .; This theory gives more accurate analysis than using a single synchronous reactance for salient-pole machines.

Phasor



Question 169

Original: 2C | EM-II ELE 2225 End Sem, 09 May 2024

A 5 MVA slow speed 3 phase synchronous generator rated at 11 kV has 32 poles. Its direct and quadrature axis reactances are 10 Ω and 4 Ω respectively. Neglecting armature resistance, determine the voltage regulation when supplying a rated load at 0.8 p.f lagging. 3

Answer 169

Use Blondel two-reaction theory: resolve armature current into direct-axis and quadrature-axis components and use X_d and X_q separately.

Step 1: the rated current is obtained from 262.43 A; $V_{ph} = 6,350.9$ V.

Step 2: $\delta = 6.8598^\circ$, $I_d = 181.41$ A, $I_q = 189.64$ A.

Step 3: $E = V \cos \delta + X_d I_d = 8,119.5$ V/phase.

Step 4: the percentage regulation = 27.848 %.

Final: The rated current is obtained from 262.43 A; $V_{ph} = 6,350.9$ V.; $\delta = 6.8598^\circ$, $I_d = 181.41$ A, $I_q = 189.64$ A.; $E = V \cos \delta + X_d I_d = 8,119.5$ V/phase.; the percentage regulation = 27.848 %.

Question 170

Original: 3A | EM-II ELE 2225 End Sem, 09 May 2024

Differentiate the power-angle characteristics of cylindrical rotor and salient pole synchronous generator. 3

Answer 170

For a cylindrical-rotor synchronous generator, the air gap is uniform and $X_d = X_q = X_s$. Neglecting resistance, the power-angle relation is $P = (3EV/X_s) \sin \delta$. The curve is a simple sine wave, and maximum power occurs at $\delta = 90^\circ$.

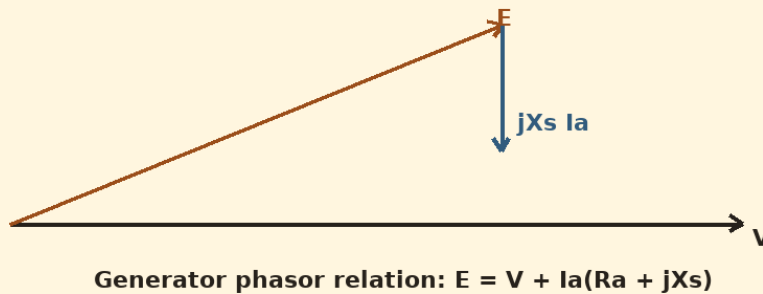
For a salient-pole generator, the air gap is non-uniform and $X_d > X_q$. The power-angle equation becomes $P = (3EV/X_d) \sin \delta + (3V^2/2)(1/X_q - 1/X_d) \sin 2\delta$.

The first term is excitation power and the second term is reluctance power due to saliency. Because of the $\sin 2\delta$ term, the power-angle curve is not a pure sine wave, and the maximum power generally occurs at an angle less than 90° .

The salient-pole machine therefore has additional torque-producing capability from reluctance variation, while the cylindrical machine depends only on excitation emf interaction with terminal voltage.

Final: Cylindrical rotor: single sine power-angle curve; salient pole: sine plus reluctance-power term.

Phasor



Question 171

Original: 1C | EM-II ELE 2225 Makeup, 26 June 2024

A three-phase, 20 MVA, 11 kV, 50 Hz star-connected alternator has $X_d = 4 \, \Omega$ and $X_q = 3 \, \Omega$. Armature resistance is negligibly small. At full load, 0.8 lagging power factor, determine: (a) Direct and quadrature axes components of the armature current. (b) Excitation emf. 4

Answer 171

Use Blondel two-reaction theory: resolve armature current into direct-axis and quadrature-axis components and use X_d and X_q separately.

Step 1: the rated current is obtained from 1,049.7 A, $V_{ph} = 6,350.9 \, \text{V}$, $\phi = 36.87^\circ$.

Step 2: $\tan \delta = I_a X_q \cos \phi / (V + I_a X_q \sin \phi)$, so $\delta = 17.000^\circ$.

Step 3: $I_d = I_a \sin(\phi + \delta) = 847.85 \, \text{A}$; $I_q = I_a \cos(\phi + \delta) = 618.94 \, \text{A}$.

Step 4: Excitation emf $E = V \cos \delta + X_d I_d = 9,464.7 \, \text{V/phase}$.

Step 5: the percentage regulation = 49.031 %.

Step 6: Electromagnetic power = 16.000 MW; reluctance power = 2.8193 MW.

Final: $I_d = I_a \sin(\phi + \delta) = 847.85 \, \text{A}$; $I_q = I_a \cos(\phi + \delta) = 618.94 \, \text{A}$; Excitation emf $E = V \cos \delta + X_d I_d = 9,464.7 \, \text{V/phase}$; the percentage regulation = 49.031 %; Electromagnetic power = 16.000 MW; reluctance power = 2.8193 MW.

Question 172

Original: 2B | EM-II ELE 2225 Makeup, 26 June 2024

What is the significance of synchronizing power coefficient for a non-salient pole alternator? 2

Answer 172

Use Blondel two-reaction theory: resolve armature current into direct-axis and quadrature-axis components and use X_d and X_q separately.

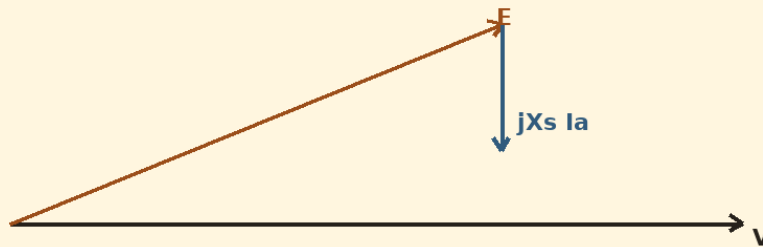
Step 1: the synchronising power coefficient denotes the incremental restoring power developed for one electrical radian change in load angle. It measures the stiffness of synchronism between an alternator and the bus. $\text{Psy} = dP/d(\delta)$ (W per electrical radian) For cylindrical rotor with R_a neglected: $P = (3EV/X_s) \sin(\delta)$ $\text{Psy} = (3EV/X_s) \cos(\delta)$ - Large Psy means a small angular displacement produces strong restoring power, so the machine is more stable.

Step 2: Small Psy means weak synchronising torque and greater tendency to hunt or lose synchronism.

Step 3: at δ near 90° for a cylindrical rotor, P_{sy} approaches zero; this is the steady-state stability limit.

Final: At δ near 90° for a cylindrical rotor, P_{sy} approaches zero; this is the steady-state stability limit.

Phasor



Generator phasor relation: $E = V + I_a(R_a + jX_s)$

Question 173

Original: 3A | EM-II ELE 2251 End Sem. 31 May 2023

Draw the per phase equivalent circuit and write the expressions relating air-gap voltage, terminal voltage, and excitation emf in an alternator. Compare the steady state mathematical model of salient-pole rotor and non-salient-pole rotor alternators with the help of phasor diagrams relating the induced voltage and terminal voltages. Also, suggest suitable methods to determine the reactance of these machines. (05)

Answer 173

For a non-salient cylindrical alternator, the per-phase model is $E = V + I_a R_a + jX_s I_a$. If R_a is neglected, $E = V + jX_s I_a$. The air-gap voltage is the voltage behind leakage reactance: $E_g = V + I_a R_a + jX_{l1} I_a$, and the excitation emf is $E = E_g + \text{armature-reaction reactance drop}$.

For a salient-pole alternator, armature reaction is resolved along direct and quadrature axes. Neglecting resistance, $E = V + jX_d I_d + jX_q I_q$, where I_d is the direct-axis current component and I_q is the quadrature-axis component. Since $X_d > X_q$, the voltage drop depends on rotor position and load angle.

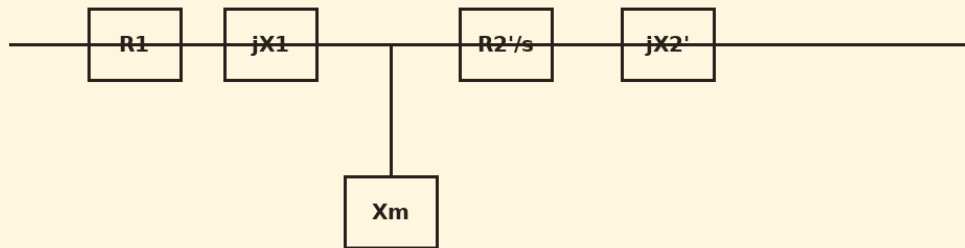
In the phasor diagram, take V as reference, draw I at the load power-factor angle, resolve I into I_d and I_q with respect to the rotor d-q axes, add $jX_q I_q$ and $jX_d I_d$ components, and locate E along the direct axis. For a cylindrical rotor, there is no need to split I because $X_d = X_q = X_s$.

Reactance determination: non-salient synchronous reactance is obtained from OCC and SCC by the EMF method. Salient-pole reactances X_d and X_q are determined by slip test. Leakage reactance and armature reaction separation can be obtained from ZPF/Potier method.

Final: Cylindrical rotor model uses one synchronous reactance X_s ; salient-pole model uses two reactances X_d and X_q with current resolved along d and q axes.

Equivalent Circuit

Approximate per-phase induction motor circuit



Question 174

Original: 4B | EM-II ELE 2251 End Sem, 31 May 2023

A 5 MVA, 6.6 kV, 3-phase, 50 Hz, star-connected salient pole alternator with direct and quadrature axes reactance of 12.5 Ω and 10.5 Ω respectively is connected to an infinite bus. When the generator is operating at its rated MVA and power factor of 0.9 lagging, calculate the generated EMF per phase & the synchronizing power per mechanical degree of displacement. (04)

Answer 174

Use Blondel two-reaction theory: resolve armature current into direct-axis and quadrature-axis components and use X_d and X_q separately.

Step 1: the rated current is obtained from 437.39 A; $V_{ph} = 3,810.5$ V.

Step 2: using $\tan \delta = I_a X_q \cos \phi / (V + I_a X_q \sin \phi)$, $\delta = 35.418^\circ$.

Step 3: $I_d = 383.50$ A, $I_q = 210.31$ A; $E = 7,899.2$ V/phase.

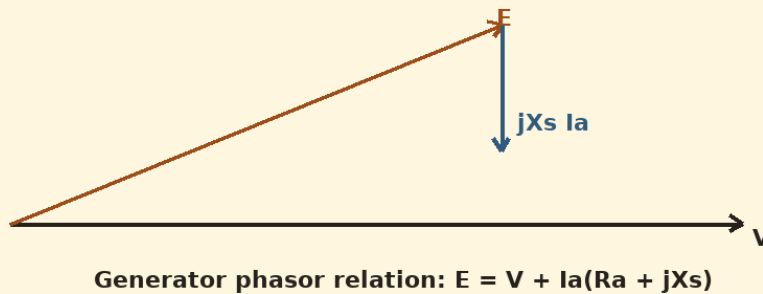
Step 4: Synchronizing power coefficient for salient pole machine is $dP/d(\delta) = 3EV/X_d \cos(\delta) + 3V^2(X_d - X_q)/(X_d X_q) \cos(2\delta)$.

Step 5: This gives 106,554 W per electrical degree.

Step 6: the question asks per mechanical degree but does not specify number of poles; multiply the electrical-degree value by $P/2$ when pole number P is known.

Final: This gives 106,554 W per electrical degree.; the question asks per mechanical degree but does not specify number of poles; multiply the electrical-degree value by $P/2$ when pole number P is known.

Phasor



Question 175

Original: 4C-trim | EM-II ELE 2251 End Sem, 31 May 2023

Differentiate the terms reluctance power and synchronizing power in the case of a synchronous generator. (part a only, 02)

Answer 175

Reluctance power is a component of electromagnetic power in a salient-pole synchronous generator. It arises because X_d and X_q are unequal, so the rotor develops torque by tending to align with the minimum-reluctance magnetic path. It is represented by the $\sin 2\delta$ term in the salient-pole power-angle equation.

Synchronizing power is different. It is the change in electrical power for a small change in load angle, i.e. $dP/d\delta$. It represents the stiffness of synchronism and determines how strongly the machine is pulled back after a small angular disturbance.

Reluctance power contributes to the total power output. Synchronizing power is the slope of the power-angle curve at the operating point. A high synchronizing power coefficient means better small-signal stability; near the maximum power point the synchronizing power becomes small.

Final: Reluctance power is a saliency power component; synchronizing power is the restoring-power coefficient $dP/d\delta$.

Question 176

Original: 3B | EM-II ELE 2251 Grade Improvement, 11 August 2021

Explain the significance of slip test in Synchronous machines. (04)

Answer 176

Use Blondel two-reaction theory: resolve armature current into direct-axis and quadrature-axis components and use X_d and X_q separately.

Step 1: Slip test is used to determine direct-axis reactance X_d and quadrature-axis reactance X_q of a salient-pole synchronous machine. Slip test connection for salient-pole synchronous machine Rotor is driven slowly near synchronous speed with field open. Maximum stator current $\rightarrow$ q-axis; minimum current $\rightarrow$ d-axis. $X = V/I$ at each position.

Step 2: the field winding is kept open-circuited.

Step 3: A low 3-phase voltage is applied to the stator.

Step 4: the rotor is driven slowly at a speed slightly different from synchronous speed.

Step 5: Because the rotor poles slowly slip past the stator mmf, stator current and voltage fluctuate between d-axis and q-axis positions.

Step 6: When stator current is minimum, reactance is maximum $\rightarrow X_d = V/I_{min}$.

Step 7: When stator current is maximum, reactance is minimum $\rightarrow X_q = V/I_{max}$. The test is significant because X_d and X_q are required for salient-pole phasor diagrams, voltage regulation and power-angle calculations.

Final: When stator current is minimum, reactance is maximum $\rightarrow X_d = V/I_{min}$.; When stator current is maximum, reactance is minimum $\rightarrow X_q = V/I_{max}$. The test is significant because X_d and X_q are required for salient-pole phasor diagrams, voltage regulation and power-angle calculations.

Slip Test



$I_{min} \rightarrow X_d = V/I_{min}$; $I_{max} \rightarrow X_q = V/I_{max}$

12. Field excitation effects, synchronous motor operation, power-angle, V-curves, starting and hunting

Syllabus area: Week 12

Question 177

Original: 4A | EM-II ELE 204 End Sem. 13 May 2014

A 3 phase, 20 MVA, star connected alternator with an impedance of $(0.5 + j6)$ per phase is operating in parallel with constant voltage 11 kV bus bars. The field current is adjusted to give a line excitation voltage of 12 kV. With constant excitation, calculate (i) maximum power output from the alternator (ii) armature current and power factor under maximum power condition. (04)

Answer 177

Use the synchronous-motor excitation or power-angle relation and separate real-power balance from reactive-power balance.

Step 1: $V = 11 / \sqrt{3} = 6.351 \text{ kV / phase}$; $E = 12 / \sqrt{3} = 6.928 \text{ kV / phase}$; $Z = 0.5 + j6 \Omega$, $|Z| = 6.021 \Omega$.

Step 2: for nonzero resistance, maximum output occurs at $\delta = \theta = \tan^{-1}(X / R) = 85.24^\circ$.

Step 3: $P_{\max} = 3EV / |Z| - 3V^2 / |Z| \cos \theta = 20.26 \text{ MW}$.

Step 4: at maximum power, $I = (E \text{ at } \delta - V) / Z = 1495 \text{ A}$.

Step 5: the operating power factor is 0.711 lagging.

Final: For nonzero resistance, maximum output occurs at $\delta = \theta = \tan^{-1}(X / R) = 85.24^\circ$; $P_{\max} = 3EV / |Z| - 3V^2 / |Z| \cos \theta = 20.26 \text{ MW}$; at maximum power, $I = (E \text{ at } \delta - V) / Z = 1495 \text{ A}$; the operating power factor is 0.711 lagging.

Question 178

Original: 5A | EM-II ELE 204 End Sem. 13 May 2014

A 3 phase, 50 Hz, 4 pole, 10 kVA, 400 V, star connected synchronous motor has a per phase synchronous impedance of $(3 + j4)$. When supplying rated current, the field current is so adjusted that the line emf induced is 451 V. Determine (a) load angle and power factor (b) active power input and gross power output (c) efficiency assuming rotational loss of 1 kW. (05)

Answer 178

Use the synchronous-motor excitation or power-angle relation and separate real-power balance from reactive-power balance.

Step 1: the rated current is obtained from $10 \text{ kVA} / (\sqrt{3} \times 400 \text{ V}) = 14.43 \text{ A}$. The phase emf = $451 / \sqrt{3} = 260.4 \text{ V}$.

Step 2: for motor operation, $I = (V - E \text{ at } -\delta) / (3 + j4)$. Solving $|I| = 14.43 \text{ A}$ gives $\delta = 15.44^\circ$.

Step 3: Current leads the terminal voltage by 53.0° , so pf = 0.602 leading.

Step 4: Active input = $\sqrt{3} V I \text{ pf} = 6.019 \text{ kW}$.

Step 5: Armature copper loss = $3I^2R = 1.875 \text{ kW}$. Gross mechanical output = 4.144 kW .

Step 6: After 1 kW rotational loss, shaft output = 3.144 kW ; efficiency = 52.2%.

Final: Current leads the terminal voltage by 53.0° , so pf = 0.602 leading; Active input = $\sqrt{3} V I \text{ pf} = 6.019 \text{ kW}$; Armature copper loss = $3I^2R = 1.875 \text{ kW}$. Gross mechanical output = 4.144 kW ; After 1 kW rotational loss, shaft output = 3.144 kW ; efficiency = 52.2%.

Question 179

Original: 5B | EM-II ELE 204 End Sem. 13 May 2014

A supply system has the following loads: 100 kW upf load and 400 kW load at 0.6 pf lag. A 50 HP synchronous motor when connected in parallel with existing load improves the overall power factor to 0.95 lag. Assuming motor efficiency as 87%, determine (a) leading kVAR supplied by the motor (b) kVA rating of motor (c) motor power factor. (03)

Answer 179

Use the synchronous-motor excitation or power-angle relation and separate real-power balance from reactive-power balance.

Step 1: Existing load: $P = 500.00 \text{ kW}$, $Q = 533.33 \text{ kVAR}$ lagging.

Step 2: Motor input active power = $50 \text{ HP} \times 0.746/0.87 = 42.874 \text{ kW}$. Final total $P = 542.87 \text{ kW}$; final Q at 0.95 lagging = 178.43 kVAR .

Step 3: Motor reactive power = -354.90 kVAR , so it supplies 354.90 kVAR leading.

Step 4: the required synchronous-motor rating is 357.48 kVA ; motor pf = 0.11993 leading.

Final: Existing load: $P = 500.00 \text{ kW}$, $Q = 533.33 \text{ kVAR}$ lagging.; Motor input active power = $50 \text{ HP} \times 0.746/0.87 = 42.874 \text{ kW}$. Final total $P = 542.87 \text{ kW}$; final Q at 0.95 lagging = 178.43 kVAR .; Motor reactive power = -354.90 kVAR , so it supplies 354.90 kVAR leading.; the required synchronous-motor rating is 357.48 kVA ; motor pf = 0.11993 leading.

Question 180

Original: 5C | EM-II ELE 204 End Sem. 13 May 2014

Mention any two applications of damper winding. (02)

Answer 180

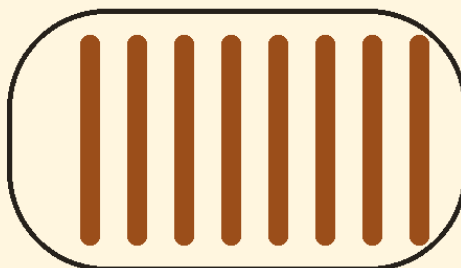
Damper windings are short-circuited copper or aluminium bars embedded in the pole faces of a salient-pole synchronous machine and connected by end rings. They act like a squirrel-cage winding.

Two important applications are: starting of synchronous motors and damping of hunting oscillations. During starting, the damper winding develops induction-motor torque and accelerates the rotor close to synchronous speed before DC field excitation is applied. During hunting, relative motion between rotor and stator field induces currents in the damper bars, producing damping torque that opposes oscillation. Damper windings also help reduce unbalanced-load oscillations and improve stability during sudden load changes.

Final: Damper windings are used mainly for induction-motor starting of synchronous motors and for damping hunting.

Damper

Pole face with damper bars



Short-circuited bars provide starting torque and damping

Question 181

Original: 4C | EM-II ELE 204 Makeup, 08 July 2014

Why synchronous motors are not self-starting? Briefly explain the starting method using an auxiliary induction motor. (03)

Answer 181

Use the synchronous-motor excitation or power-angle relation and separate real-power balance from reactive-power balance.

A synchronous motor is not self-starting because the stator produces a rotating magnetic field at synchronous speed immediately after the 3-phase supply is applied, while the rotor is initially stationary.

Step 1: During one half-cycle the rotor experiences torque in one direction; during the next half-cycle the torque reverses.

Step 2: Because of rotor inertia, the rotor cannot accelerate instantly to synchronous speed.

Step 3: the average starting torque over a cycle is zero, so the rotor only vibrates if no starting method is used. With an auxiliary induction motor/pony motor, the rotor is brought close to synchronous speed first. Then DC field excitation is applied; the rotor locks with the stator rotating field and continues at synchronous speed.

Final: Then DC field excitation is applied; the rotor locks with the stator rotating field and continues at synchronous speed.

Question 182

Original: 5B | EM-II ELE 204 Makeup, 08 July 2014

A 3 phase, 50 Hz, 400 V installation draws a current of 36.1 A at 0.8 power factor lagging. A 3 phase 10 KVA synchronous motor operating at 0.6 power factor leading is connected in parallel with the existing load. Determine (i) the overall power factor of the installation, (ii) net reactive power taken from the supply. (03)

Answer 182

Use the synchronous-motor excitation or power-angle relation and separate real-power balance from reactive-power balance.

Step 1: Existing load: $P_1 = \sqrt{3} \times 400 \times 36.1 \times 0.8 = 20.01 \text{ kW}$; $Q_1 = 15.01 \text{ kVAR}$ lagging.

Step 2: Synchronous motor: $S = 10 \text{ kVA}$ at 0.6 leading $\rightarrow P_2 = 6 \text{ kW}$ and $Q_2 = -8 \text{ kVAR}$.

Step 3: Total $P = 26.01 \text{ kW}$ and total $Q = 7.01 \text{ kVAR}$.

Step 4: Overall $kVA = 26.94 \text{ kVA}$ and overall $pf = 26.01/26.94 = 0.966$ lagging.

Step 5: Net reactive power taken from supply = 7.01 kVAR lagging.

Final: Synchronous motor: $S = 10 \text{ kVA}$ at 0.6 leading $\rightarrow P_2 = 6 \text{ kW}$ and $Q_2 = -8 \text{ kVAR}$.; Total $P = 26.01 \text{ kW}$ and total $Q = 7.01 \text{ kVAR}$.; Overall $kVA = 26.94 \text{ kVA}$ and overall $pf = 26.01/26.94 = 0.966$ lagging.; Net reactive power taken from supply = 7.01 kVAR lagging.

Question 183

Original: 4C | EM-II ELE 204 Makeup, 09 July 2015

"Synchronous motor is not self-starting". Give reasons. (02)

Answer 183

Use the synchronous-motor excitation or power-angle relation and separate real-power balance from reactive-power balance.

A synchronous motor is not self-starting because the stator produces a rotating magnetic field at synchronous speed immediately after the 3-phase supply is applied, while the rotor is initially stationary.

Step 1: During one half-cycle the rotor experiences torque in one direction; during the next half-cycle the torque reverses.

Step 2: Because of rotor inertia, the rotor cannot accelerate instantly to synchronous speed.

Step 3: the average starting torque over a cycle is zero, so the rotor only vibrates if no starting method is used. Starting method for synchronous motor With an auxiliary induction motor/pony motor, the rotor is brought close to synchronous speed first. Then DC field excitation is applied; the rotor locks with the stator rotating field and continues at synchronous speed.

Final: Then DC field excitation is applied; the rotor locks with the stator rotating field and continues at synchronous speed.

Question 184

Original: 5A | EM-II ELE 204 Makeup, 09 July 2015

A factory has an average load of 300 kW at a power factor of 0.6 lagging. A synchronous motor is used to raise the combined power factor to 0.9 lagging and at the same time to supply a mechanical load of 60 kW. Calculate KVA rating and power factor of synchronous motor. Assume 88% as the efficiency of synchronous motor (03)

Answer 184

Use the synchronous-motor excitation or power-angle relation and separate real-power balance from reactive-power balance.

Step 1: Initial load $Q = 400.00$ kVar lagging.

Step 2: Synchronous motor input $P = 60/0.88 = 68.182$ kW.

Step 3: Total P after adding motor = 368.18 kW.

Step 4: Desired total Q at 0.9 lagging = 178.32 kVar.

Step 5: the synchronous motor reactive power is -221.68 kVar -> motor supplies 221.68 kVar leading.

Step 6: the required synchronous-motor rating is 231.93 kVA; motor pf = 0.29398 leading.

Final: Total P after adding motor = 368.18 kW.; Desired total Q at 0.9 lagging = 178.32 kVar.; the synchronous motor reactive power is -221.68 kVar -> motor supplies 221.68 kVar leading.; the required synchronous-motor rating is 231.93 kVA; motor pf = 0.29398 leading.

Question 185

Original: 5C | EM-II ELE 204 Makeup, 09 July 2015

A 3 phase synchronous machine with P poles, moment of inertia J kg-m², and rotor speed ω_m rad/s has synchronising power coefficient PSY watts per electrical radian. Prove that the natural frequency of hunting is $f_n = (1/2\pi) \sqrt{(P PSY/(2 J \omega_m))}$. (03)

Answer 185

During hunting, a synchronous machine oscillates about its equilibrium load angle. For a small electrical displacement δ_e , the change in synchronizing power is approximately $P_{syn} \delta_e$, where P_{syn} is the synchronizing power coefficient in W per electrical radian.

Electrical angle and mechanical angle are related by $\delta_e = (P/2) \delta_m$. Therefore the incremental synchronizing power for a mechanical displacement δ_m is $P_{syn}(P/2)\delta_m$.

The corresponding synchronizing torque is incremental power divided by mechanical angular speed ω_m :

$$T_s = [P_{syn}(P/2)\delta_m]/\omega_m.$$

This is a restoring torque, so the rotor equation for small oscillation is $J d^2(\delta_m)/dt^2 + [P P_{syn}/(2 \omega_m)] \delta_m = 0$.

Comparing with the standard SHM equation $d^2x/dt^2 + \omega_n^2 x = 0$, $\omega_n^2 = P P_{syn}/(2 J \omega_m)$.

Therefore $f_n = (1/(2\pi)) \sqrt{(P P_{syn}/(2 J \omega_m))}$.

Final: The natural frequency of hunting oscillations is $f_n = \frac{1}{2\pi} \sqrt{\frac{P_{syn}}{2J\omega_m}}$.

Question 186

Original: 6B | EM-II ELE 204 Makeup, 09 July 2015

What is damper winding? Where is it located? Explain starting of synchronous motor using damper winding. (04)

Answer 186

Use the synchronous-motor excitation or power-angle relation and separate real-power balance from reactive-power balance.

Damper winding consists of short-circuited bars embedded in the pole faces of the rotor. During starting, the rotating stator field cuts these bars and induces rotor currents. Damper bars embedded in pole faces and short-circuited at ends - With DC field initially unexcited or shorted through resistance, the motor behaves like a squirrel-cage induction motor.

Step 1: the damper-bar currents produce induction torque and accelerate the rotor close to synchronous speed.

Step 2: Then DC excitation is applied to the field winding.

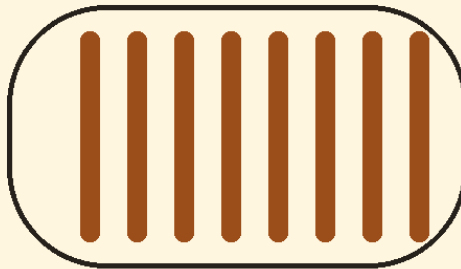
Step 3: the rotor locks with the rotating stator field and runs at synchronous speed.

Step 4: Once synchronized, relative speed between stator field and damper bars is nearly zero, so no damper current flows in steady state.

Final: During starting, the rotating stator field cuts these bars and induces rotor currents. Damper bars embedded in pole faces and short-circuited at ends - With DC field initially unexcited or shorted through resistance, the motor behaves like a squirrel-cage induction motor.

Damper

Pole face with damper bars



Short-circuited bars provide starting torque and damping

Question 187

Original: 3B | EM-II ELE 2202 End Sem, 10 May 2016

A turbo alternator with synchronizing reactance of 10Ω , is delivering 200A at up to an infinite bus at a voltage of 11kV. If the prime mover input is kept constant & excitation is increased by 20%, find the new current and power factor. Keeping the excitation at the increased level, find the maximum power output of the alternator and the corresponding current and power factor. (06)

Answer 187

Use the synchronous-motor excitation or power-angle relation and separate real-power balance from reactive-power balance.

Step 1: Initially $I = 200$ A at upf. $E = V + j10I = 6.658$ kV/phase at $\delta = 17.48^\circ$.

Step 2: Excitation increased by 20%: $E' = 7.990$ kV / phase. Prime mover input is constant, so $P = \sqrt{3} \times 11$ kV $\times 200 = 3.811$ MW.

Step 3: New load angle $= \sin^{-1}(PX / (3E'V)) = 14.50^\circ$.

Step 4: New current $= 243.3$ A at pf $= 0.822$ lagging.

Step 5: Keeping E' constant, maximum power $= 3E'V / X = 15.22$ MW.

Step 6: at maximum power, current $= 1020.7$ A at pf $= 0.783$ leading.

Final: New load angle $= \sin^{-1}(PX / (3E'V)) = 14.50^\circ$; New current $= 243.3$ A at pf $= 0.822$ lagging.; Keeping E' constant, maximum power $= 3E'V / X = 15.22$ MW.; at maximum power, current $= 1020.7$ A at pf $= 0.783$ leading.

Question 188

Original: 5A | EM-II ELE 2202 End Sem. 23 April 2018

The excitation of a 415 V, 3 phase star connected synchronous motor is such that the induced emf is 520 V. The impedance per phase is $0.5 + j4 \Omega$. The friction and iron losses are constant at 1,000 W. Calculate the horse power output, line current and efficiency for a) Maximum power output b) Maximum power input (06)

Answer 188

Use the synchronous-motor excitation or power-angle relation and separate real-power balance from reactive-power balance.

Step 1: Taking 520 V as line emf: The phase emf $= 520/\sqrt{3} = 300.2$ V, V phase $= 415/\sqrt{3} = 239.6$ V.

Step 2: for each load angle, $I = (V - E \text{ at } -\delta)/(0.5 + j4)$.

Step 3: Maximum shaft output occurs near $\delta = 82.88^\circ$: line current $= 89.34$ A, pf $= 0.891$, input $= 57.19$ kW.

Step 4: Gross mechanical power $=$ input - armature copper loss $= 45.21$ kW; shaft output after 1 kW loss $= 44.21$ kW $= 59.27$ HP; efficiency $= 77.3\%$.

Step 5: Maximum input occurs near $\delta = 97.13^\circ$: line current $= 100.88$ A, pf $= 0.811$, input $= 58.83$ kW.

Step 6: Corresponding shaft output $= 42.57$ kW $= 57.06$ HP; efficiency $= 72.4\%$.

Step 7: If 520 V is interpreted as per-phase emf, all numerical values increase; the line-emf interpretation is used here.

Final: Maximum input occurs near $\delta = 97.13^\circ$: line current $= 100.88$ A, pf $= 0.811$, input $= 58.83$ kW.; Corresponding shaft output $= 42.57$ kW $= 57.06$ HP; efficiency $= 72.4\%$.

Question 189

Original: 5B | EM-II ELE 2202 End Sem. 23 April 2018

What is hunting in synchronous motor? How is it mitigated? (04)

Answer 189

Use the synchronous-motor excitation or power-angle relation and separate real-power balance from reactive-power balance.

Hunting is the oscillation of the rotor of a synchronous motor about its new equilibrium load angle after a sudden change in load, excitation or supply condition.

Step 1: When load suddenly changes, the rotor cannot instantly take the new load angle due to inertia.

Step 2: It accelerates or decelerates around the equilibrium position, causing oscillations in torque, current and power angle.

Step 3: Severe hunting may cause loss of synchronism. Damper bars embedded in pole faces and short-circuited at ends Mitigation methods include damper/amortisseur windings, flywheel effect for

fluctuating loads, proper excitation control, avoiding sudden load changes, and using fast voltage regulators/power-system stabilizers.

Final: Hunting is the oscillation of the rotor of a synchronous motor about its new equilibrium load angle after a sudden change in load, excitation or supply condition.; When load suddenly changes, the rotor cannot instantly take the new load angle due to inertia.

Question 190

Original: 2B | EM-II ELE 2202 End Sem. 24 April 2017

What is hunting phenomenon in a synchronous motor? Explain the functions of Amortisseur windings. (03)

Answer 190

Use the synchronous-motor excitation or power-angle relation and separate real-power balance from reactive-power balance.

Hunting is the oscillation of the rotor of a synchronous motor about its new equilibrium load angle after a sudden change in load, excitation or supply condition.

Step 1: When load suddenly changes, the rotor cannot instantly take the new load angle due to inertia.

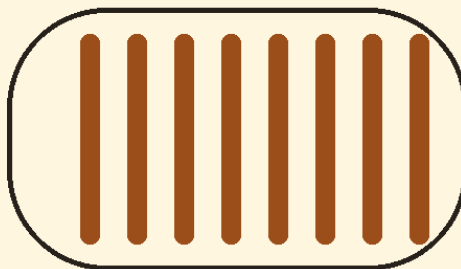
Step 2: It accelerates or decelerates around the equilibrium position, causing oscillations in torque, current and power angle.

Step 3: Severe hunting may cause loss of synchronism. Damper bars embedded in pole faces and short-circuited at ends Mitigation methods include damper/amortisseur windings, flywheel effect for fluctuating loads, proper excitation control, avoiding sudden load changes, and using fast voltage regulators/power-system stabilizers.

Final: Hunting is the oscillation of the rotor of a synchronous motor about its new equilibrium load angle after a sudden change in load, excitation or supply condition.; When load suddenly changes, the rotor cannot instantly take the new load angle due to inertia.

Damper

Pole face with damper bars



Short-circuited bars provide starting torque and damping

Question 191

Original: 5A | EM-II ELE 2202 End Sem. 24 April 2017

An industrial plant is supplied with 875 kVA of electrical power at 0.8 pf lagging from a three-phase, 50 Hz, 11 kV substation. A synchronous motor of rating 100 kVA operating at a leading power factor of 0.6 is added during the expansion. Calculate the new kVA supplied by the substation and the overall power factor of the plant. (03)

Answer 191

Use the synchronous-motor excitation or power-angle relation and separate real-power balance from reactive-power balance.

Step 1: Initial plant: $P_1 = 875 \times 0.8 = 700 \text{ kW}$; $Q_1 = 875 \times 0.6 = 525 \text{ kVAr}$.

Step 2: Synchronous motor: $P_2 = 100 \times 0.6 = 60 \text{ kW}$; $Q_2 = -100 \times 0.8 = -80 \text{ kVAr}$.

Step 3: New $P = 760 \text{ kW}$, $Q = 445 \text{ kVAr}$.

Step 4: New kVA supplied = $\sqrt{760^2 + 445^2} = 880.8 \text{ kVA}$.

Step 5: Overall pf = $760/880.8 = 0.863$ lagging.

Final: Synchronous motor: $P_2 = 100 \times 0.6 = 60 \text{ kW}$; $Q_2 = -100 \times 0.8 = -80 \text{ kVAr}$.; New $P = 760 \text{ kW}$, $Q = 445 \text{ kVAr}$.; New kVA supplied = $\sqrt{760^2 + 445^2} = 880.8 \text{ kVA}$.; Overall pf = $760/880.8 = 0.863$ lagging.

Question 192

Original: 3B | EM-II ELE 2202 End Sem. 29 April 2019

What is the role of damper winding in synchronous machines? Where is it placed in the machine? (03)

Answer 192

Use the synchronous-motor excitation or power-angle relation and separate real-power balance from reactive-power balance.

Damper windings are short-circuited copper/aluminium bars embedded in the pole faces of a salient-pole synchronous machine and connected by end rings, similar to a squirrel-cage winding.

Step 1: They provide starting torque in synchronous motors by allowing the motor to start as an induction motor.

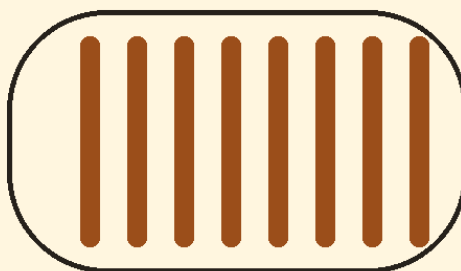
Step 2: They damp rotor oscillations or hunting by producing currents that oppose speed/angle oscillations.

Step 3: They improve stability during sudden load changes and faults.

Final: Damper windings are short-circuited copper/aluminium bars embedded in the pole faces of a salient-pole synchronous machine and connected by end rings, similar to a squirrel-cage winding.

Damper

Pole face with damper bars



Short-circuited bars provide starting torque and damping

Question 193

Original: 4B | EM-II ELE 2202 Makeup, 02 July 2016

Explain the different methods employed to start synchronous motor. 5M

Answer 193

Use the synchronous-motor excitation or power-angle relation and separate real-power balance from reactive-power balance.

Common starting methods for synchronous motors are used to bring the rotor near synchronous speed before applying/locking field excitation.

Step 1: Damper-winding starting: motor starts as an induction motor using amortisseur bars; DC field is applied near synchronous speed.

Step 2: Pony motor starting: an auxiliary motor drives the rotor close to synchronous speed; then the synchronous motor is synchronized to supply.

Step 3: Variable-frequency starting: a VFD starts at low frequency and gradually increases frequency while maintaining synchronism.

Step 4: Reduced-voltage methods may be used along with damper winding to limit starting current.

Final: Variable-frequency starting: a VFD starts at low frequency and gradually increases frequency while maintaining synchronism.

Question 194

Original: 5B | EM-II ELE 2202 Makeup, 02 July 2016

A factory has an average load of 250 kW at a power factor of 0.7 lagging. In addition, a synchronous motor with an efficiency of 88 % supplies a mechanical load of 60 kW. Determine the kVA rating and power factor of synchronous motor if it is used to raise the combined power factor to 0.9 lag. 5M

Answer 194

Use the synchronous-motor excitation or power-angle relation and separate real-power balance from reactive-power balance.

Step 1: Initial load $Q = 255.05$ kVar lagging.

Step 2: Motor input active power = $60/0.88 = 68.182$ kW. Final total $P = 318.18$ kW; required Q at 0.9 lagging = 154.10 kVar. - The synchronous motor reactive power is -100.95 kVar; it therefore supplies 100.95 kVar leading.

Step 3: the required synchronous-motor rating is 121.82 kVA; motor pf = 0.55971 leading.

Final: Initial load $Q = 255.05$ kVar lagging.; Motor input active power = $60/0.88 = 68.182$ kW. Final total $P = 318.18$ kW; required Q at 0.9 lagging = 154.10 kVar. - The synchronous motor reactive power is -100.95 kVar; it therefore supplies 100.95 kVar leading.; the required synchronous-motor rating is 121.82 kVA; motor pf = 0.55971 leading.

Question 195

Original: 5B | EM-II ELE 2202 Makeup, 13 June 2019

Draw and explain the significance of V curve in synchronous motors. (03)

Answer 195

Use the synchronous-motor excitation or power-angle relation and separate real-power balance from reactive-power balance.

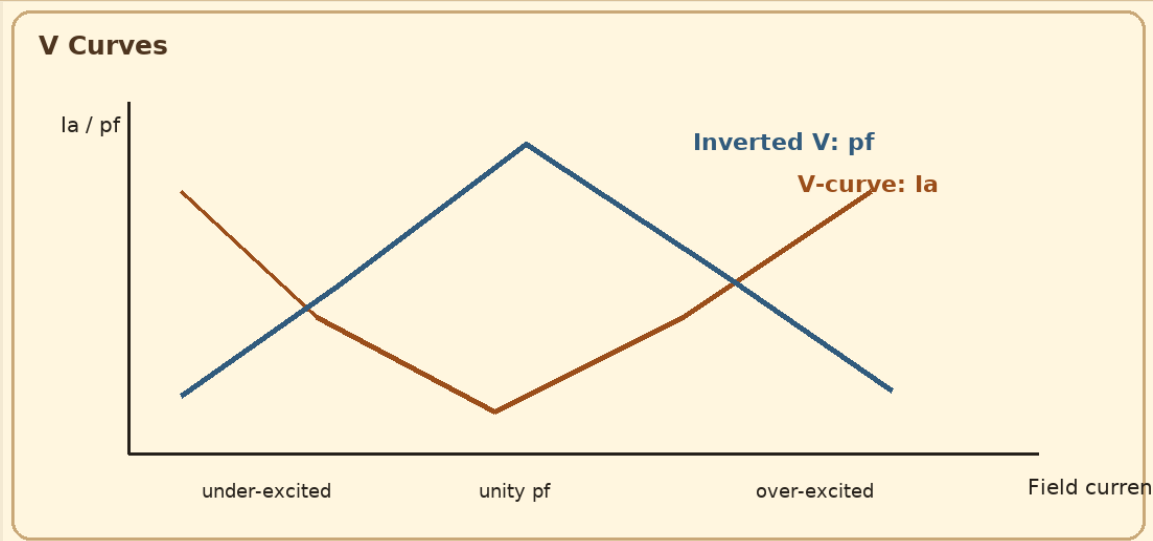
V-curves and inverted V-curves show how a synchronous motor behaves when field current is varied at constant terminal voltage and constant mechanical load.

Step 1: V-curve: plot of armature current I_a against field current I_f . I_a is high at low excitation, falls to a minimum near unity power factor, and rises again at over-excitation.

Step 2: Inverted V-curve: plot of power factor against I_f . The power factor improves from lagging to unity and then becomes leading as I_f is increased.

Step 3: at low I_f , the machine takes lagging reactive power. At high I_f , it supplies leading reactive power.

Final: V-curves and inverted V-curves show how a synchronous motor behaves when field current is varied at constant terminal voltage and constant mechanical load.; V-curve: plot of armature current I_a against field current I_f . I_a is high at low excitation, falls to a minimum near unity power factor, and rises again at over-excitation.



Question 196

Original: 4C | EM-II ELE 2202 Makeup, 16 June 2017

Why a synchronous motor is not self-starting? Explain any two methods of starting synchronous motor. (03)

Answer 196

A synchronous motor is not self-starting because the stator rotating magnetic field immediately rotates at synchronous speed, while the rotor is stationary at the instant of switching on. During successive half cycles the rotor experiences torque first in one direction and then in the opposite direction. Because of inertia, the rotor cannot accelerate instantly to synchronous speed, so the average starting torque is zero.

Method 1: damper-winding or induction-motor starting. Damper bars are embedded in the pole faces and short-circuited by end rings. With the field winding initially unexcited or suitably protected, the machine starts like a squirrel-cage induction motor. When the speed reaches nearly synchronous speed, DC excitation is applied and the rotor pulls into synchronism.

Method 2: pony-motor starting. A small auxiliary motor brings the rotor close to synchronous speed in the correct direction. The field is then excited and the stator is connected to the supply under synchronizing conditions. After the synchronous motor is locked in step, the pony motor is disconnected.

Another modern method is variable-frequency starting, where the supply frequency is raised gradually from a low value so that the rotor remains in synchronism during acceleration.

Final: A synchronous motor has zero average starting torque at standstill; it is commonly started by damper-winding induction start, pony motor, or variable-frequency starting.

Question 197

Original: 5A | EM-II ELE 2202 Makeup, 16 June 2017

An industrial plant is supplied with 850kVA of electrical power at 0.7pf lagging from a 3-phase 50Hz, 11kV substation. A synchronous motor of rating 110kVA operating at a leading power factor of 0.5 is added during expansion. Calculate the new kVA supplied and overall power factor of the plant. (05)

Answer 197

Use the synchronous-motor excitation or power-angle relation and separate real-power balance from reactive-power balance.

Step 1: Initial plant: $P_1 = 850 \times 0.7 = 595 \text{ kW}$; $Q_1 = 850 \sin(\cos^{-1} 0.7) = 607 \text{ kVAr}$.

Step 2: Motor: $P_2 = 110 \times 0.5 = 55 \text{ kW}$; $Q_2 = -110 \sin(\cos^{-1} 0.5) = -95.3 \text{ kVAr}$.

Step 3: Total $P = 650 \text{ kW}$; total $Q = 511.6 \text{ kVAr}$.

Step 4: New substation $\text{kVA} = \sqrt{(P^2 + Q^2)} = 827.1 \text{ kVA}$.

Final: Initial plant: $P_1 = 850 \times 0.7 = 595 \text{ kW}$; $Q_1 = 850 \sin(\cos^{-1} 0.7) = 607 \text{ kVAr}$.; Motor: $P_2 = 110 \times 0.5 = 55 \text{ kW}$; $Q_2 = -110 \sin(\cos^{-1} 0.5) = -95.3 \text{ kVAr}$.; Total $P = 650 \text{ kW}$; total $Q = 511.6 \text{ kVAr}$.; New substation $\text{kVA} = \sqrt{(P^2 + Q^2)} = 827.1 \text{ kVA}$.

Question 198

Original: 5B | EM-II ELE 2202 Makeup, 19 June 2018

An industrial load of 200 kW is supplied at 11 kV, the power factor being 0.8 lagging. A synchronous motor is used to meet an additional load of 50 kW and at the same time, it is used to raise the overall power factor to 0.9 lagging. Find the kVA capacity of synchronous motor and the power factor at which it operates. (06)

Answer 198

Use the synchronous-motor excitation or power-angle relation and separate real-power balance from reactive-power balance.

Step 1: for the original plant load, $P_1 = 200 \text{ kW}$, $Q_1 = 200 \tan(\cos^{-1} 0.8) = 150 \text{ kVAr}$.

Step 2: Since motor efficiency is not stated, take its active input as 50 kW because efficiency is not specified. The combined active power after adding the motor is 250 kW. The required total reactive power at 0.9 lagging = $250 \tan(\cos^{-1} 0.9) = 121.1 \text{ kVAr}$. - The synchronous motor reactive power is $121.1 - 150 = -28.9 \text{ kVAr}$ leading.

Step 3: the required synchronous-motor rating is $\sqrt{50^2 + 28.9^2} = 57.8 \text{ kVA}$.

Step 4: the motor power factor is $50/57.8 = 0.865$ leading.

Final: The required synchronous-motor rating is $\sqrt{50^2 + 28.9^2} = 57.8 \text{ kVA}$.; the motor power factor is $50/57.8 = 0.865$ leading.

Question 199

Original: 2B | EM-II ELE 2225 End Sem, 09 May 2024

A sugarcane industry is supplied with 92.5 kW of electrical power at 0.83 pf lagging from a three-phase, 50 Hz, 11 kV substation. A synchronous motor of rating 30 kVA operating at a leading power factor of 0.63 is added during the expansion. a) Calculate the real and reactive supplied by the substation, after expansion. b) Will the overall power factor of the industry improve after the addition of the synchronous motor? Justify your answer.

Answer 199

Use the synchronous-motor excitation or power-angle relation and separate real-power balance from reactive-power balance.

Step 1: Original load $Q = 62.160 \text{ kVAr}$ lagging.

Step 2: Synchronous motor: $P = 30 \times 0.63 = 18.900 \text{ kW}$; $Q = -30 \sin(\cos^{-1} 0.63) = -23.298 \text{ kVAr}$.

Step 3: After expansion: $P_{\text{total}} = 111.40 \text{ kW}$; $Q_{\text{total}} = 38.863 \text{ kVAr}$.

Step 4: Substation $\text{kVA} = 117.98 \text{ kVA}$; overall $\text{pf} = 0.94419$ lagging.

Step 5: Yes, the overall power factor improves from 0.83 to about 0.944 because the over-excited synchronous motor supplies leading kVAr.

Final: Synchronous motor: $P = 30 \times 0.63 = 18.900 \text{ kW}$; $Q = -30 \sin(\cos^{-1} 0.63) = -23.298 \text{ kVAr}$.; After expansion: $P_{\text{total}} = 111.40 \text{ kW}$; $Q_{\text{total}} = 38.863 \text{ kVAr}$.; Substation $\text{kVA} = 117.98 \text{ kVA}$; overall $\text{pf} = 0.94419$ lagging.; Yes, the overall power factor improves from 0.83 to about 0.944 because the over-excited synchronous motor supplies leading kVAr.

Question 200

Original: 2A | EM-II ELE 2225 Makeup, 26 June 2024

An industrial load of 500 kW, 0.707 pf lagging is required to be improved to 600 kW, 0.95 pf lagging by connecting a synchronous motor in parallel. Determine the kVA rating and power factor at which it operates. 4

Answer 200

Initial plant load is $P_1 = 500$ kW at 0.707 lagging power factor. Since $\cos \varphi_1 = 0.707$, φ_1 is approximately 45° and $Q_1 = P_1 \tan \varphi_1 = 500 \times 1 = 500$ kVar lagging.

After improvement, total real power is $P_2 = 600$ kW at 0.95 lagging power factor. For $\cos \varphi_2 = 0.95$, $\tan \varphi_2 = \sqrt{(1 - 0.95^2)/0.95} = 0.3287$. Therefore final reactive power drawn from supply is $Q_2 = 600 \times 0.3287 = 197.2$ kVar lagging.

The synchronous motor therefore changes the real power by $P_m = 600 - 500 = 100$ kW. Its reactive power is $Q_m = Q_2 - Q_1 = 197.2 - 500 = -302.8$ kVar. The negative sign means that the motor supplies leading reactive power.

The kVA rating of the synchronous motor is $S_m = \sqrt{(P_m^2 + Q_m^2)} = \sqrt{(100^2 + 302.8^2)} = 318.9$ kVA.

Motor power factor = $P_m/S_m = 100/318.9 = 0.314$. Since Q_m is negative, the motor operates at leading power factor.

Final: Synchronous motor rating = 319 kVA approximately, operating at 0.314 leading power factor.

Question 201

Original: 2C | EM-II ELE 2225 Makeup, 26 June 2024

A 3 phase, 20 MVA, star connected alternator with an impedance of $(0.5 + j6) \Omega$ per phase is operating in parallel with constant voltage 11 kV bus bars. The field current is adjusted to give a line excitation voltage of 12 kV. With constant excitation, calculate (a) Maximum power output from the alternator (b) Armature current and power factor under maximum power condition. 4

Answer 201

Use the synchronous-motor excitation or power-angle relation and separate real-power balance from reactive-power balance.

Step 1: $V = 11 / \sqrt{3} = 6.351$ kV / phase; $E = 12 / \sqrt{3} = 6.928$ kV / phase; $Z = 0.5 + j6 \Omega$, $|Z| = 6.021 \Omega$.

Step 2: for nonzero resistance, maximum output occurs at $\delta = \theta = \tan^{-1}(X / R) = 85.24^\circ$.

Step 3: $P_{\max} = 3EV / |Z| - 3V^2 / |Z| \cos \theta = 20.26$ MW.

Step 4: at maximum power, $I = (E \text{ at } \delta - V)/Z = 1495$ A.

Step 5: the operating power factor is 0.711 lagging.

Final: For nonzero resistance, maximum output occurs at $\delta = \theta = \tan^{-1}(X / R) = 85.24^\circ$.; $P_{\max} = 3EV / |Z| - 3V^2 / |Z| \cos \theta = 20.26$ MW.; at maximum power, $I = (E \text{ at } \delta - V)/Z = 1495$ A.; the operating power factor is 0.711 lagging.

Question 202

Original: 2C | EM-II ELE 2251 End Sem. 31 May 2023

A 1,000 kW, 3.3 kV, 24 poles, 50 Hz, 3-phase star-connected synchronous motor has a synchronous reactance of 3.4Ω per phase and the resistance is negligible. The motor is fed from an infinite bus bar at 3.3 kV. Its field excitation is adjusted to result in unity power factor operation at rated load. At this excitation, estimate the maximum power and torque this motor can deliver. Explain with the help of power-angle characteristics. (04)

Answer 202

At rated load and unity power factor, motor input current is $I = 1000 \text{ kW}/(\sqrt{3} \times 3.3 \text{ kV}) = 174.95 \text{ A}$.
Per-phase terminal voltage is $V = 3.3 \text{ kV}/\sqrt{3} = 1905.3 \text{ V}$.

For a synchronous motor with negligible resistance, $E = V - jX_s I$. At unity power factor, I is in phase with V . Therefore $E = 1905.3 - j(3.4 \times 174.95) = 1905.3 - j594.8 \text{ V}$. Hence $|E| = 1996 \text{ V}$ per phase and load angle is 17.34° .

With excitation kept constant, maximum power from the cylindrical-rotor power-angle relation is $P_{\max} = 3EV/X_s = 3 \times 1996 \times 1905.3/3.4 = 3.36 \text{ MW}$.

For 24 poles and 50 Hz, synchronous speed is $N_s = 120f/P = 250 \text{ rpm}$. Mechanical angular speed is $\omega_s = 2\pi \times 250/60 = 26.18 \text{ rad/s}$.

Maximum torque is $T_{\max} = P_{\max}/\omega_s = 3.36 \times 10^6/26.18 = 128 \text{ kN m}$.

Final: Maximum power = 3.36 MW and maximum torque = 128 kN m at the same excitation.

Question 203

Original: 3B | EM-II ELE 2251 End Sem. 31 May 2023

Write a technical note on hunting in synchronous motors and how it can be mitigated. (02)

Answer 203

Hunting is the oscillation of the rotor of a synchronous motor about its steady load angle after a sudden change in load, excitation, or supply condition. Because of rotor inertia, the rotor cannot immediately settle at the new torque angle; it overshoots and oscillates around the new equilibrium position.

During hunting, load angle, stator current, electromagnetic torque, and power input fluctuate. If the oscillation is severe, the motor may pull out of synchronism.

Hunting is mitigated by damper or amortisseur windings placed in the pole faces. Relative motion between the rotor and rotating field induces currents in these short-circuited bars, and the resulting torque opposes the oscillation. Other measures are adequate flywheel effect for pulsating loads, avoiding sudden load changes, proper excitation control, and power-system stabilizers.

Final: Hunting is rotor-angle oscillation about synchronism; damper windings are the main practical damping method.